CHAPTER 4
CHEMICAL BONDING

4.1
Lewis Structure

INTRODUCTION

Chemical bonding refers to the theories, ideas and models chemist use to explain how atoms are held together to form molecule and compound.

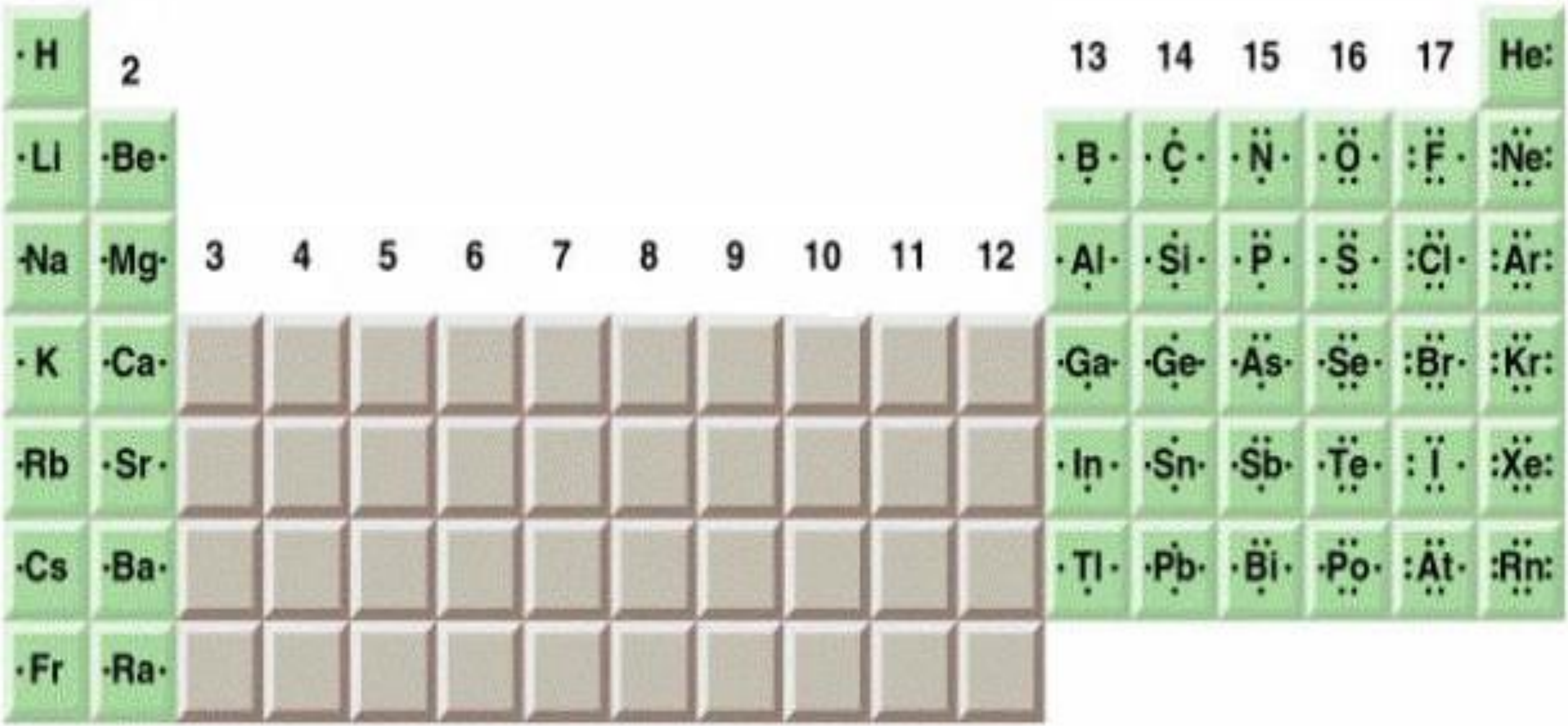
Lewis Dot Symbol

- ✔ A Lewis dot symbol of an element consists of a chemical symbol with one or more dots placed around it.
- ✔ Each dot represents a valence electron.

Example:
Sodium atom
$1s^2\ 2s^2\ 2p^6\ \underline{3s^1}$

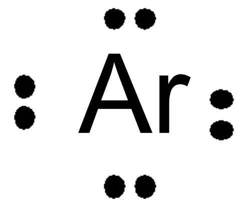
Example:
Sulphur atom
$1s^2\ 2s^2\ 2p^6\ \underline{3s^2\ 3p^4}$

Lewis Dot Symbol of Elements



OCTET RULE

- **Octet rule** states that **atoms** tend to **lose, gain or share electrons** in order to **attain the stable electronic configuration** of a noble gas or surrounded by eight valence electrons.
- In terms of Lewis dot symbols, an **octet consists of 4 pairs of valence electrons.**



- Octet rule is obeyed by almost all **elements of *s* and *p* blocks** with a few **exceptional cases**.
- For example :
  - **H atoms achieve duplet** (surrounded by 2 valence electrons) instead of octet.
  - Transition metals **do not** obey octet rule.

- **Atoms** achieve stability by attaining **stable configurations** as follows:

Group 1, 2 and 13 (metal) — lose e^- to form cation → **NOBLE GAS CONFIGURATION**

Group 15, 16 and 17(non-metal) — gain e^- to form anion → **NOBLE GAS CONFIGURATION**

***d* block elements (transition metal)**

- lose e^- to form cation → **PSEUDO-NOBLE GAS CONFIGURATION**
- lose e^- to form cation → **HALF-FILLED ORBITAL**

i. Stability of Noble Gas Configuration

- In order to achieve octet, **metal atoms tend to lose electrons** while **non-metal atoms gain electrons**.

Element	Electronic Configuration	Cation/ Anion	Electronic Configuration
$_{3}Li$	$1s^2 2s^1$	Li^+	$1s^2$
$_{13}Al$	$1s^2 2s^2 2p^6 3s^2 3p^1$@ $[Ne]\ 3s^2 3p^1$	Al^{3+}	$1s^2 2s^2 2p^6$@ [Ne]
$_{16}S$	$1s^2 2s^2 2p^6 3s^2 3p^4$@ $[Ne]\ 3s^2 3p^4$	S^{2-}	$1s^2 2s^2 2p^6 3s^2 3p^6$@ [Ar]

ii. Stability of Pseudo-noble Gas Configuration

- This stability is for ***d* block elements (transition metal)** of which the **electrons from *4s* orbital are transferred first before the *3d* electrons.**
- Pseudo means **"*false*"**.
- Pseudo-noble gas configuration **does not** have the configuration of a noble gas.
- The valence electronic configuration attained is ***ns^2 np^6 nd^{10}*.**

Example:

Zn $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$

Zn^{2+} $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$ @ [Ar] $3d^{10}$

iii. Stability of Half-filled Orbital

- ***d* block elements** can donate electrons to obtain the stability of the **half-filled orbitals.**
- The valence electronic configuration attained is $\boldsymbol{ns^2\ np^6\ nd^5}$

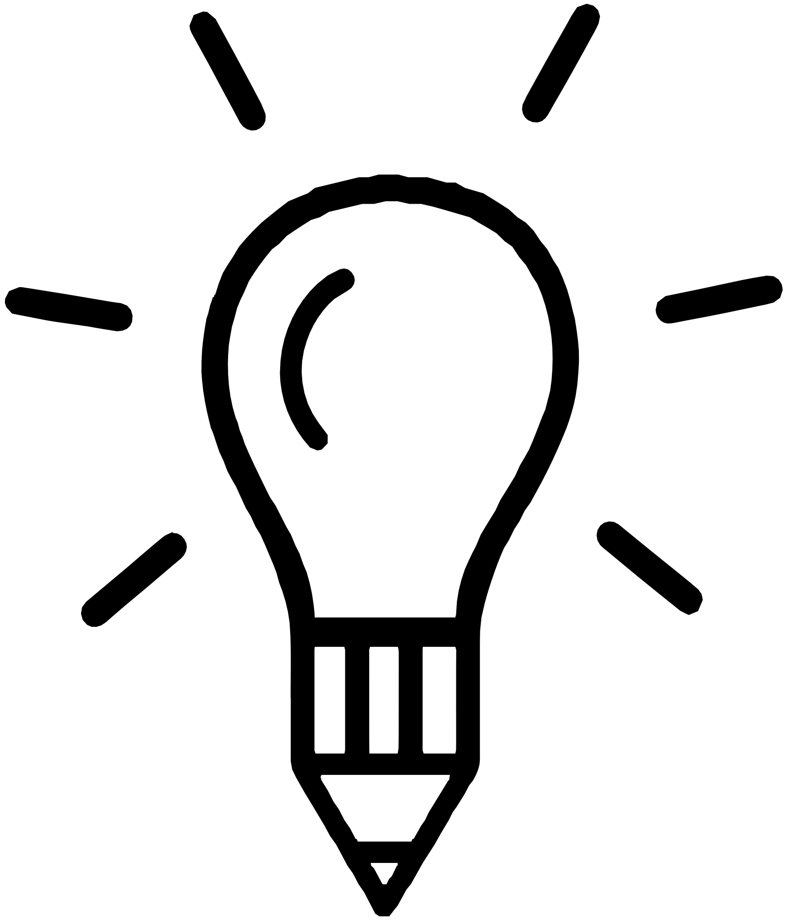
Example:

Fe $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$

Fe^{3+} $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$

Half-filled *d* orbital

FORMATION OF BONDS USING LEWIS DOT SYMBOL FOR:



01 Ionic or electrovalent bond

02 Covalent bond

03 Dative or coordinate bond

1. IONIC / ELECTROVALENT BOND

Definition

The **electrostatic force of attraction** between the **positive** and **negative ions** in an ionic compound.

Formation

Metal atoms lose ē to form **cations**
Non-metal atoms gain ē to form **anions.**

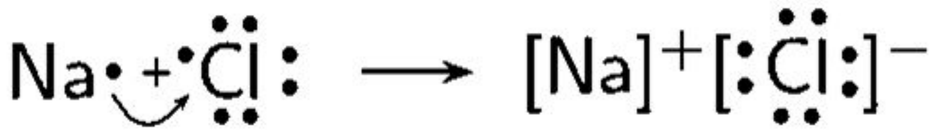
Cation

Valence electron in neutral atom is **removed** to achieve stable electronic configuration.

Anion

The valence shell is **completely-filled** with valence ē (octet).

Example:



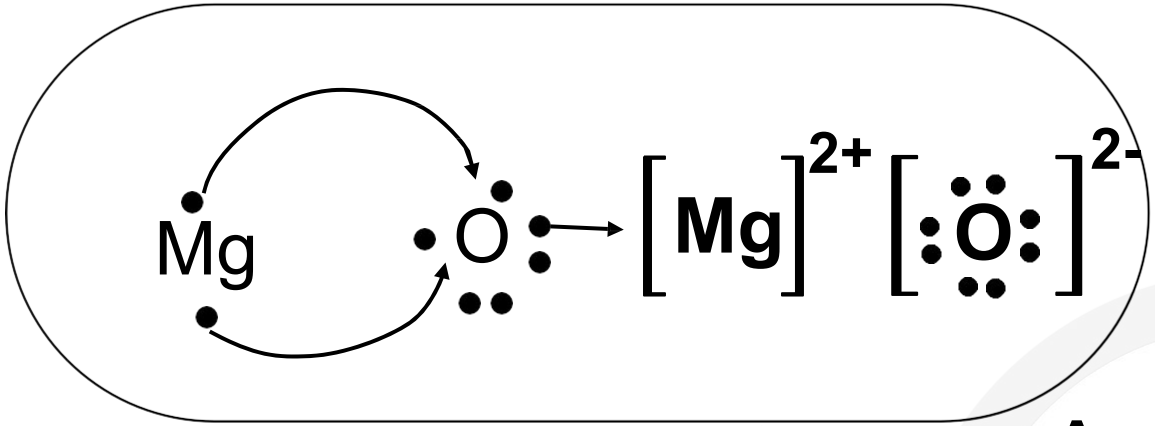
Lewis dot symbol

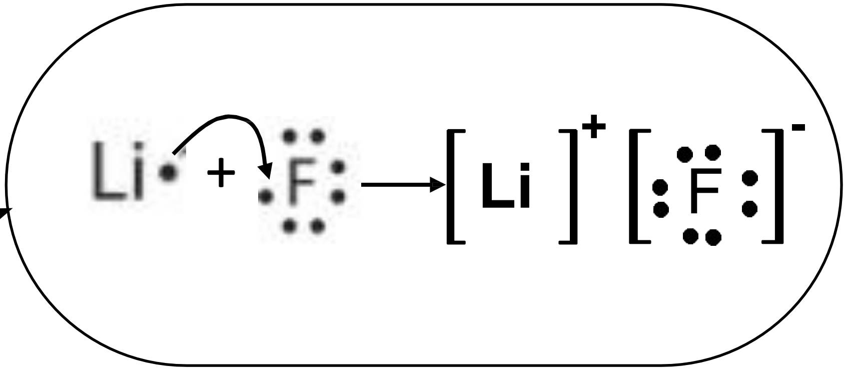
Lewis structure

NaCl

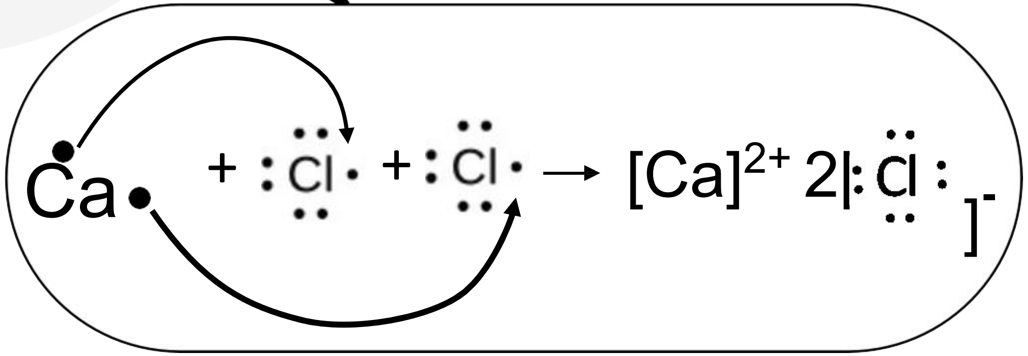
Formula unit of ionic compound

EXAMPLE:





Arrows show the transfer of electrons from metal atom to non-metal atoms



2. COVALENT BOND

Definition

A covalent bond is the bond formed between atoms by a **sharing of electrons**.

Formation

Atoms **share electrons** so that the **stable electronic configuration** of the noble gas can be obtained.

Involves atoms with **small difference in electronegativity**,

i.e. non-metal atom and non-metal atom.

(some exceptional cases : $AlCl_3$, $BeCl_2$)

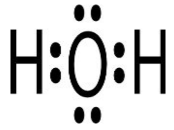
COVALENT MOLECULES WITH SINGLE, DOUBLE OR TRIPLE BOND

<u>Single bond</u>
Two atoms share one pair of electrons.

<u>Double bond</u>
Two atoms share two pairs of electrons.

<u>Triple bond</u>
Two atoms share three pairs of electrons.

single covalent bonds



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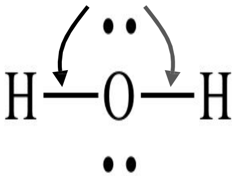

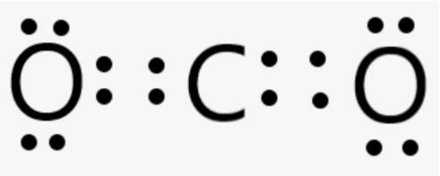

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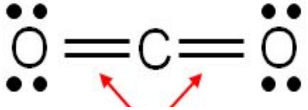
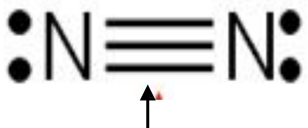


double bond

or



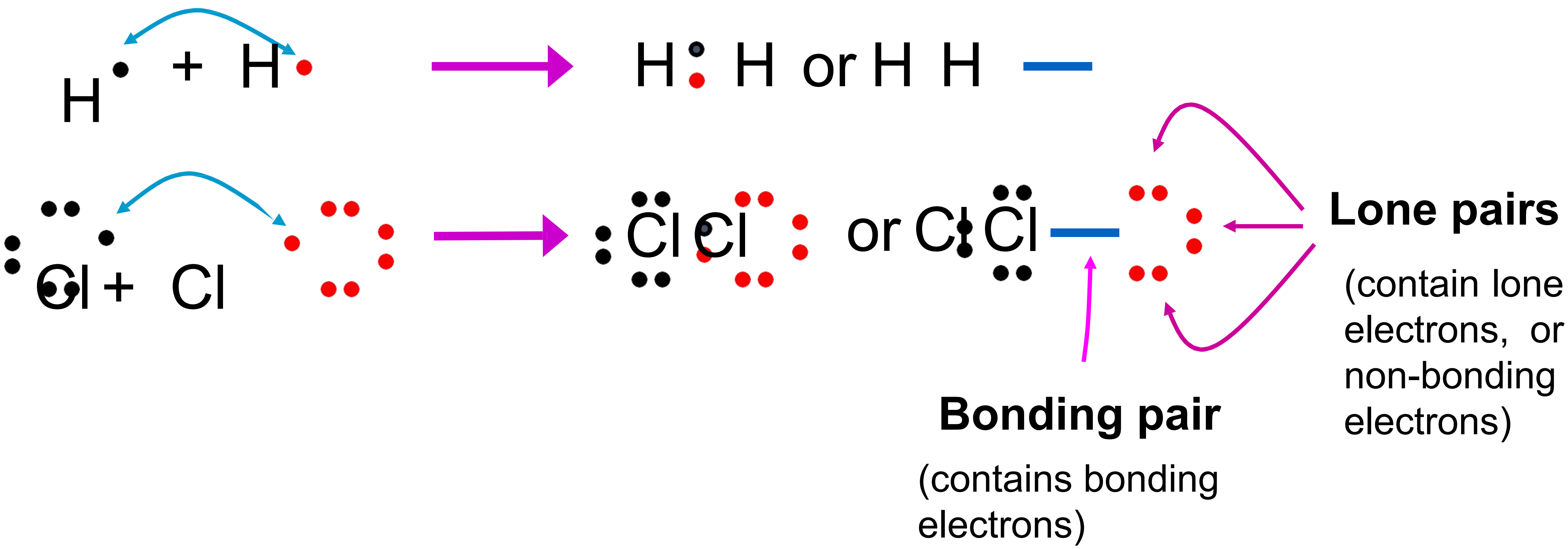
triple bond

i. Single bond

A single bond is formed when a pair of electrons are shared between two atoms.

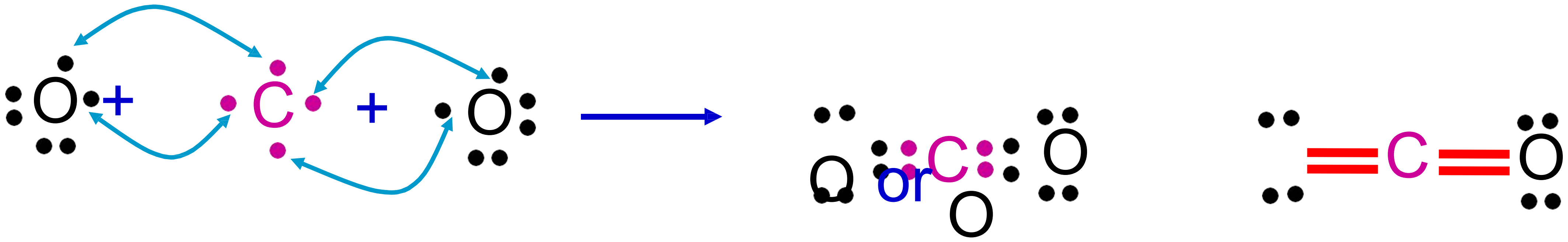
Example:

<u>Lewis dot symbol</u> **<u>Lewis structure of covalent molecules</u>**



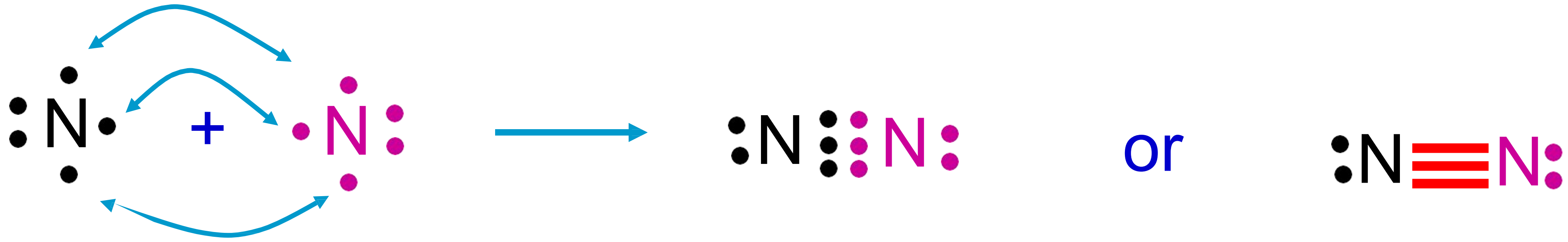
ii. Double bond

A double bond is formed when two electron pairs are shared between two atoms.



iii. Triple bond

A triple bond is formed when three electron pairs are shared between two atoms.

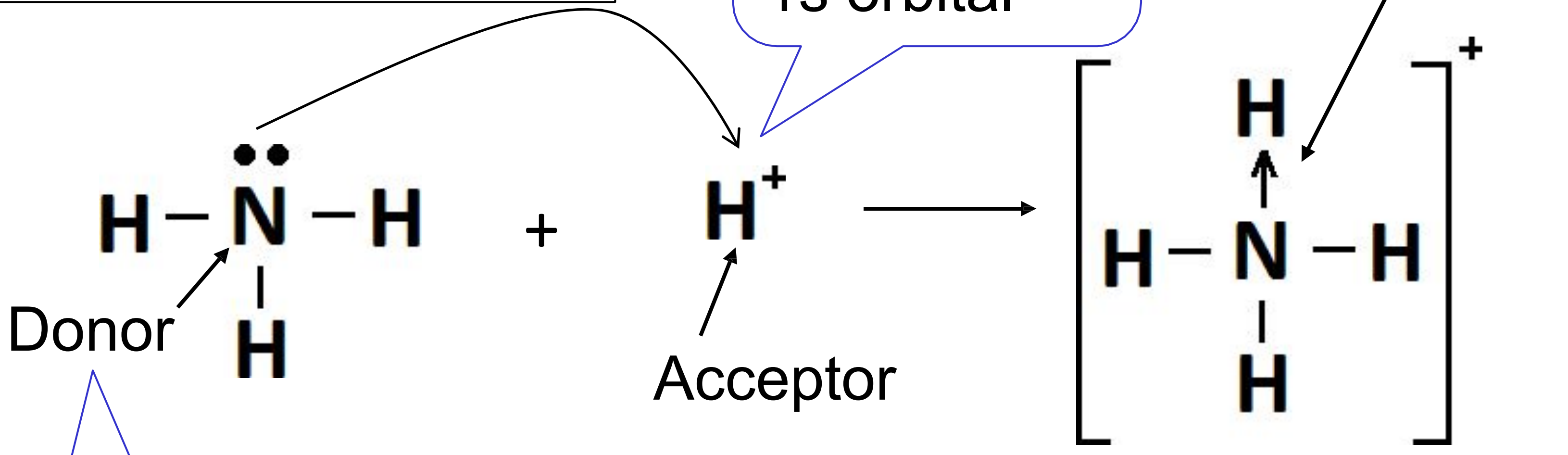


3. COORDINATE BOND / DATIVE BOND

A dative bond is a covalent bond in which the shared pair of bonding electrons is provided by only one of the bonding atoms.

EXAMPLE 1:
Ammonium ion, NH_4^+

the H^+ ion has an empty 1s orbital



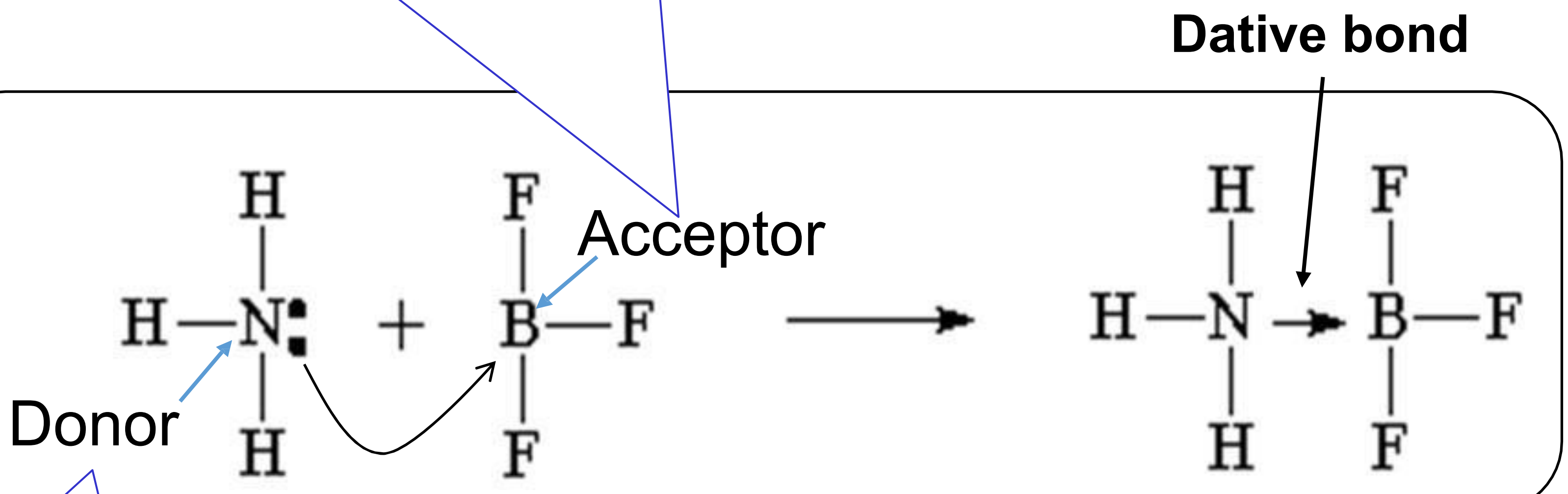
The N atom in the NH_3 molecule has one lone pair of electrons.

- **The N atom donates its lone pair for sharing with the empty 1s orbital of H^+ ion** to form a **dative bond**.
- H atom achieve **duplet** in NH_4^+ ion.

COORDINATE BOND / DATIVE BOND

EXAMPLE 2: Ammonia-boron trifluoride compounds

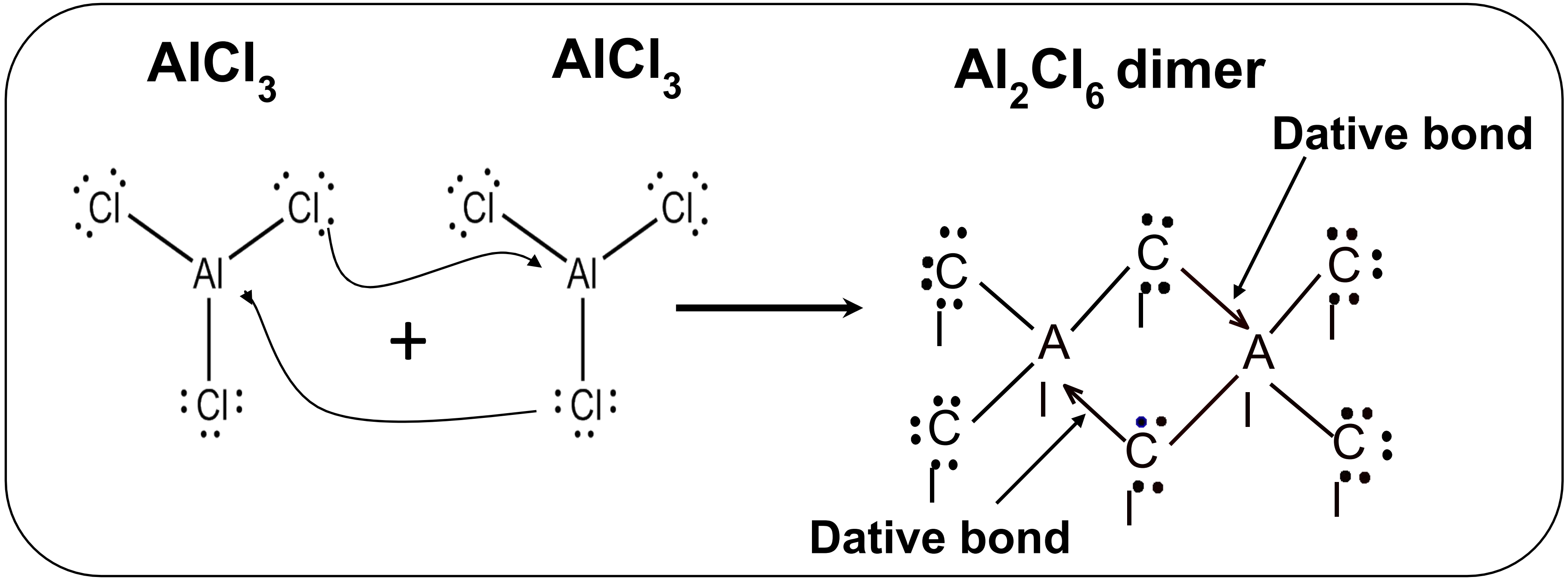
The **B atom** in BF_3 molecule has only **6 valence electrons** and **requires 2 electrons to complete octet configuration**.



The **N atom** in the NH_3 molecule **has one lone pair** of electrons.

- The **N atom** of NH_3 donates its **lone pair** for sharing with the **B atom** of BF_3 to form a **dative bond**.
- Thus, B atom achieves **octet.**

EXAMPLE 2: Al_2Cl_6

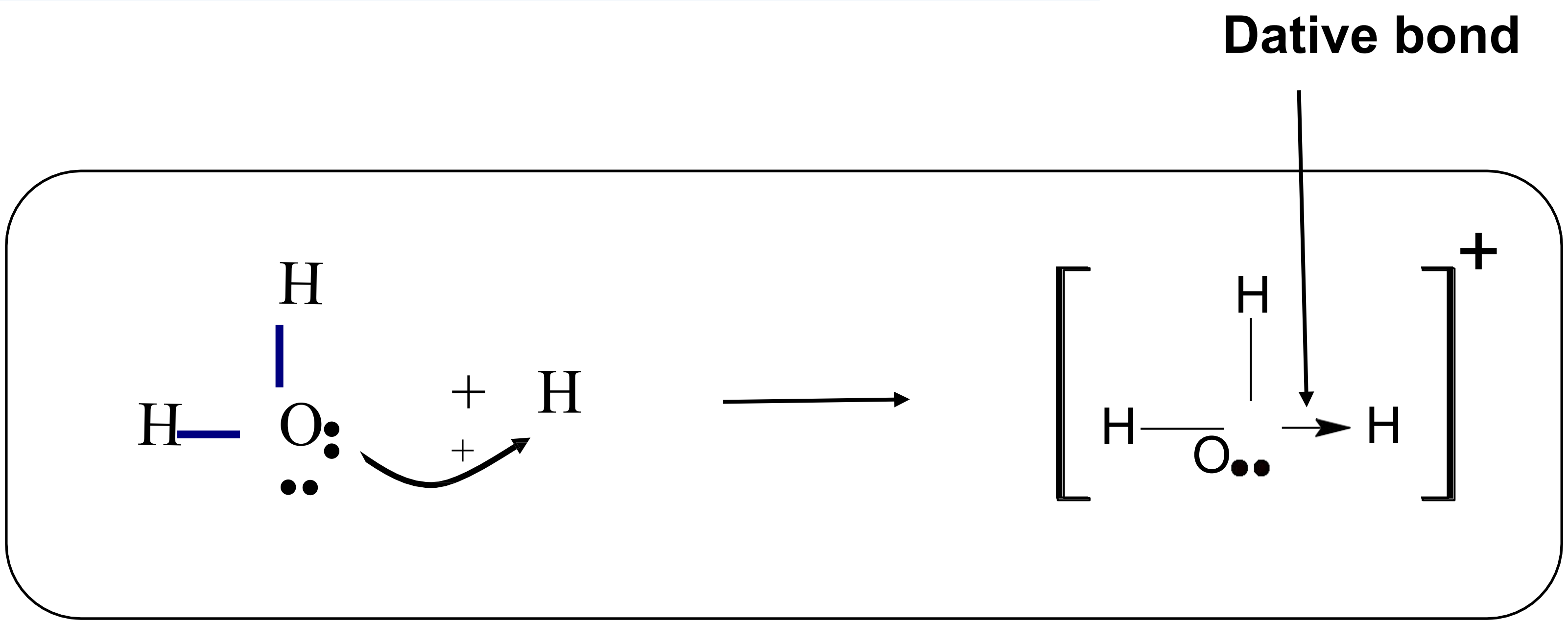


- Each Al atom in $AlCl_3$ molecule has **only 6 valence electrons** and requires 2 electrons to complete octet configuration.
- Each Cl atom has **3 lone pairs**.
- Two $AlCl_3$ molecules will be held together by **two dative bonds** to form a Al_2Cl_6 molecule, known as **dimer.**
- The **bonding electrons** in the dative bonds are **provided by only Cl atoms**.
- Al atoms achieve **octet** after sharing the electrons.

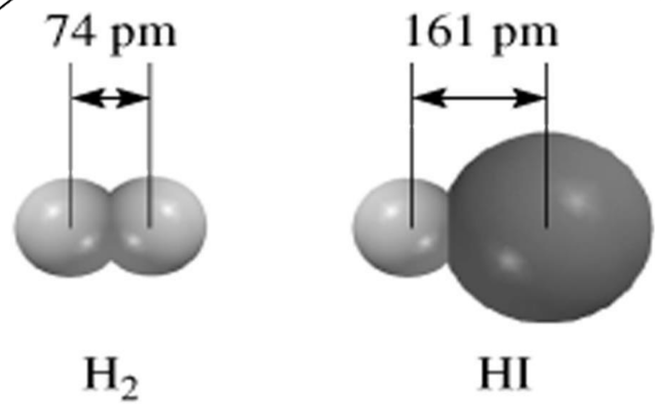
COORDINATE BOND / DATIVE BOND

Example 4:

Hydronium ion, H_3O^+



- The O atom in H_2O has 2 lone pairs while H^+ ion has an empty 1s orbital.
- The O atom can donate one of its lone pairs for sharing with the empty 1s orbital of H^+ ion to form a dative bond.
- H atom achieve duplet in H_3O^+ ion.



Bond length is the **distance between the nuclei of two covalently bonded atoms** in a molecule

Comparison Of Bond Length

Bond Type	Bond Length (pm)
C-C	154
C=C	133
C≡C	120
C-N	143
C=N	138
C≡N	116

increasing bond length →

triple bond ,double bond , single bond

← increasing number of bonds

WRITING THE LEWIS STRUCTURE OF COVALENT COMPOUNDS

1.

- Determine the **central atom** (the least electronegative atom).
- **Hydrogen is never a central atom**.

2.

- Count the **total number of valence ē** from all atoms in the molecule or ion.
- The number of valence ē is **deduced from the group** in the periodic table to which the element belong.
- For ions, add 1 for each negative charge and subtract 1 for each positive charge.

3.

- Draw a **skeletal structure** by joining the atoms together.

4.

- Draw a **single covalent bond** between the central atom and each of the terminal atoms.

WRITING LEWIS STRUCTURE OF MOLECULES / POLYATOMIC ION

5.

- Calculate the number of **non-bonding electrons.**
- Since 1 covalent bond is formed by using 2 electrons. Thus, **Non-bonding ē = Valence ē - bonding ē**

6.

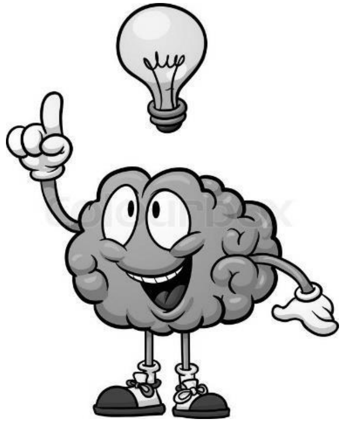
- Assign the remaining electrons to the **terminal atom** so that each terminal atom has **8 electrons.**
- Take note that if **hydrogen** is the terminal atom, it can only have **2 electrons(duplet)**.

7.

- Place the **balance** of the electrons **to the central** atom.

8.

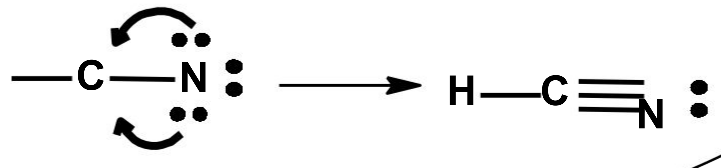
- If the central atom is from the **Period 3** elements, it may have **more than 8 electrons.**
- If the octet rule is not satisfied for the central atom, try **adding double or triple bonds** between the



LEWIS STRUCTURES

***Central atom** is the least electronegative atom (except for H atom)*

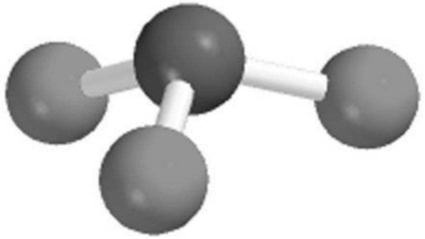
If the central atom still has less than an octet, try to form double or triple bonds by using the lone pairs from the terminal atoms.

Example: 

*Elements that form **double bond** :*
C, N, O and S.

*Elements that form **triple bond**:*
C and N

DRAWING LEWIS STRUCTURE



Example 1: NF_3

STEP 1

N is less electronegative than F

N is the **central** atom.

STEP 2

Count total number **valence** electrons

N : 1 x 5e
F : 3 x 7e

26 e

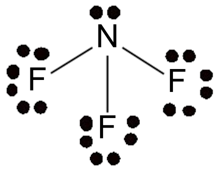
STEP 3

Draw single bonds between N and F atoms (skeletal) and calculate the non-bonding e^-.

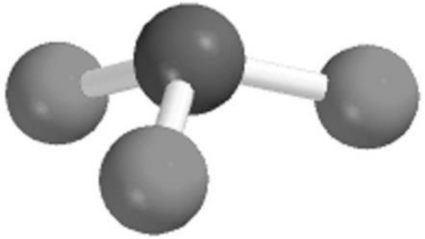
Non-bonding e^-
= 26 – 3 (2)
= 20

STEP 4

Complete the **octet the terminal** atom first , then **remaining** electron put on **central** atom



DRAWING LEWIS STRUCTURE



Example 2 : NH_4^+

STEP 1

N is less electronegative than H

N is the **central** atom.

STEP 2

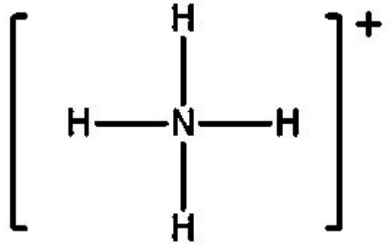
Count total number of valence electrons.
For **polyatomic ions, subtract one electron** for **each positive charge**

N : 1 x 5e
H : 4 x 1e
+ve charge : -1e
8 e

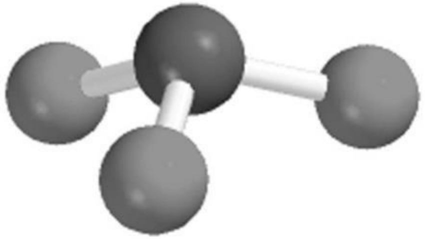
STEP 3

Draw single bonds (**skeletal**)between **N and H** atoms (H only duplet),and calculate the non-bonding e$^-$.

Non-bonding e$^-$
8 – 4 (2) = 0



DRAWING LEWIS STRUCTURE



Example 3: CO_3^{2-}

STEP 1

C is less electronegative than O — **C** is the **central** atom.

STEP 2

Count total number valence electrons.
For **polyatomic ions, add one electron** for **each negative charge.**

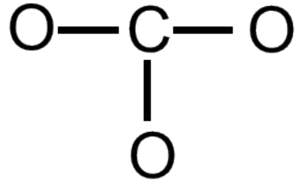
C : 1 x 4e
O : 3 x 6e
-ve charge: + 2e
24e

STEP 3

Draw single bonds (**skeletal**) between **C and O atoms** and calculate the non-bonding e^-.

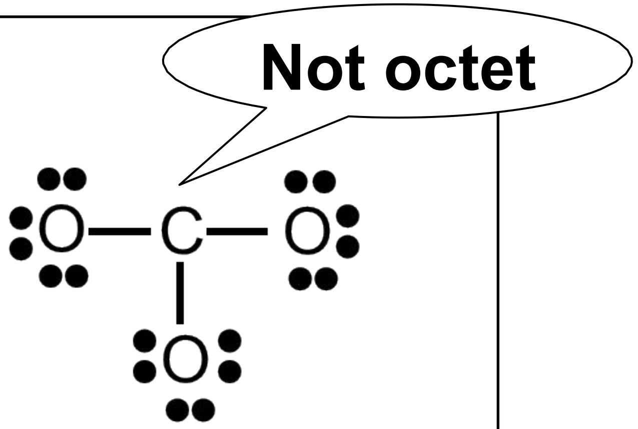
Non-bonding e^-

24 – 3 (2) =18



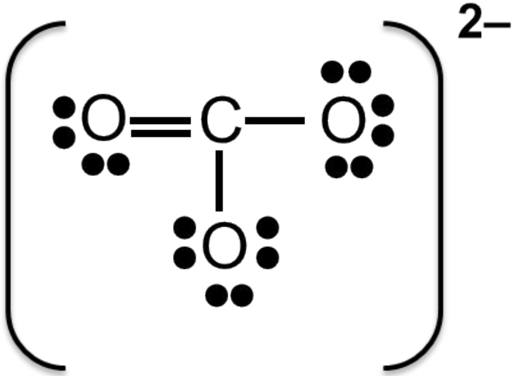
STEP 4

Complete the **octet** on **terminal atom** first (**oxygen atom**). As we can see there is **no remaining electron,** but the **C atom doesn't achieve octet**



STEP 5

If the octet rule is **not satisfied** for the central atom, try **adding double or triple bonds** between the terminal and the central atom.



FORMAL CHARGE

Formal charge is the ***apparent charge*** assigned to a bonded atom in a Lewis structure.

Formal charge is used to determine the ***most stable*** or the ***most plausible Lewis structure***.

$\sum$ ***Formal charge*** **= Charge on the molecule or ion**

Formal charge =

No. of valence ē – [(Non-bonding ē) + ½ (No. of bonding ē)]

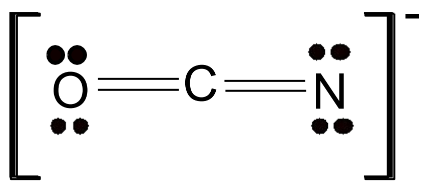
The most stable or plausible Lewis structure is the one with

1. the **lowest formal charges** (i.e. formal charge of each atom is <u>zero</u> or <u>close to zero</u>).
2. the **negative formal charge** appears or resides on the **most electronegative atom.**

Example :

Draw all Lewis structure of NCO^- and determine the best structure.

SOLUTION:



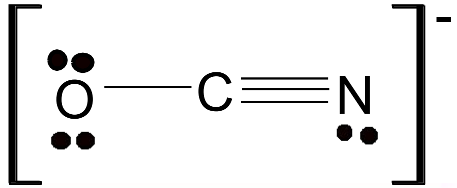
Structure (A)

Formal charges :

$C = 4 - [0 + ½(8)] = 0$

$N = 5 - [4 + ½(4)] = -1$

$O = 6 - [4 + ½(4)] = 0$



Structure (B)

Formal charges :

$C = 4 - [0 + ½(8)] = 0$

$N = 5 - [2 + ½(6)] = 0$

$O = 6 - [6 + ½(2)] = -1$

The most plausible structure is **structure (B). O atom is more electronegative than N atom**, hence **O atom should take up formal charge of -1**.

EXCEPTION TO THE OCTET RULE

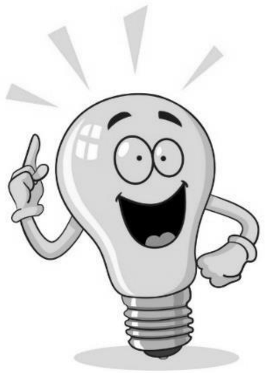
There are covalent molecules that disobey / violate the octet rule.

Exception to the octet rule

1) Incomplete Octet

2) Expanded Octet

3) Odd number electrons

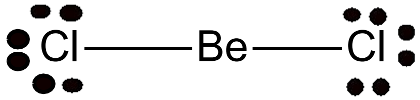


1) INCOMPLETE OCTET

A central atom has **<u>less than 8 electrons</u>** in its **valence shell**.

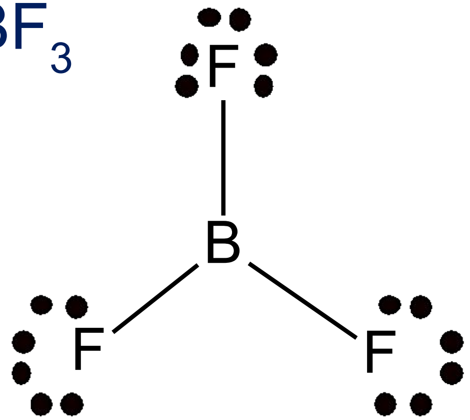
Example: **Beryllium, boron** and **aluminum** do not achieve octet configuration even after sharing electron with other atoms.

$BeCl_2$

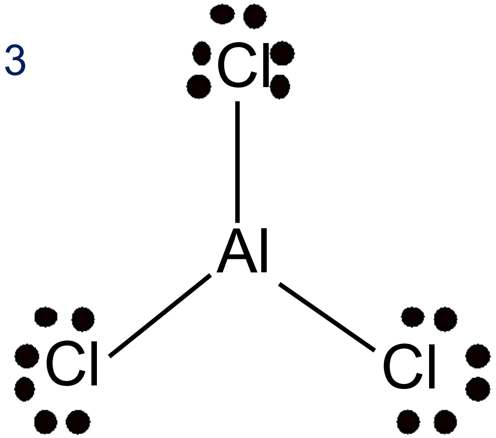


Be surrounded by only **four** electrons and **not eight.**

BF_3



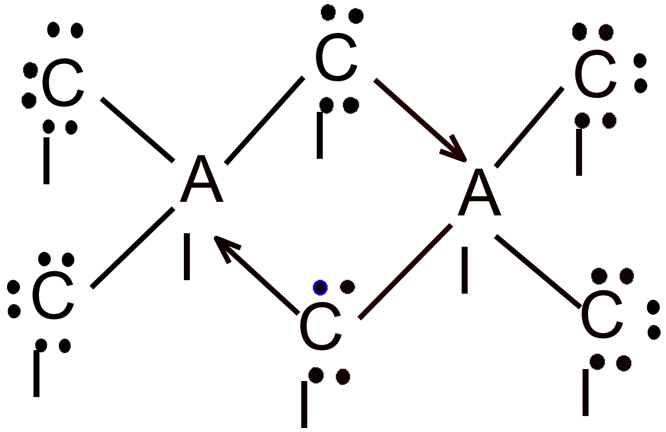
$AlCl_3$



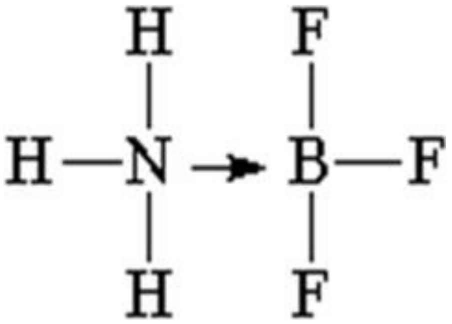
For cases of molecules with incomplete octet, there is a high tendency to achieve octet through dimerisation or bridging involving the formation of dative bond.

Example :

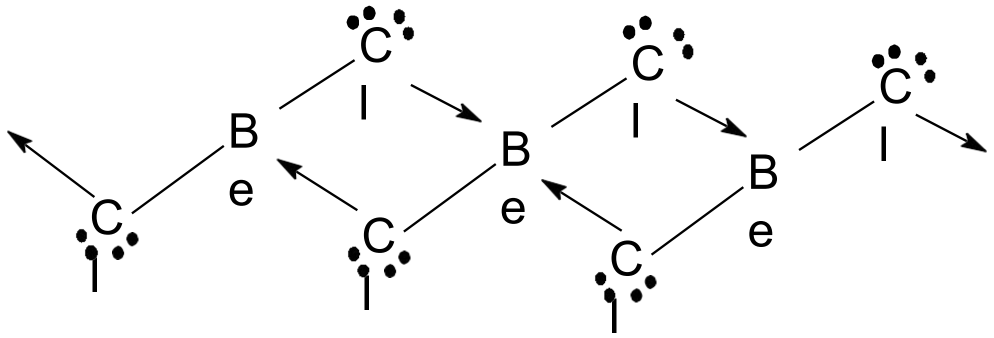
Al_2Cl_6 dimer



$NH_3.BF_3$ compound



$BeCl_2$



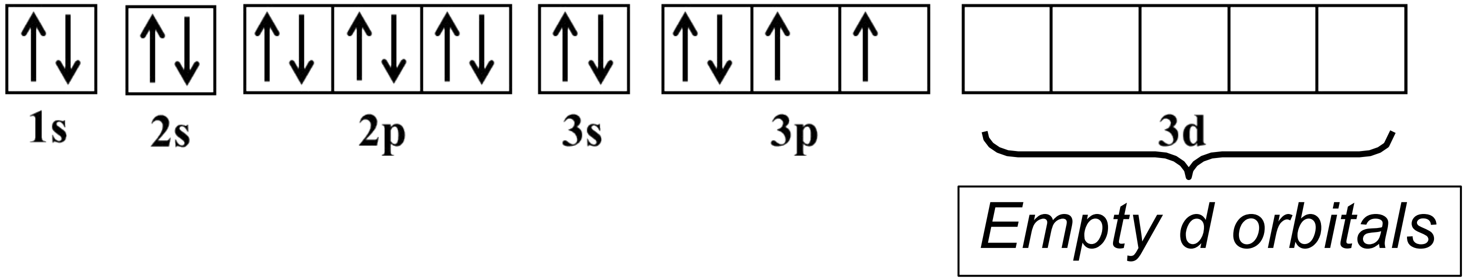
Remark :
Bridging in $BeCl_2$ is only for reference (not in CS)

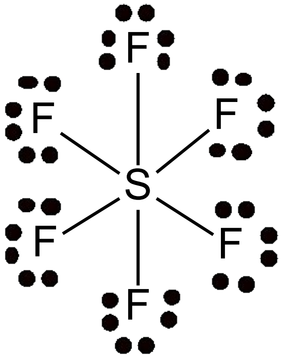
2) EXPANDED OCTET

A central atom has **more than 8 electrons** in its **valence shell**.

When expanding the octet, the central atom uses the **empty *d* orbital** available at its valence shell (**non-metal elements in Period 3 and above**).

The electronic configuration for atom of S :

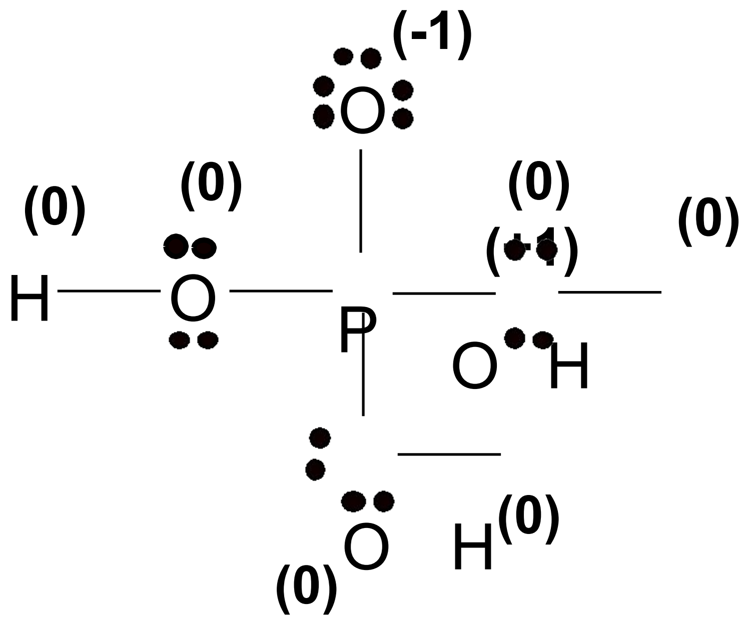




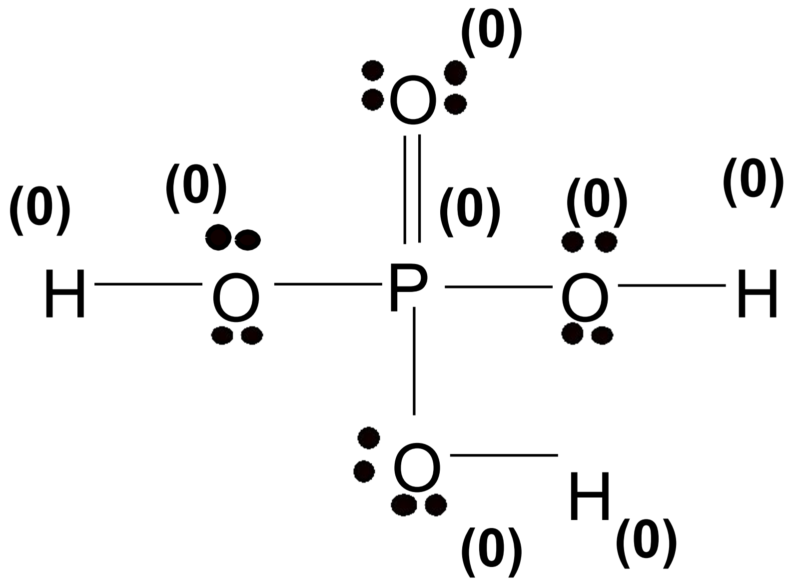
The central atom S can accommodate **12 valence ē** due to the presence of the empty *d* orbitals.

Example:

Draw all possible Lewis structures for the H_3PO_4 molecule and determine the more stable structure.



STRUCTURE I

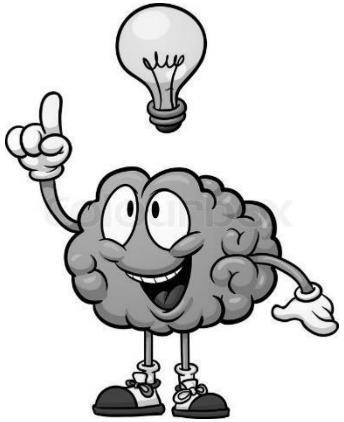


STRUCTURE II

- Note that H_3PO_4 molecule contains **P which is a Period 3 element** that can have **<u>expanded octet</u>.**

- The **most stable Lewis structure** is structure II because all of the atoms bear the lowest formal charges@ have formal charge of zero.

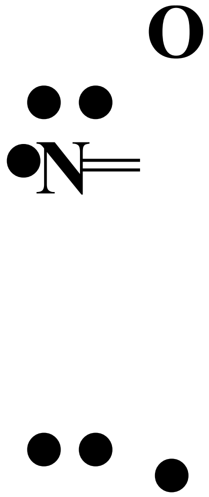
3) ODD NUMBER ELECTRONS



Odd-electron molecules have a **central atom from an odd-numbered group**, such as nitrogen, N (Group15) and chlorine, Cl (Group 17)

Example : NO

The number of valence e^- at central atom, N **is seven (odd number)**



CONCEPT OF RESONANCE/RESONANCE STRUCTURES

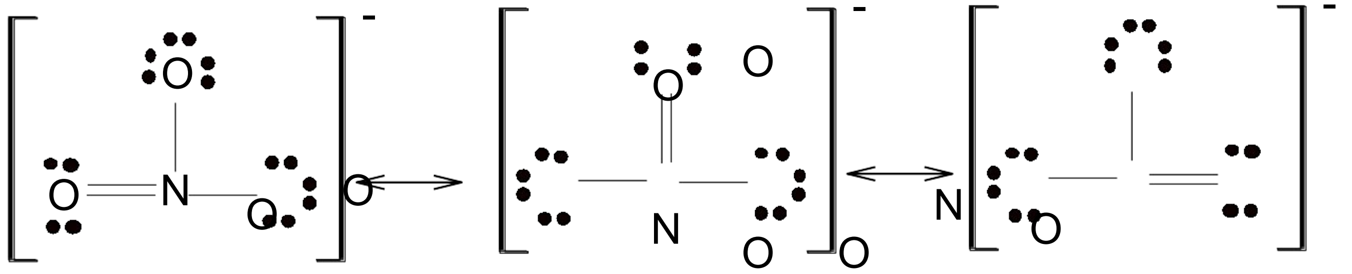
Resonance structures are two or more Lewis structures having the **same arrangement of atoms** but **differ from one another in the position of their electrons.**

None of the individual Lewis structure can accurately represent the actual structure of the molecule or polyatomic ion.

Resonance occurs as a result of the delocalisation of pi electrons

The **double headed arrow** (↔) indicates that the structures shown are resonance structures.

Example: NO_3^-

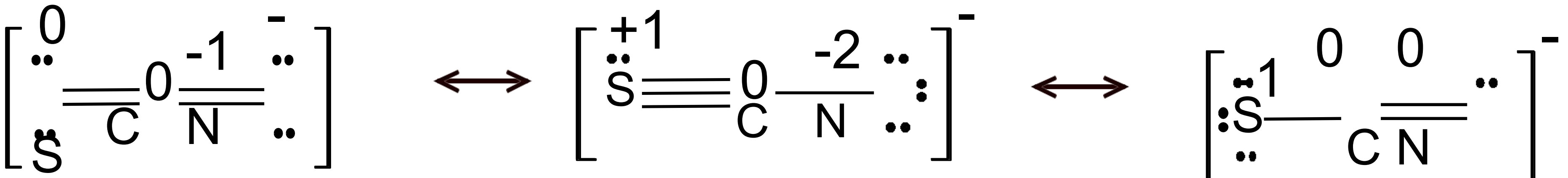


All the **N-O bond lengths** are the **intermediate between a single N-O bond and a double N-O bond**.

Some resonance structures have different terminal atoms.

Thus, the resonance structures do not contribute equally to the actual structure of the molecule or ion.

Example: SCN^- ion



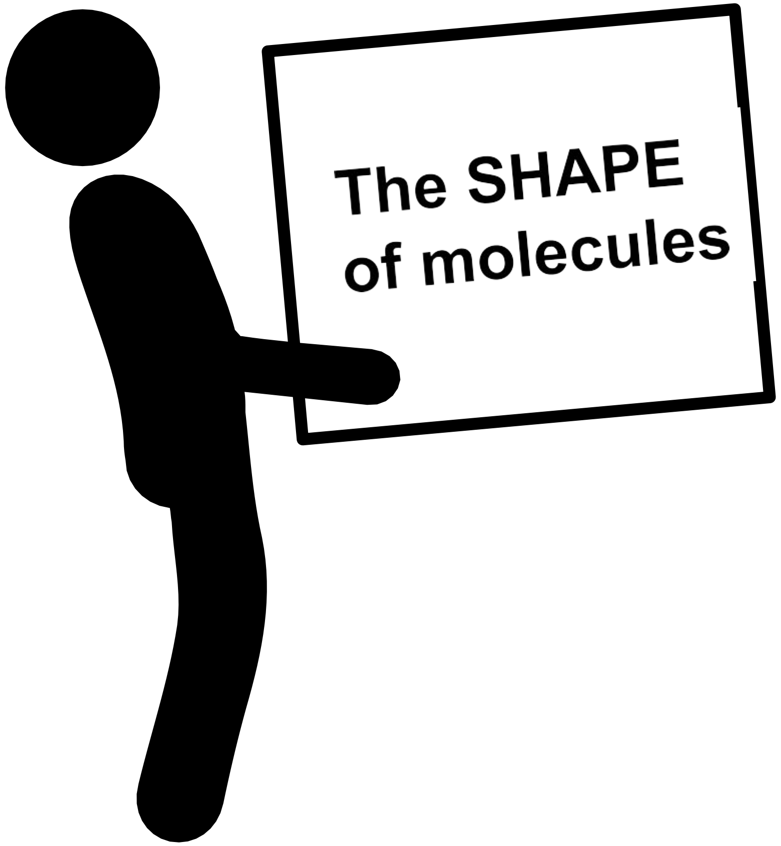
Structure I **Structure II** **Structure III**

Structure I is the **most stable Lewis structure** because it has the **lowest formal charges** and the **negative formal charge** (-1) resides on the **most electronegative atom**, N.

(note: N is more electronegative than S)

4.2
Molecular Shape And Polarity

MOLECULAR SHAPE AND POLARITY



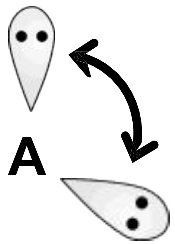
- ✓ Molecules are three-dimensional objects.
- ✓ Although Lewis structures show which atoms are connected to which and even provide a basis estimate the distances between bonded atoms, they reveal nothing about the way the atoms are situated in space.
- ✓ An extension of Lewis theory is required to predict the angles defined at an atom by two or more covalently bonded neighbours ~ VSEPR.

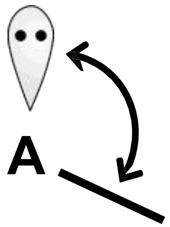
① **The electron pairs in a valence shell surrounding a central atom repel one another.** Thus, the bonding pair and lone pair electron arranged themselves to be as far apart as possible to minimise the repulsion.

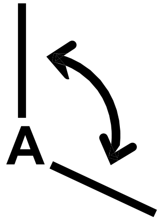
② **A lone pair repels other electron pairs more strongly than a bonding pair.**

The repulsion between electron pairs decreases in the order:

L.P – L.P repulsion **>** ***L.P – B.P repulsion*** **>** ***B.P – B.P repulsion***







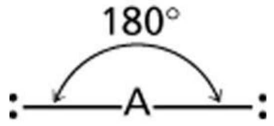
Valence
Shell
Electron
Pair
Repulsion
(**VSEPR**)
theory.

L.P denotes lone pair
B.P denotes bonding pair

ELECTRON PAIR GEOMETRY

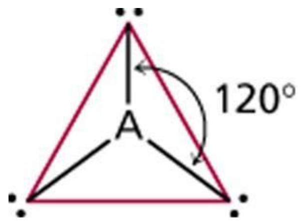
The arrangement of electron pairs about the central atom of an **AX_n** **molecule or ion** is known as **electron pair geometry.**

linear



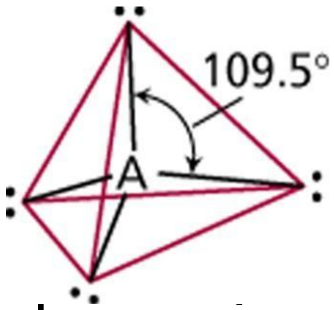
2 electron pairs

trigonal planar



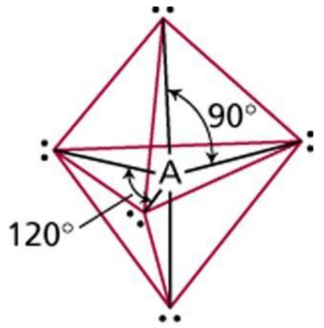
3 electron pairs

tetrahedral

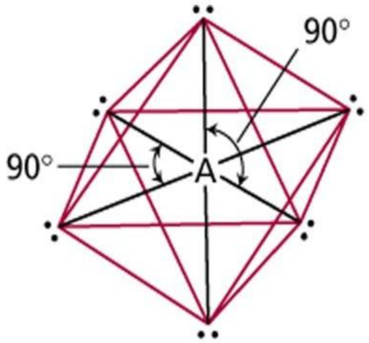


4 electron pairs

trigonal bipyramidal



5 electron pairs



octahedral

6 electron pairs

A balloon analogy for electron pair geometry

linear

2 electron pairs

trigonal planar

3 electron pairs

tetrahedral

4 electron pairs

trigonal bipyramidal

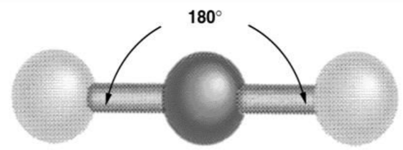
5 electron pairs

octahedral

6 electron pairs

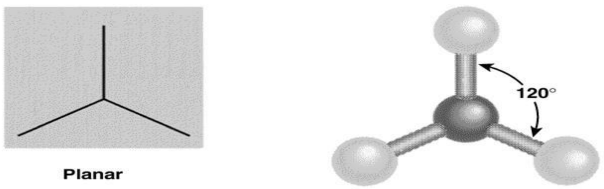
5 BASIC MOLECULAR GEOMETRY (without lone pairs around central atom)

1 Linear



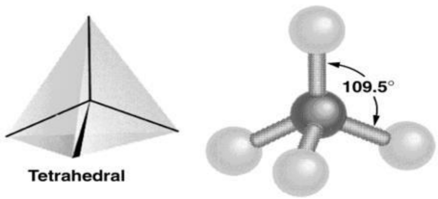
Bond angle = 180°
Two bonding pair electrons.
General formula : AX_2

2 Trigonal Planar



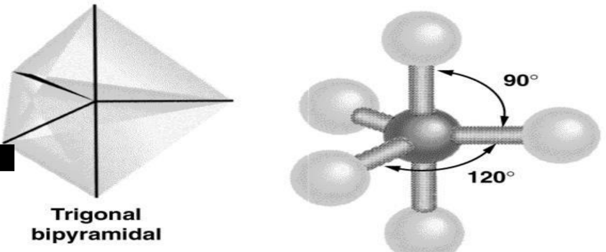
Bond angle = 120°
Three bonding pair electrons.
General formula : AX_3

3 Tetrahedral



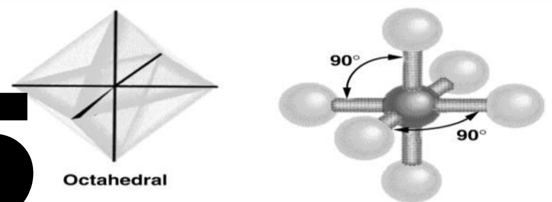
Bond angle = 109.5°
Four bonding pair electrons.
General formula : AX_4

4 Trigonal bipyramidal



Bond angle = 120°, 90°
Five bonding pair electrons.
General formula : AX_5

5 Octahedral



Bond angle = 90°
Six bonding pair electrons.
General formula : AX_6

EFFECTS OF LONE PAIRS ON BOND ANGLE

- When all the **electron pairs are bonding pairs**,

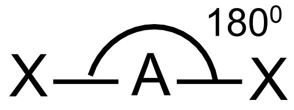
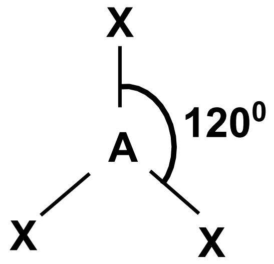
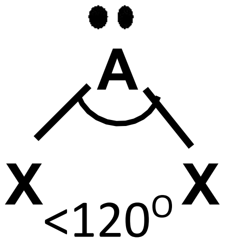
  the molecular geometry / shape is the same as electron pair geometry

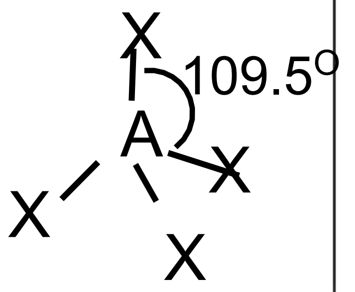
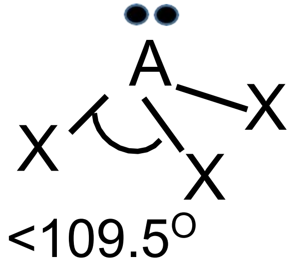
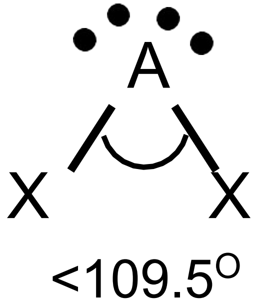
  $\Rightarrow$ The shape is called **basic shape.**

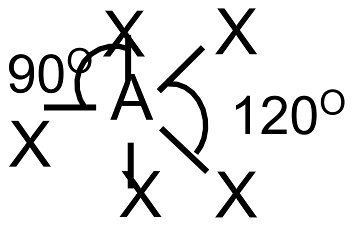
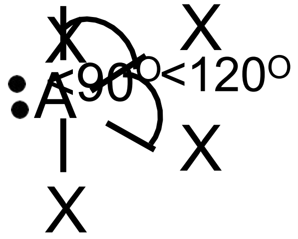
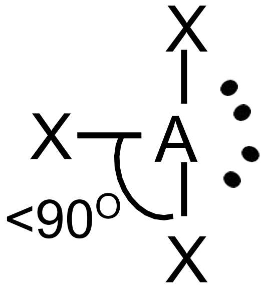
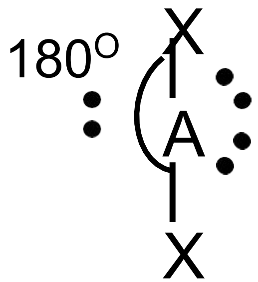
- When the central atom contains lone pairs and bonding pairs, the **lone pairs will influence (compress) the bond angle** although do not account for the shape.

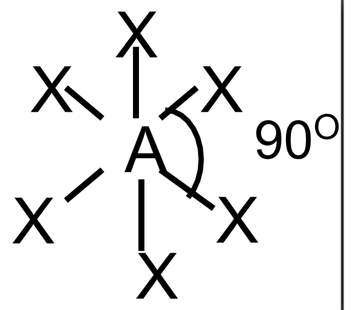
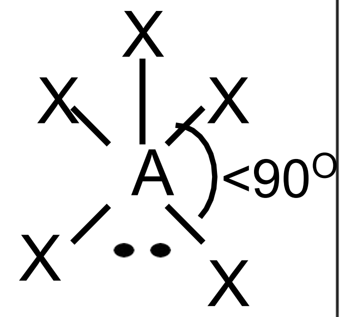
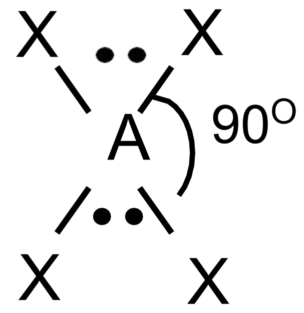
- A **lone pair exerts greater repulsion than a bonding pair** because lone pair experiences less nuclear attraction, and its electron cloud is spread out more in space than that for a bonding pair.

MOLECULAR GEOMETRY WITH LONE PAIR ELECTRONS

Class	No. of bonding pairs at central atom	No. of lone pairs at central atom	Electron Pair Geometry	Molecular Geometry	Molecular Geometry
AX_2	2	0	linear	linear	
AX_3	3	0	Trigonal planar	Trigonal planar	
AX_2E	2	1	Trigonal planar	Bent @ V-shaped	

Class	No. of bonding pairs at central atom	No. of lone pairs at central atom	Electron Pair Geometry	Molecular Geometry	Molecular Geometry
AX_4	4	0	tetrahedral	tetrahedral	
AX_3E	3	1	tetrahedral	trigonal pyramidal	
AX_2E_2	2	2	tetrahedral	bent @ V-shaped	

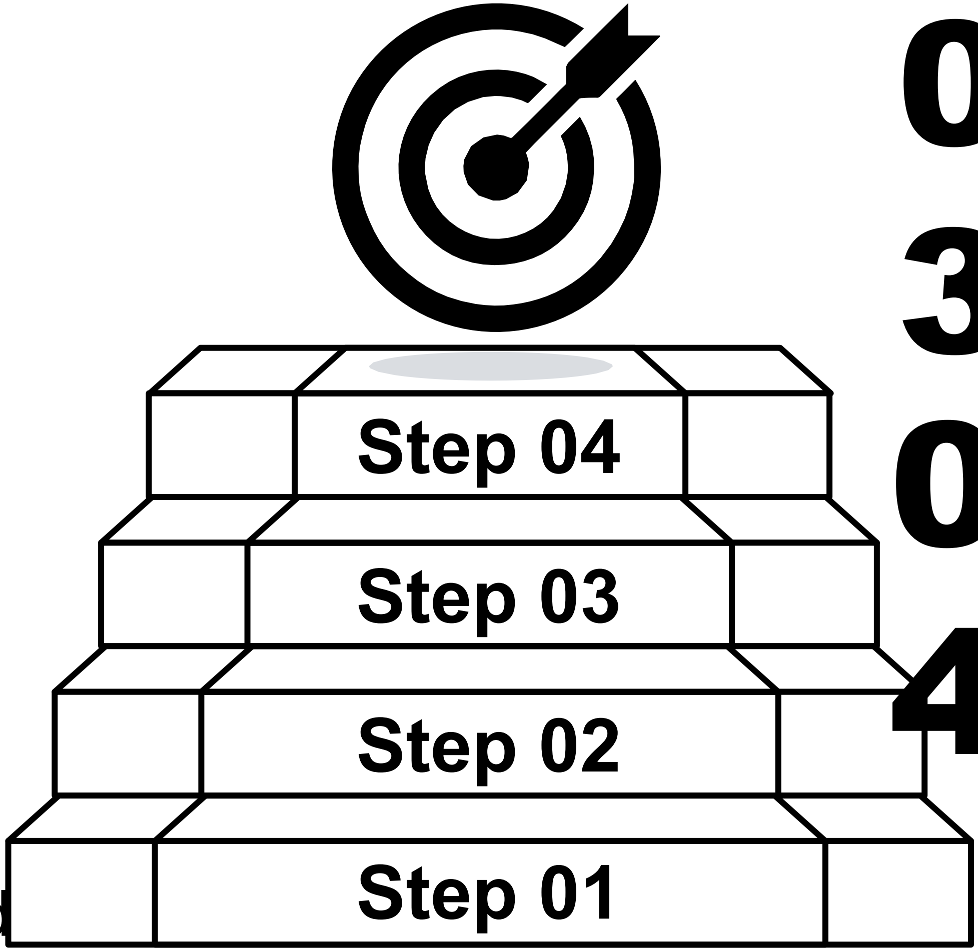
Class	No. of bonding pairs at central atom	No. of lone pairs at central atom	Electron Pair Geometry	Molecular Geometry	Molecular Geometry
AX_5	5	0	Trigonal bipyramidal	trigonal bipyramidal	
AX_4E	4	1	Trigonal bipyramidal	Seesaw @ distorted tetrahedron	
AX_3E_2	3	2	Trigonal bipyramidal	T-shaped	
AX_2E_3	2	3	Trigonal bipyramidal	linear	

Class	No. of bonding pairs at central atom	No. of lone pairs at central atom	Electron Pair Geometry	Molecular Geometry	Molecular Geometry
AX_6	6	0	octahedral	octahedral	
AX_5E	5	1	octahedral	square pyramidal	
AX_4E_2	4	2	octahedral	square planar	

GUIDELINES IN PREDICTING MOLECULAR GEOMETRY / SHAPE OF A MOLECULE

01 Draw a plausible **Lewis structure**

02 **Count the total number of electron pairs** (bonding pairs + lone pairs) at a central atom (A double/ triple bond is counted as one bonding pair when predicting geometry)



03 **Establish the electron pair geometry** around the central atom

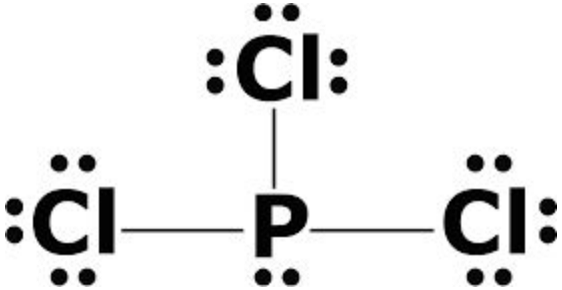
04 **Use VSEPR** to predict a molecular geometry / shape.

EXAMPLE 1: Draw and name the molecular geometry for PCl_3

P = 5 e^-

3 x Cl = 21 e^-

Total = 26 e^-



Step 1 : Draw a plausible Lewis Structure.

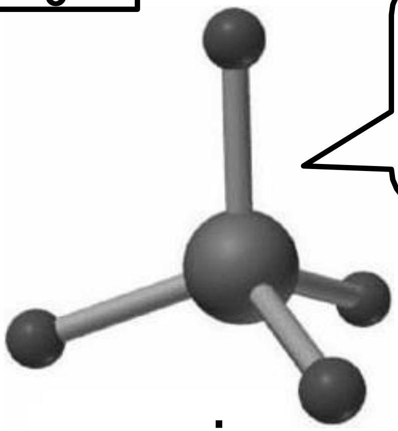
$\boxed{\text{P has 4 } e^- \text{ pairs}}$

One lone pair

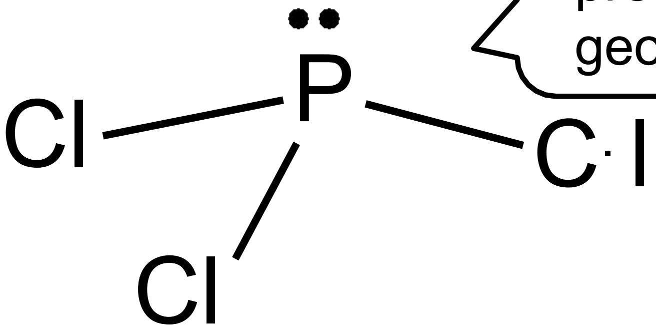
Three bonding pairs

Step 2 : Count total number of electron pairs around central atom.

Class $\boxed{AX_3E}$



Step 3 : Establish the electron pair geometry around the central atom



Step 4 : Use VSEPR to predict a molecular geometry/shape.

Electron-pair geometry :

tetrahedral

Molecular geometry :

Trigonal pyramidal

EXAMPLE 2: Draw and name the molecular geometry for H_2S

2 x H = 2 é

S = 6 é

Total = 8 é

H – S̈ – H

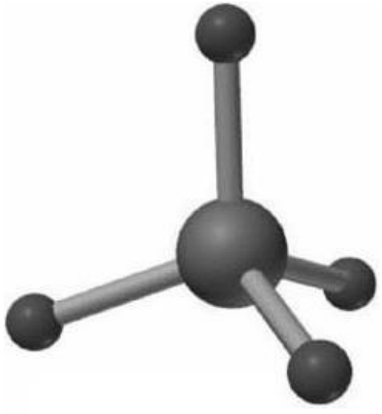
S has 4 é pairs

2 lone pairs

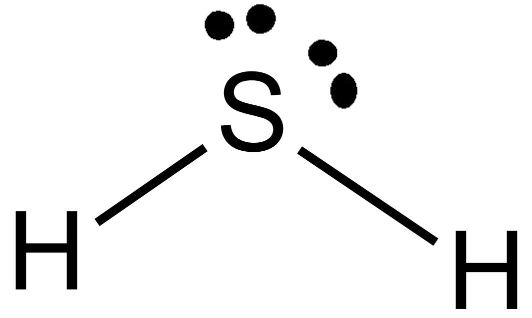
2 bonding pairs

Class

$\boxed{\mathbf{AX_2E_2}}$

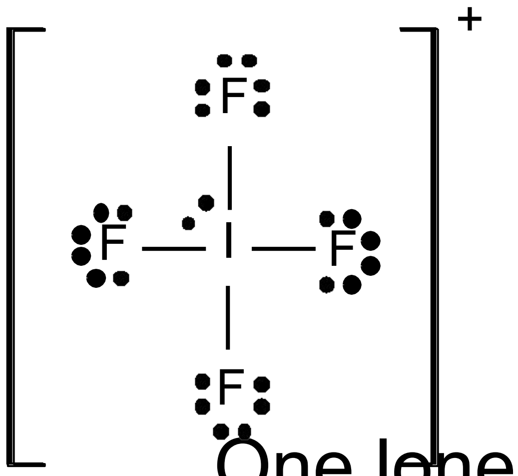


Electron-pair geometry : tetrahedral



Molecular geometry : V-shaped @ bent

EXAMPLE 3: Predict the shape of IF_4^+ ion



$I = 7\ e^-$

$4 \times F = 28\ e^-$

$+ve = -1\ e^-$

$Total = 34\ e^-$

$\boxed{\text{I has 5 } e^- \text{ pairs}}$

One lone pair

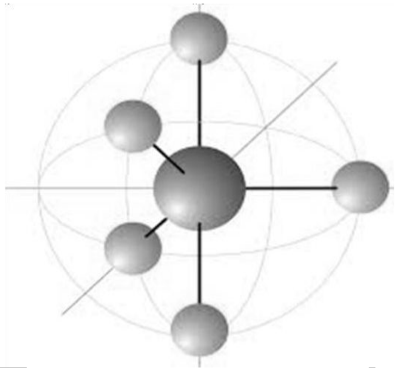
Four bonding pairs

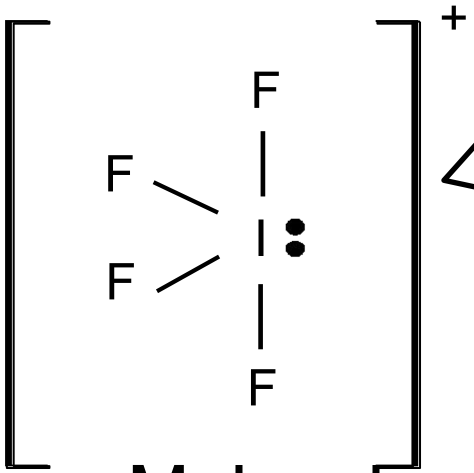
Step 1 : Draw a plausible Lewis Structure of polyatomic ion.

Step 2 : Count total number of electron pairs around central atom.

Class $\boxed{\mathbf{AX_4E}}$

Step 3 : Establish the electron pair geometry around the central atom





Step 4 : Use VSEPR to predict a molecular geometry/shape.

Electron-pair geometry :

Trigonal bipyramidal

Molecular geometry :

Seesaw @ distorted tetrahedron

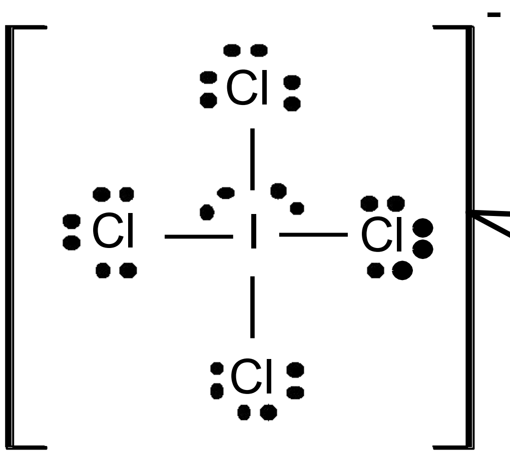
EXAMPLE 4: Predict the shape of ICl_4^- ion.

$I = 7e^-$

$4 \times Cl = 28\ e^-$

$-ve = +1\ e^-$

Total = $36\ e^-$



Step 1 : Draw a plausible Lewis Structure of polyatomic ion.

I has 6 e^- pairs

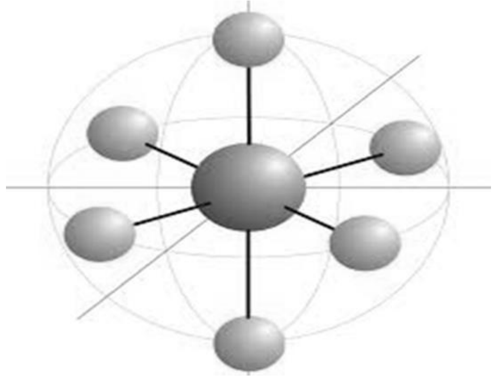
Two lone pairs

Four bonding pairs

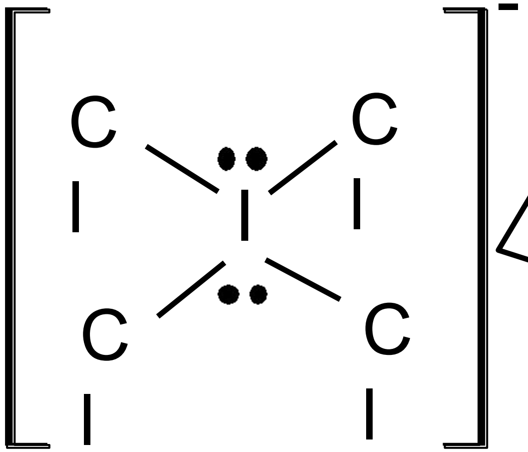
Step 2 : Count total number of electron pairs around central atom.

Class $\boxed{AX_4E_2}$

Step 3 : Establish the electron pair geometry around the central atom.



Electron-pair geometry : **octahedral**

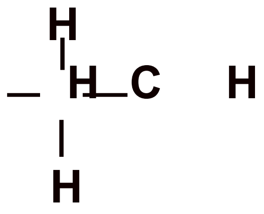
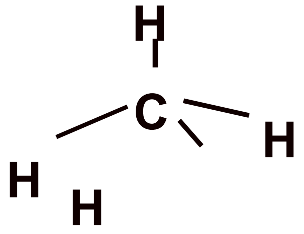
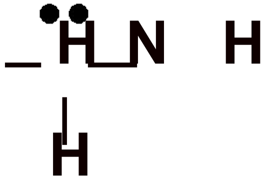
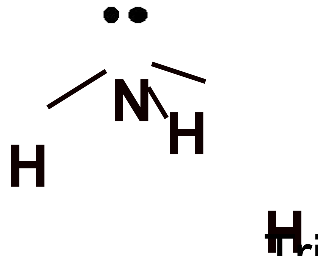
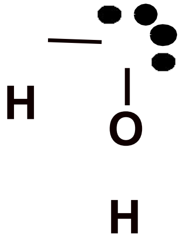
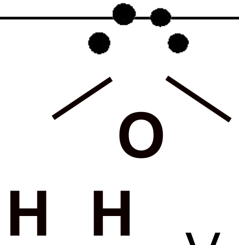


Step 4 : Use VSEPR to predict a molecular geometry/shape.

Molecular geometry : **Square planar**

EXAMPLE 5 :

Based on **valence shell electron pair repulsion (VSEPR) theory**, predict the **molecular geometry** of methane, CH_4, ammonia, NH_3 and water, H_2O molecules. **Compare and explain** the **differences in the bond angles**.

Molecule	Lewis Structure	No. of electron pairs	Electron pair geometry	Molecular geometry
CH_4		4 bonding pairs	Tetrahedral	 tetrahedral
NH_3		3 bonding pairs and 1 lone pair	Tetrahedral	 Trigonal pyramidal
H_2O		2 bonding pairs and 2 lone pairs	Tetrahedral	 V-shaped

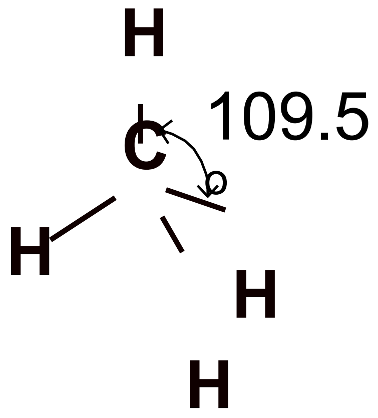
The **Valence Shell Electron Pair Repulsion (VSEPR) theory** states that the electron pairs in the valence shells of the atoms will **repel** one another and **arrange** themselves **as far apart as possible**.

Order of force of repulsion:

Lone pair - lone pair > lone pair - bonding pair > bonding pair - bonding pair

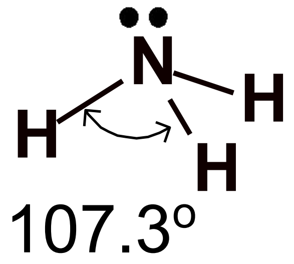
For CH_4, the central C atom has only **4 bonding pairs** and no lone pair. There is only bonding pair-bonding pair repulsion present. Therefore, the **repulsion** between the bonding pairs is **equivalent**.

The angle of H-C-H is **109.5º**.



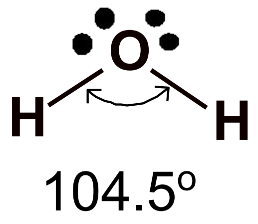
tetrahedral

- For NH_3 ,it has **three bonding pairs** and **one lone pair.**
- The **force of repulsion between lone pair and bonding pairs is stronger** than that between bonding pairs. As a result, the **H-N-H angle** is **reduced to approximately 107.3º**.



Trigonal pyramidal

- For H_2O, there are **two bonding pairs and two lone pairs**.
- There are **lone pair-lone pair, lone pair-bonding pair** and **bonding pair- bonding pair repulsions** present.
- **The force of repulsion is the greatest between the two lone pairs**
- Hence, the **H-O-H angle is reduced to a greater extent** which is approximately **104.5º**.



V-shaped / bent

BOND POLARITY

✔ Bond polarity is due to **differences in electronegativity** of the bonded atoms.

✔ Electronegativity is defined as the **ability of an atom in a particular molecule to attract bonding electrons to itself.**

✔ The **greater the difference in electronegativity (ΔEN)**, the more polar the bond.

Polar covalent bonds

- Two bonded atoms have **different electronegativities.**
- **Unequal electron sharing**.
- Example: HCl

H — Cl

Non-polar covalent bonds

- Two bonded atoms have **similar electronegativities (ΔEN**≈0)
- **Equal electron sharing**.
- Example: Cl_2

Cl — Cl

BOND POLARITY AND DIPOLE MOMENT

The polarity of bond is shown by using δ + for partially positive charge (less electronegative atom) and δ – for partially negative charge (more electronegative atom)

Dipole moment (μ) is defined as the product of the **magnitude** of the charge (δ) and the **distance** (d) that separates the centre of positive and negative charge. Dipole moment is a *vector quantity.*

$$\mu = \delta \, \mathbf{d}$$

Molecules that possess a dipole moment ($\mu \neq 0$) are called **polar molecules**. Molecules without a dipole moment ($\mu = 0$) are **non-polar molecules**.

Bond dipole is shown by using a cross-based arrow ($+\!\!\!\longrightarrow$) to indicates the direction of electron displacement, from less electronegative to more electronegative atom.

POLAR AND NON-POLAR MOLECULES

1. Diatomic Molecules

✔ Dipole moment of diatomic molecule = Dipole moment of the bond @ bond dipole

✔ Thus, the polarity of molecule depends on the polarity of the only one bond.

EXAMPLE 6 : Explain the polarity of H_2 and ICl molecules

H—H

- There is no difference in electronegativity between the two H atoms.
- H-H bond is nonpolar bond.

$\therefore \mu = 0 \Rightarrow H_2$ is **Non-polar molecule**.

I—Cl

- Cl atom is more electronegative than I atom.
- I – Cl bond is polar bond.

$\therefore \mu \neq 0 \Rightarrow$ ICl is **polar molecule.**

2. Polyatomic Molecules

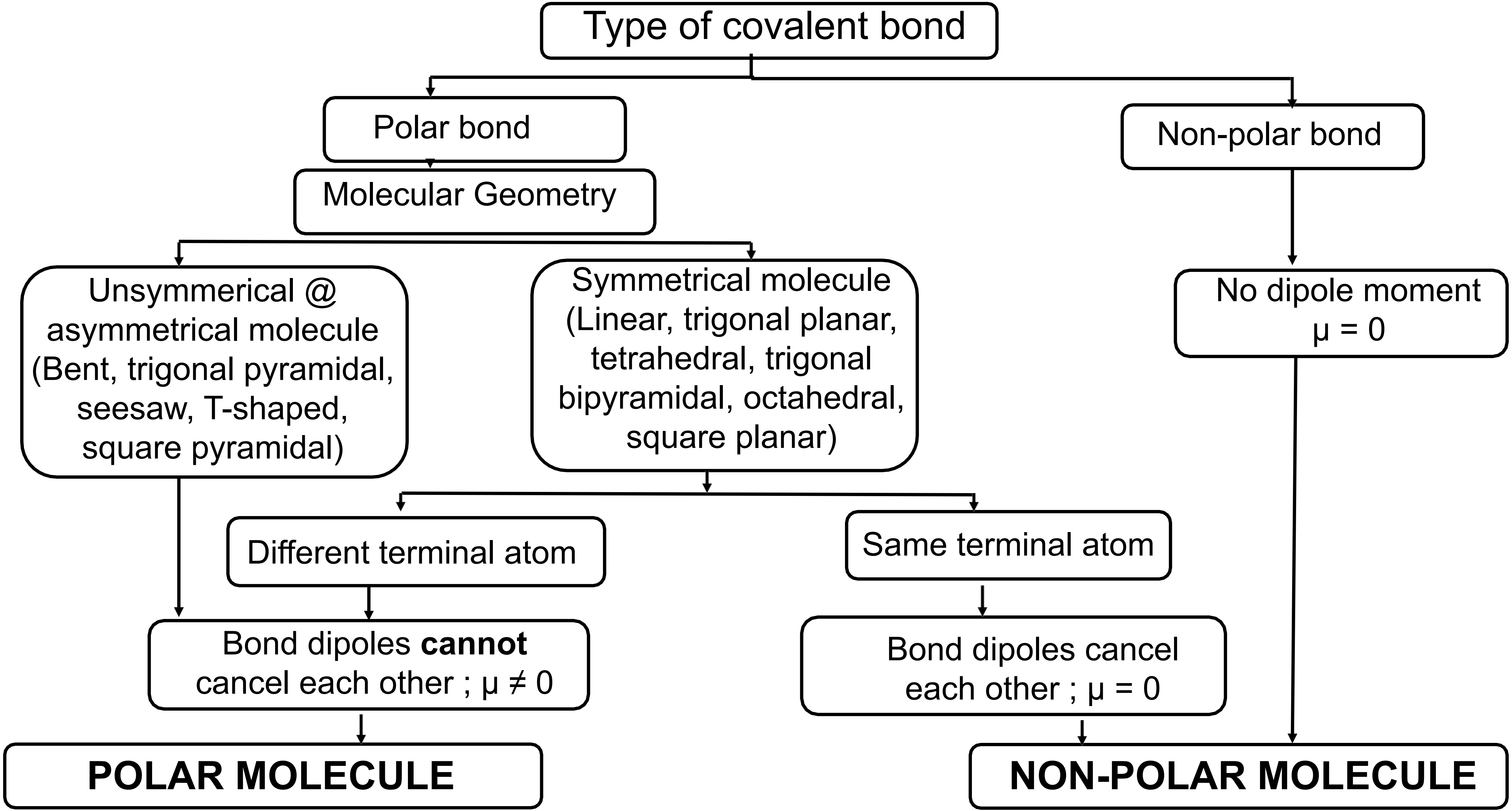
✔ The **dipole moment** of a polyatomic molecule is the **sum of all bond dipoles** in the molecule.

✔ The **polarity of the molecule** depends on the

1. **bond polarity**
2. **molecular shape**

✔ **Molecules with two or more polar bonds** can be **nonpolar molecule because its bond dipoles can cancel each other** to give a **resultant dipole moment of zero (μ = 0)** if the bonds are arranged **symmetrically.**

BOND POLARITY, MOLECULAR SHAPE AND MOLECULAR POLARITY



Guideline to deduce the polarity of molecule

01

Draw the molecular geometry

02

Identify polar bonds in the molecule. Indicate each bond dipole using a vector quantity

03

Determine sum of vectors of bond dipoles:

If $\mu = 0$ □ non-polar molecule.

If $\mu \neq 0$ □ polar molecule.

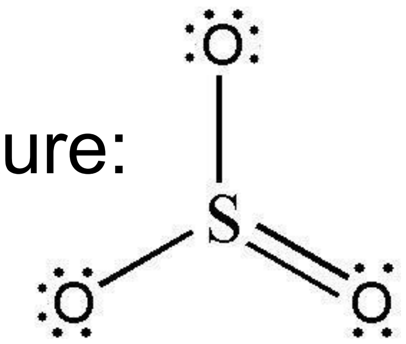
EXAMPLE 7: Deduce the polarity of molecule SO_3

3 x O = 18 é
1 x S = 6 é

Total = 24 é
- 6 é

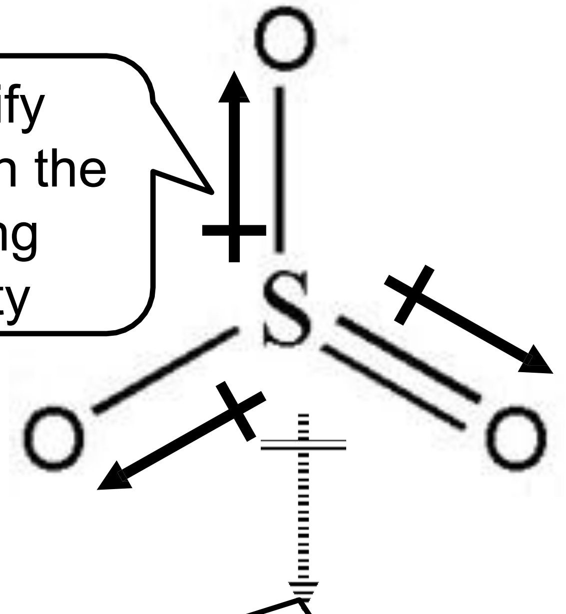
18 é
- 18 é

Lewis structure:



Step 1: Draw the molecular geometry

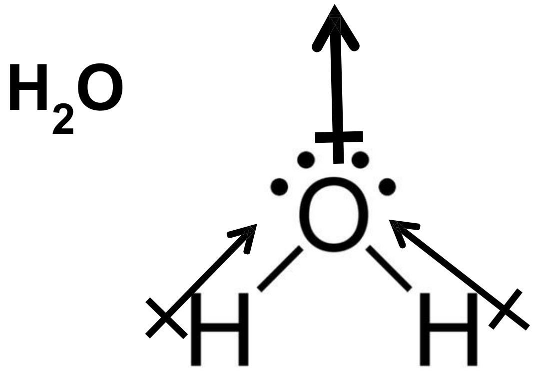
Step 2: Identify polar bonds in the molecule using vector quantity



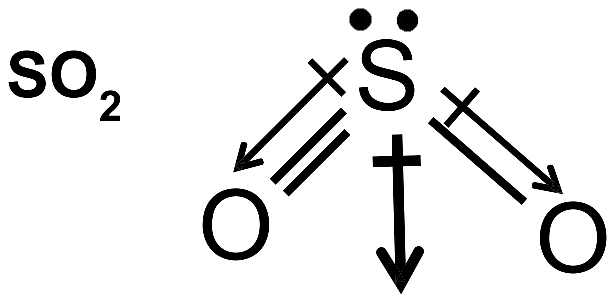
Step 3: Determine sum of vectors of bond dipole. (μ)

- Molecular shape : Trigonal planar
- S → O dipoles are in opposite directions.
- Bond dipoles cancel one another.
- No dipole moment (μ = 0).
- SO_3 is **non-polar molecule.**

EXAMPLE 8: Deduce the polarity of molecule $\mathbf{H_2O}$ and $\mathbf{SO_2}$



- ✔ Molecular geometry : **Bent**
- ✔ **Bond dipoles do not cancel** each other.
- ✔ $\mu \neq 0$
- ✔ $\therefore$ $\mathbf{H_2O}$ is polar molecule

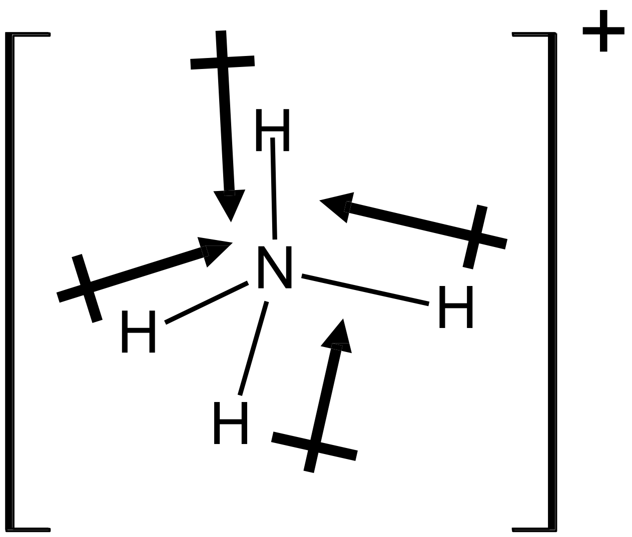


- ✔ Molecular geometry : **Bent**
- ✔ **Bond dipoles do not cancel** each other
- ✔ $\mu \neq 0$
- ✔ $\therefore$ $\mathbf{SO_2}$ is **polar** molecule

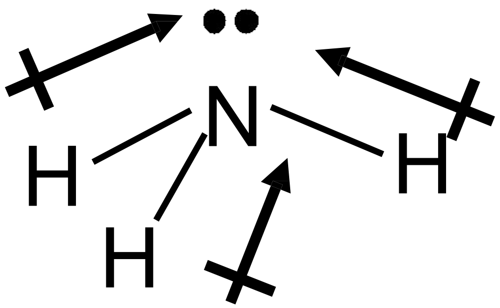
- ✔ Molecules with **lone pairs** are usually **polar (except for $\mathbf{AX_2E_3}$ and $\mathbf{AX_4E_2}$ with same terminal atoms)**
- ✔ Because of the molecular geometry, the bond dipoles do not cancel each other, but combine to give a resultant dipole moment for the molecule($\mu \neq 0$).

EXAMPLE 9: Explain why the NH_3 molecule is polar whereas the NH_4^+ ion is non-polar.

Solution :



NH_4^+



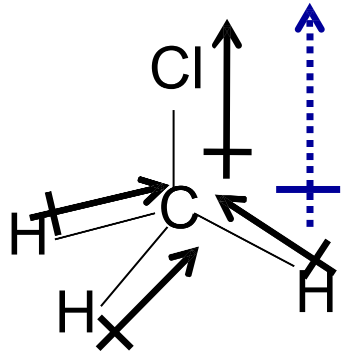
NH_3

- ✔ Molecular geometry :**Tetrahedral**
- ✔ **Bond dipoles cancel** each other.
- ✔ μ **= 0**
- ✔ ∴ NH_4^+ ion is **non-polar.**

- ✔ Molecular geometry :**Trigonal pyramidal**
- ✔ **Bond dipoles does not cancel** each other.
- ✔ $\mu \neq$ **0**
- ✔ ∴ NH_3 molecule is **polar**.

EXAMPLE 10: Determine whether Chloromethane, CH_3Cl molecule is **polar or non-polar**. Explain.

Solution :



Resultant dipole moment

Chloromethane, CH_3Cl

✔ Although the molecular geometry is tetrahedral, the terminal atoms are not identical.

✔ Cl atom is more electronegative than C atom, so the C—Cl bonds is polar. The C−H bond has very low polarity because the electronegativity difference between C and H atom is very small.

✔ The **bond dipoles do not cancel**.

✔ Thus, the CH_3Cl molecule has a resultant dipole moment; $\mu \neq \mathbf{0}$

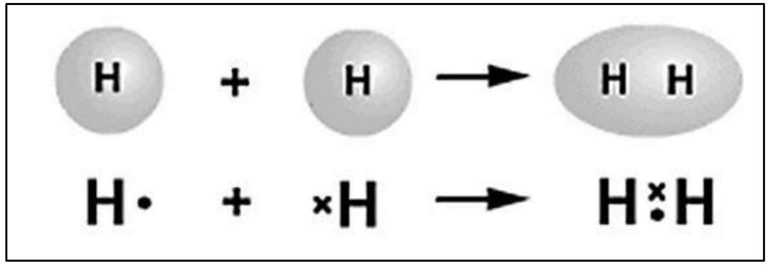
✔ Chloromethane, CH_3Cl is **polar molecule.**

4.3
Orbital Overlap & Hybridisation

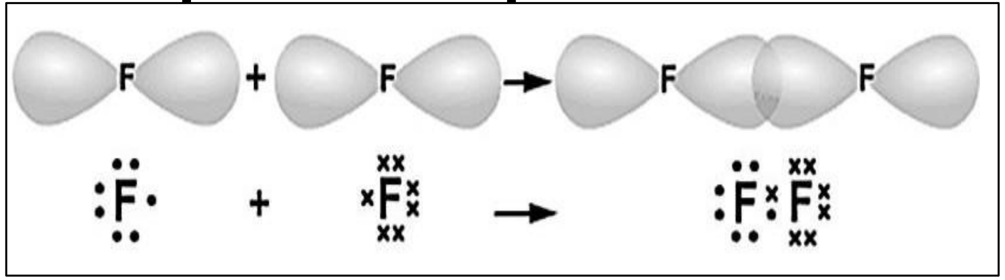
VALENCE BOND THEORY

When atoms **share a pair of electrons to form a bond**, the overlapping of valence orbitals occurs.

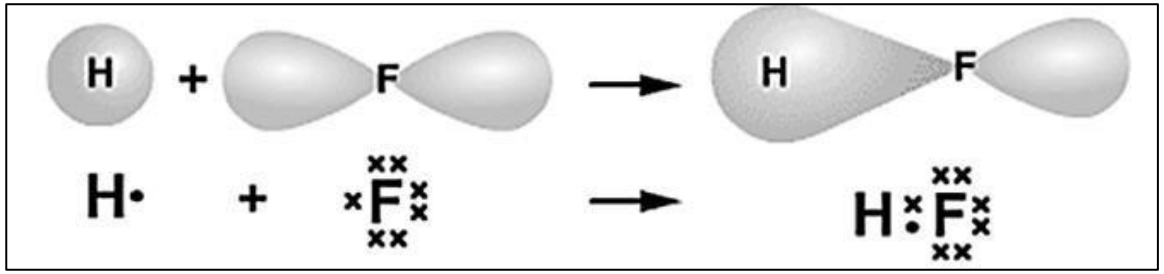
s orbital + s orbital



p orbital + p orbital



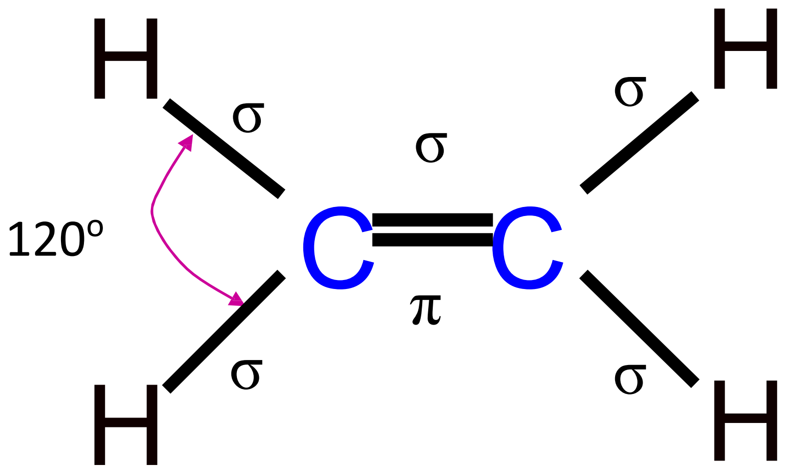
s orbital + p orbital

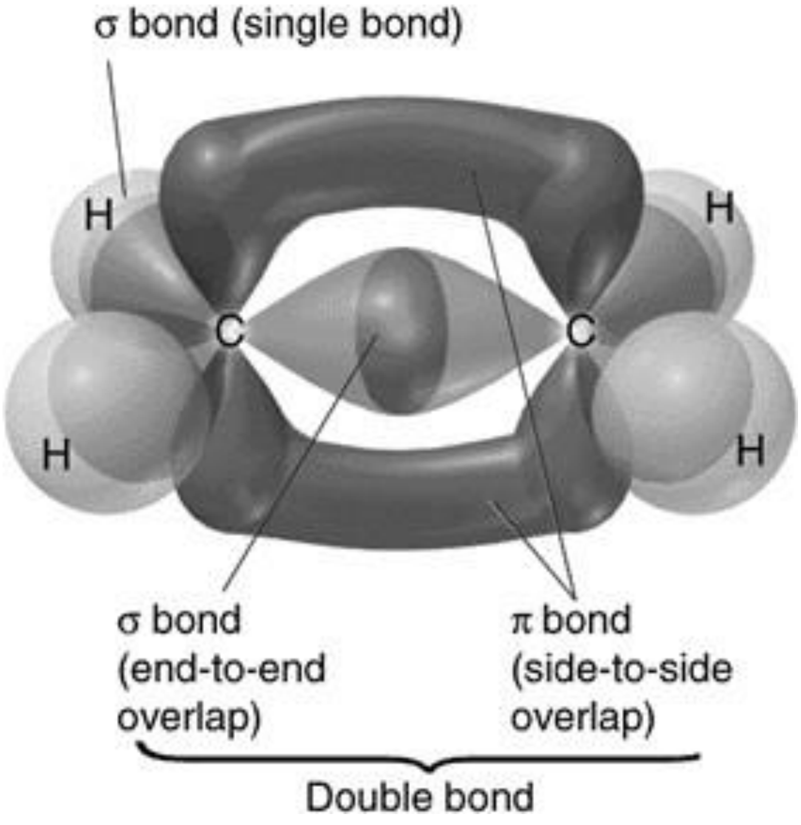


VALENCE BOND THEORY

Types of covalent bond

- Sigma bond (σ)
- Pi bond (π)

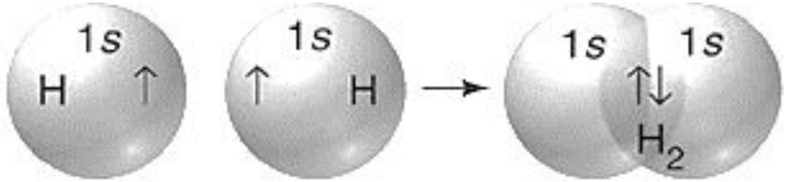




SIGMA BOND (σ)

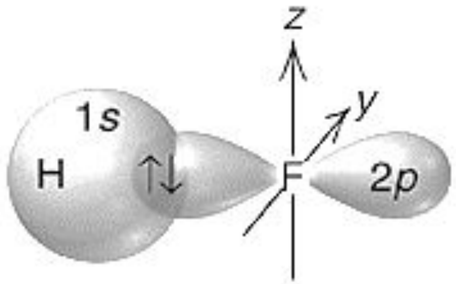
A **covalent bond** formed by the **head-to-head** (or end-to-end) **overlapping** of

- ❯ **two *s* orbitals**



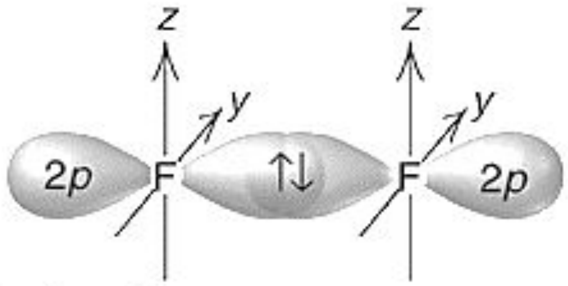
A Hydrogen, H_2

- ❯ **one *s* and one *p* orbitals**



B Hydrogen fluoride, HF

- ❯ **two *p* orbitals**

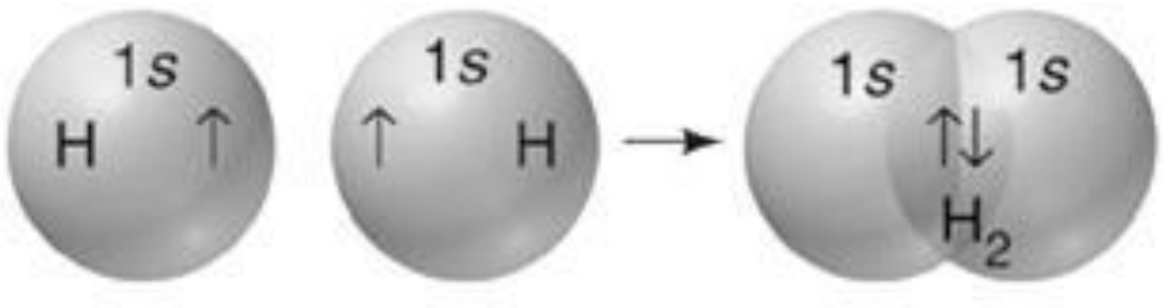


C Fluorine, F_2

<u>Sigma bond (σ): Overlapping of **two *s* orbitals**</u>

Example: H_2

H : $\underline{\uparrow}$
$1s$



Sigma bond (σ): Overlapping of **one *s* and one *p* orbitals**

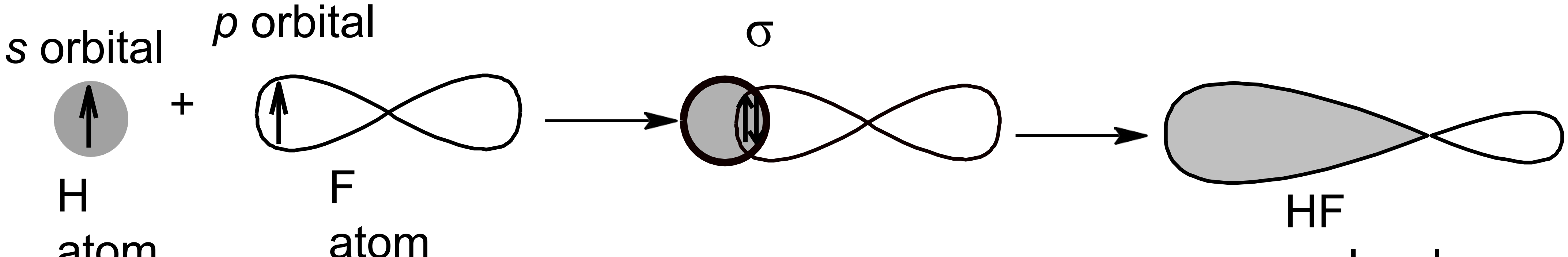
Example: HF

H : $\underline{\uparrow}$
$1s$

F : $\underline{\uparrow\downarrow}$ $\underline{\uparrow\downarrow}$ $\underline{\uparrow\downarrow}$ $\underline{\uparrow}$
$2s$ $2p$

(Valence electronic configuration)

- ☑ One of the $2p$ orbital of F is **singly occupied.**
- ☑ The **1*s* orbital of hydrogen** overlaps with the **2*p* orbital of fluorine** to form a **sigma bond.**



Sigma bond (σ): Overlapping of **two *p* orbitals**

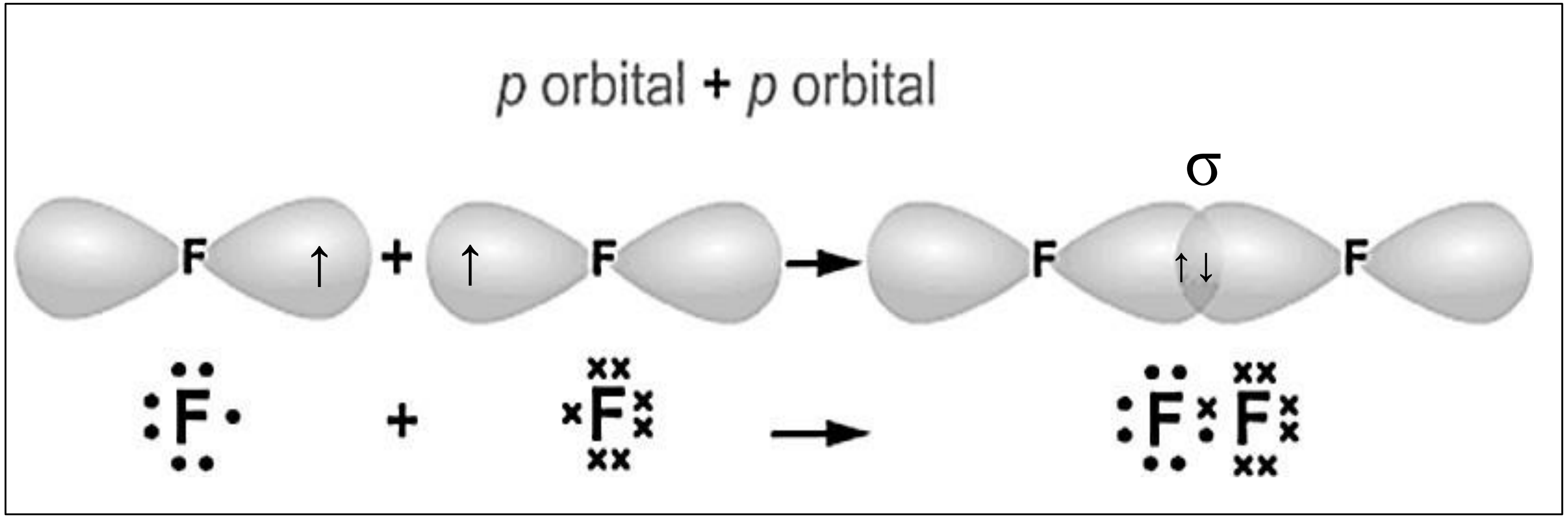
Example: F_2

F : ↑↓ ↑↓ ↑↓ ↑↓ ↑

1*s* 2*s* 2*p*

F : ↑↓ ↑↓ ↑↓ ↑↓ ↑

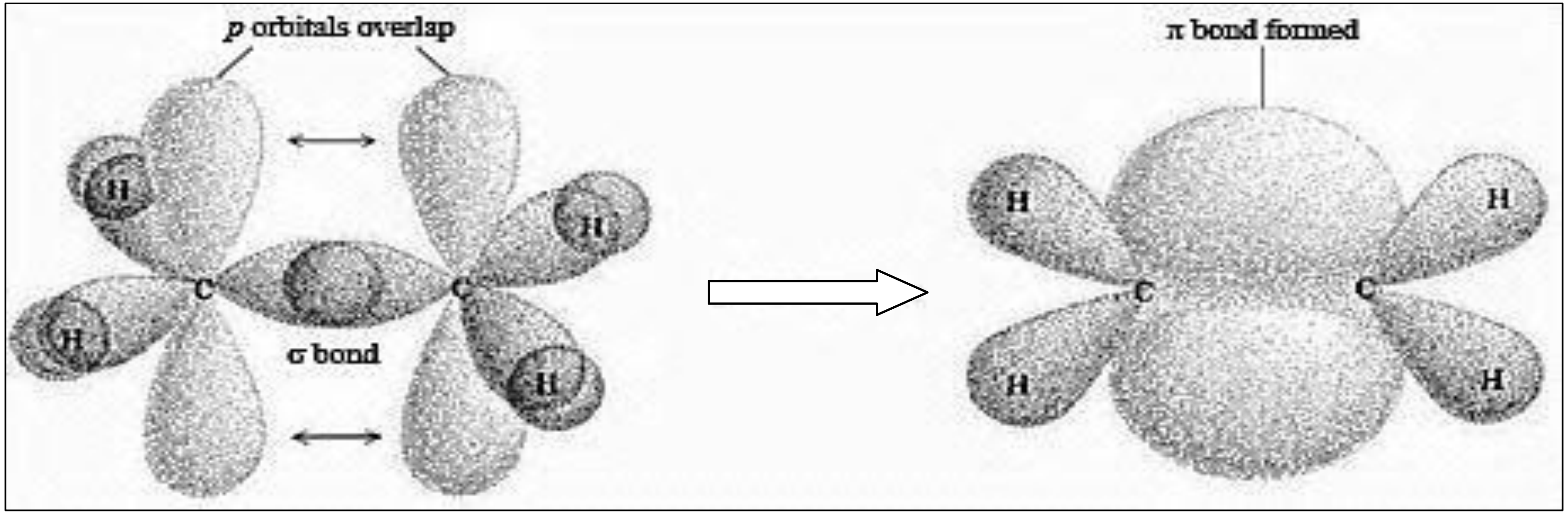
1*s* 2*s* 2*p*



PI BOND (π)

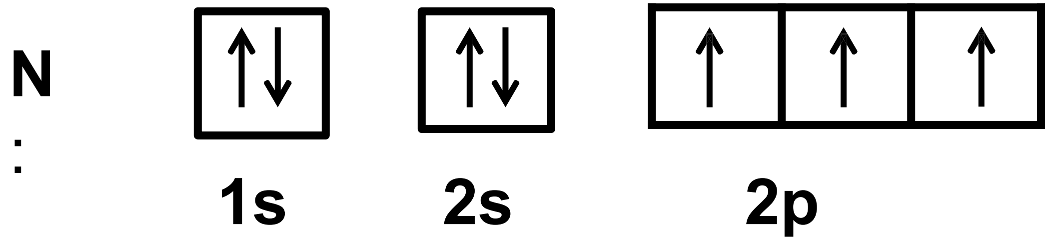
- A covalent bond formed by the **side-by-side overlapping of two *p* orbitals.**

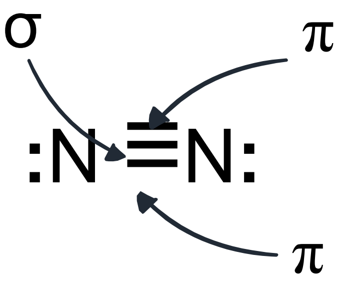
- It occurs in molecules with **multiple** (**double or triple) bonds**.



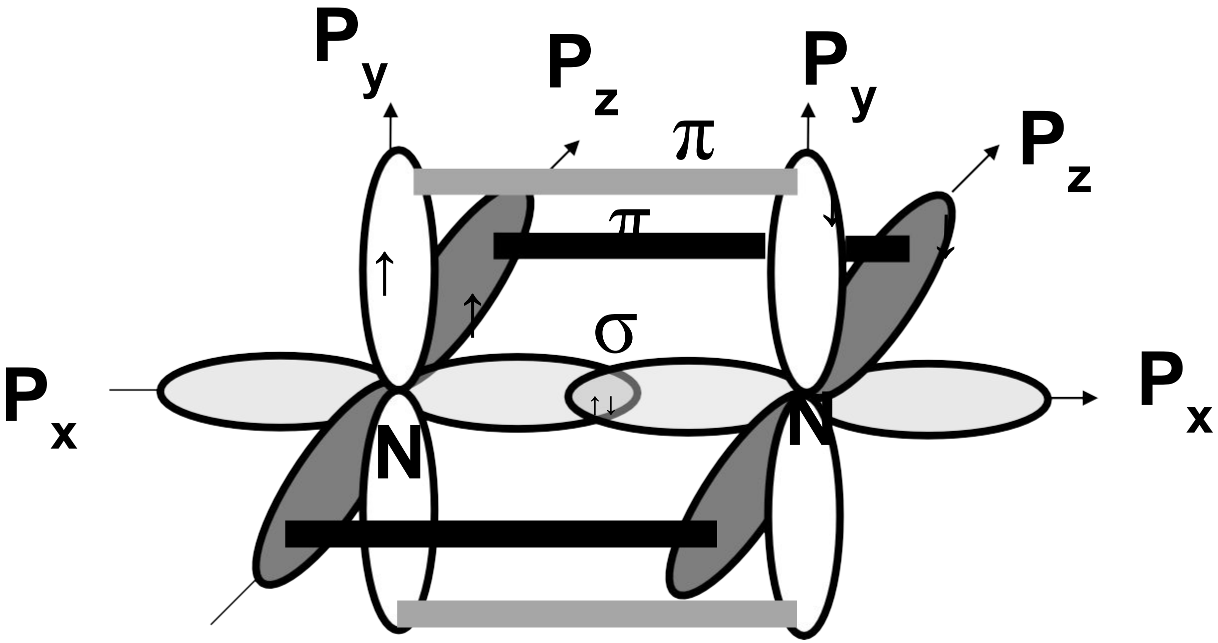
EXAMPLE: N_2

N :$1s^2\ 2s^2\ 2p_x^{\ 1}\ 2p_y^{\ 1}\ 2p_z^{\ 1}$





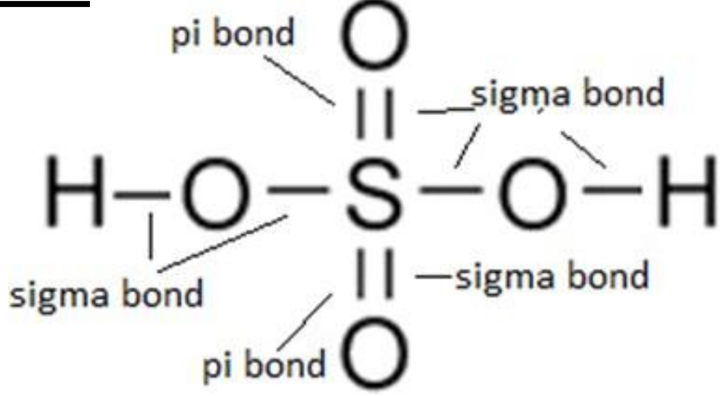
There are one **σ** bond and two bonds π molecule of N_2 in a



Types of normal covalent bonds

- Single bond
  - 1 σ bond
- Double bond
  - 1 σ bond
  - 1 π bond
- Triple bond
  - 1 σ bond
  - 2 π bond

EXAMPLE



LIMITATION OF VALENCE BOND THEORY

- The theory assumes a simple overlap of adjacent atomic orbitals in the formation of a covalent bond.
- The theory predicts correctly the structures of some simple diatomic molecules like H_2 , F_2 and HCl.

But it fails when applied to polyatomic molecules or ions where sometimes the no. of bonds formed exceeds the no. of unpaired electrons.

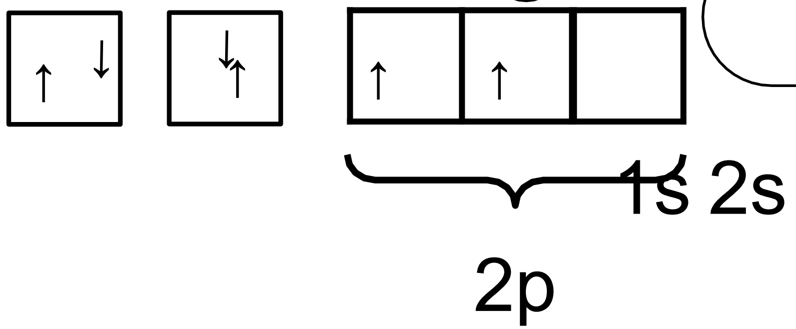
LIMITATION OF VALENCE BOND THEORY

Example: Carbon atom

It shows that C has only **2 unpaired ē**

This indicates that C can form **only 2 covalent bonds**

$_6C : 1s^2\,2s^2\,2p^2$



However, carbon can form 4 bonds in all of its compounds such as CH_4 , CH_3Cl....

To explain this abnormalities, the **hybridisation theory was introduced.**

4.3 ORBITAL OVERLAP & HYBRIDISATION

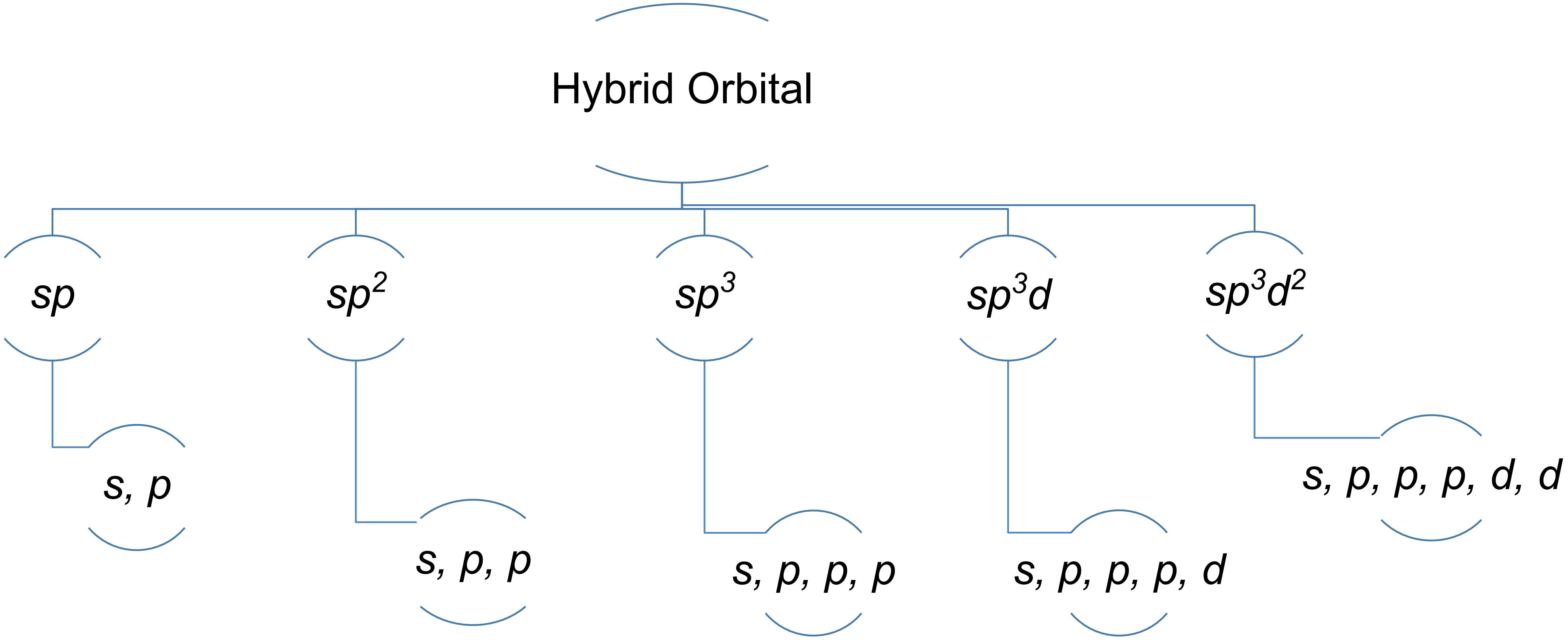
HYBRIDISATION

- The process of **mixing two or more atomic orbitals on an atom** to generate **a new set of hybrid orbitals** of **identical shape and energy.**

HYBRID ORBITAL

- The **new atomic orbitals** produced by hybridization process.
- The **number of hybrid orbitals** obtained is the **same as the number of atomic orbitals mixed.**
- The **type of hybrid orbitals obtained varies** with the types of atomic orbitals mixed.

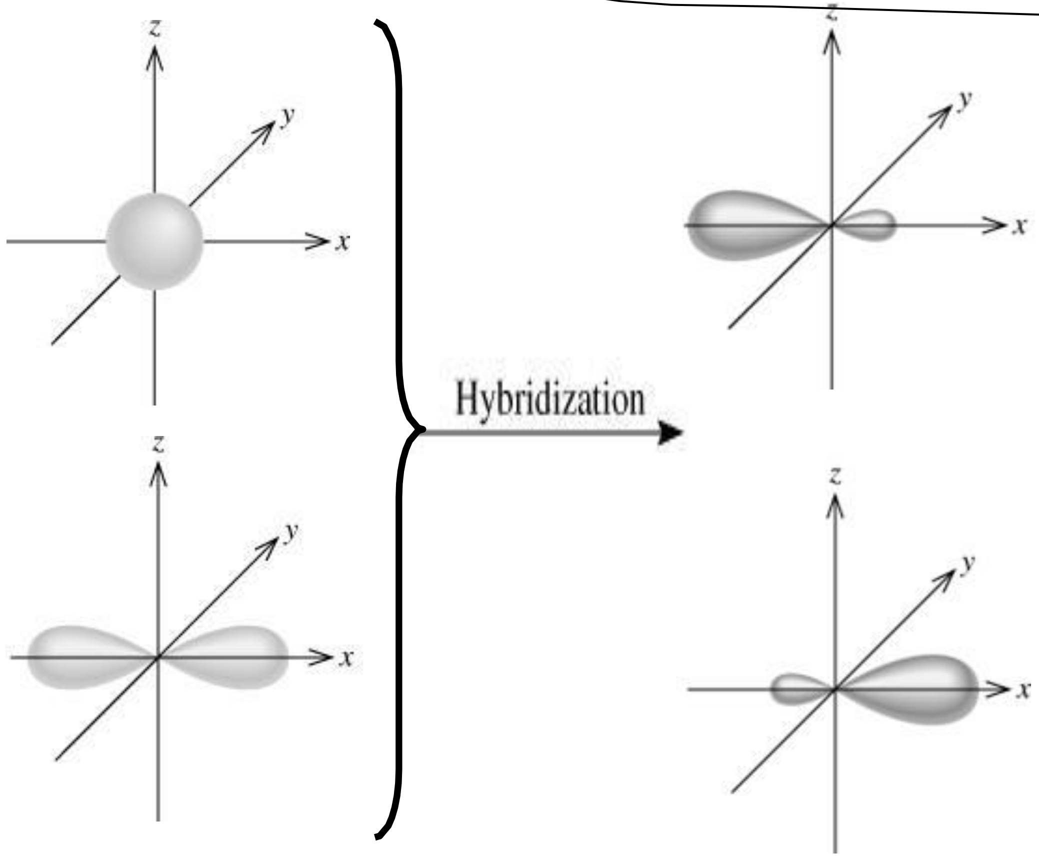
TYPE OF HYBRIDISATION

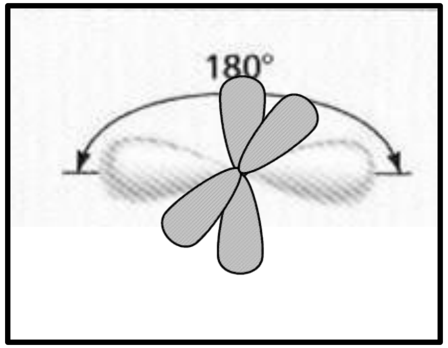


HYBRIDISATION: ***sp***

***sp* hybridisation**

The mixing of **one *s*** and **one *p*** orbital to produce **two** equivalent ***sp* hybrid orbitals** arranged in a **linear geometry**.



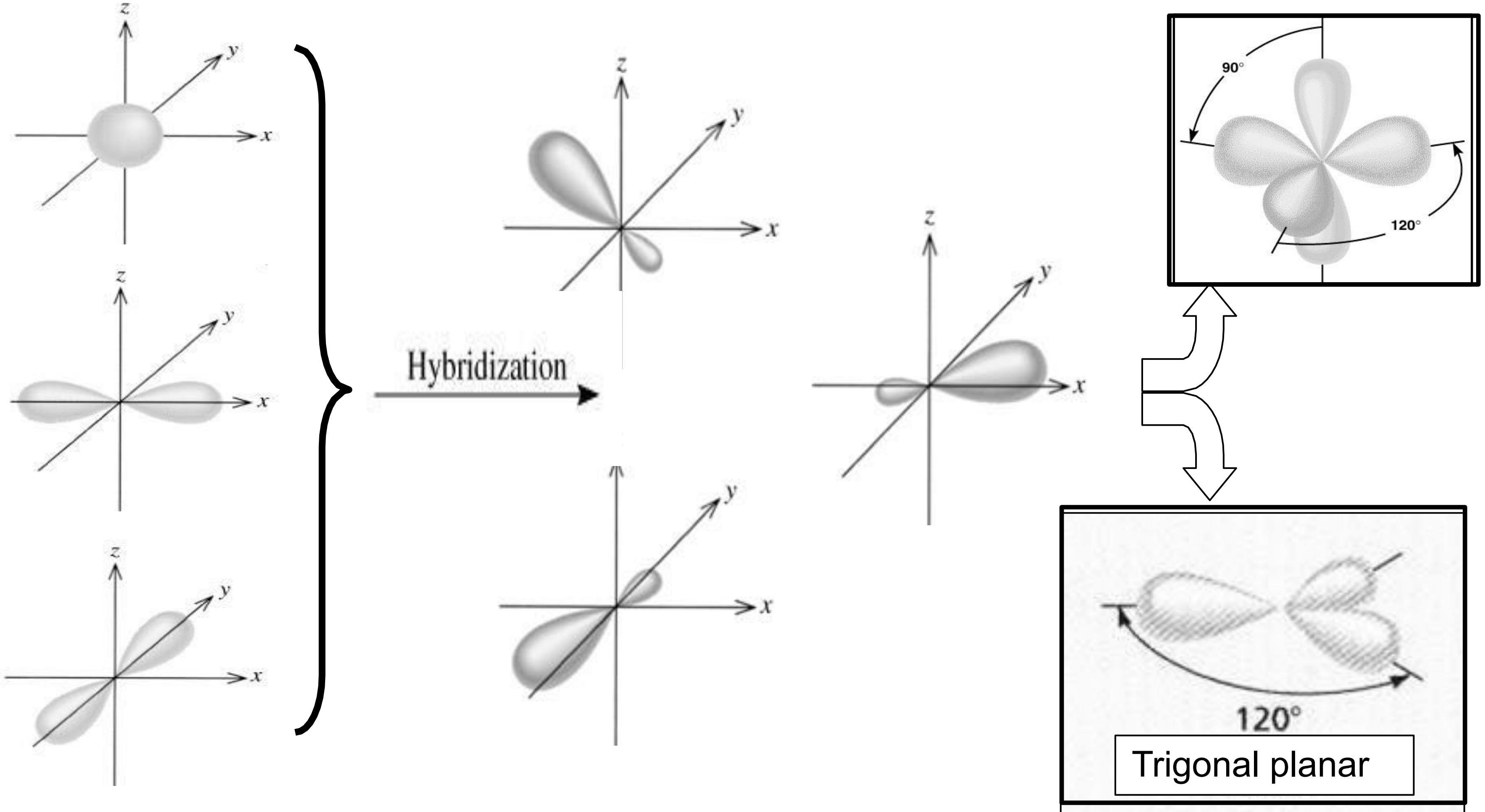


NOTE: There are **two unhybridised p orbitals** which are perpendicular to the hybrid orbitals.

HYBRIDISATION: ***sp***^{2}

sp^2 hybridisation

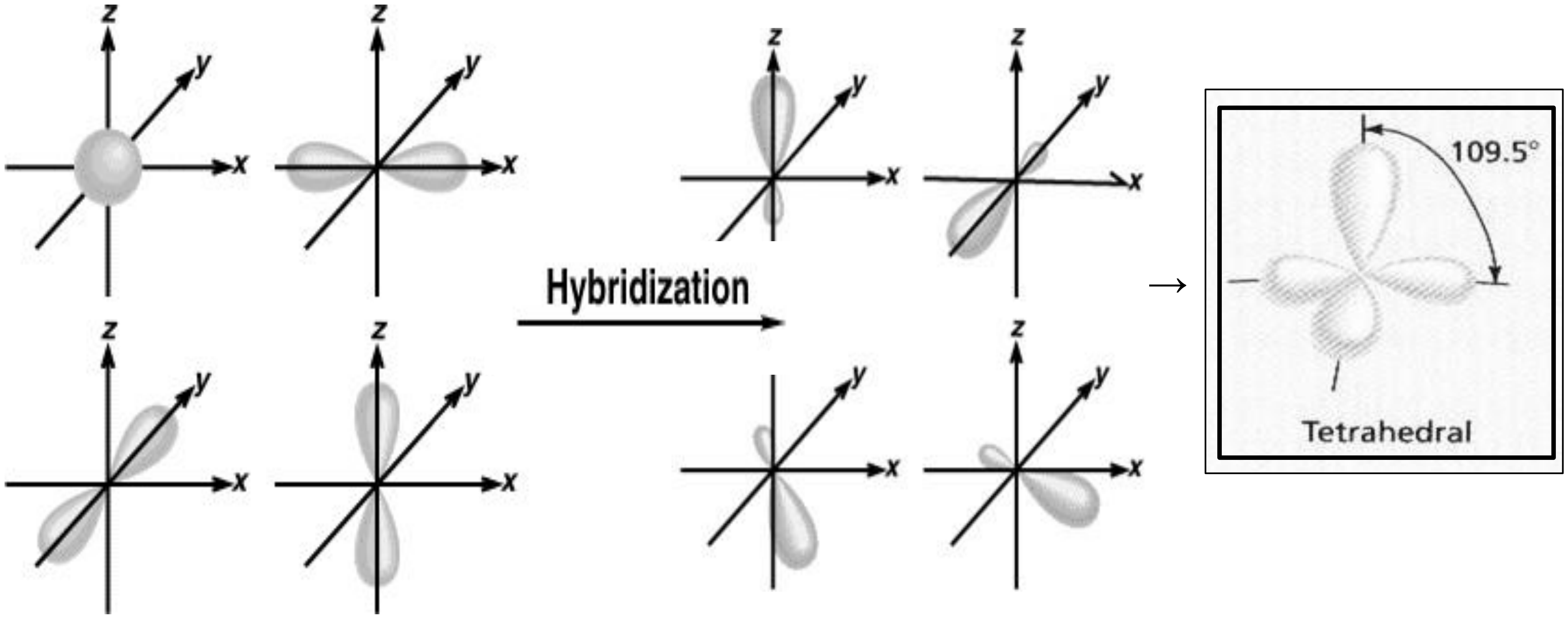
The mixing of **one *s*** and **two *p*** orbitals to produce **three** equivalent ***sp***^{2} **hybrid orbitals** arranged in a **trigonal planar geometry.**



HYBRIDISATION: ***sp***^{3}

sp^{3} hybridisation

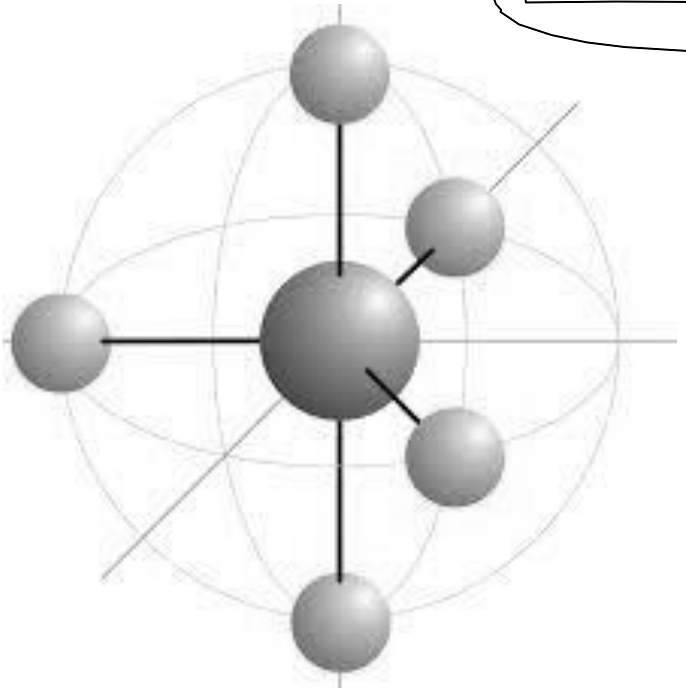
The mixing of **one *s*** and **three *p*** orbitals to produce **four** equivalent ***sp***^{3} **hybrid orbitals** arranged in a **tetrahedral geometry.**



HYBRIDISATION: ***sp³d***

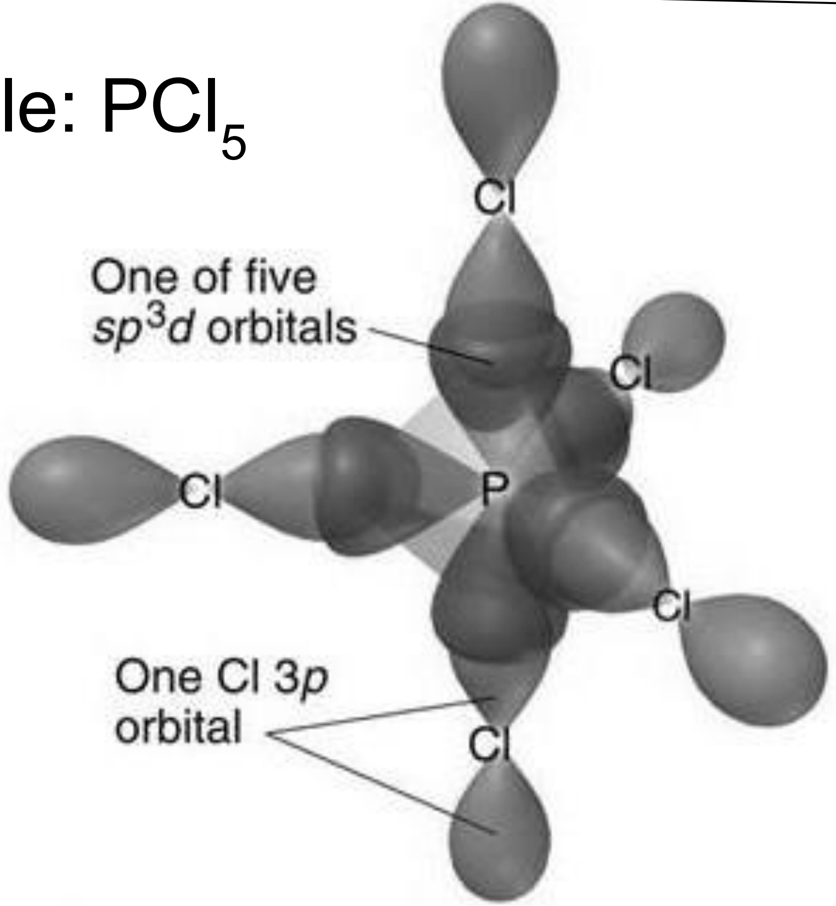
sp^3d hybridisation

The mixing of **one *s*, three *p*** and **one *d*** orbitals to produce **five equivalent sp^3d hybrid orbitals** arranged in the **trigonal bipyramidal** geometry.



Trigonal bipyramidal

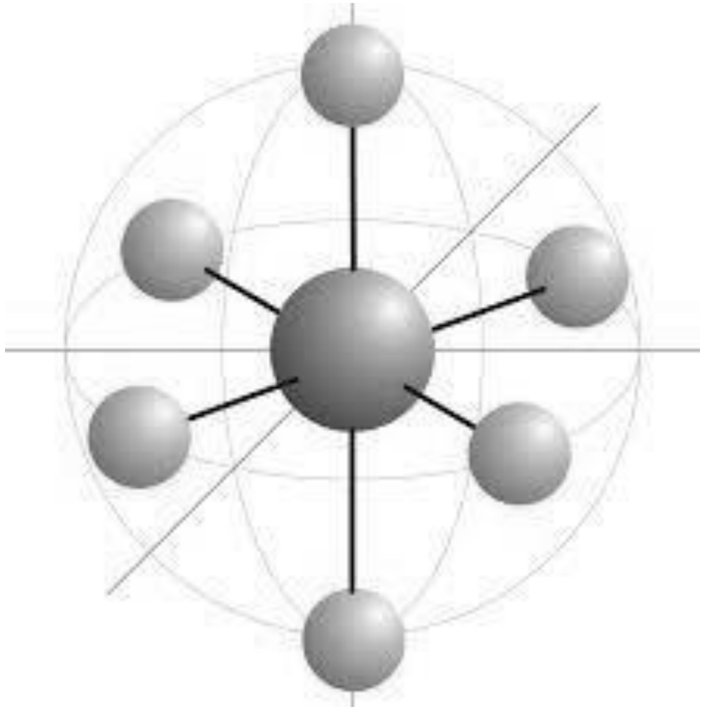
Example: PCl_5



HYBRIDISATION: sp^3d^2

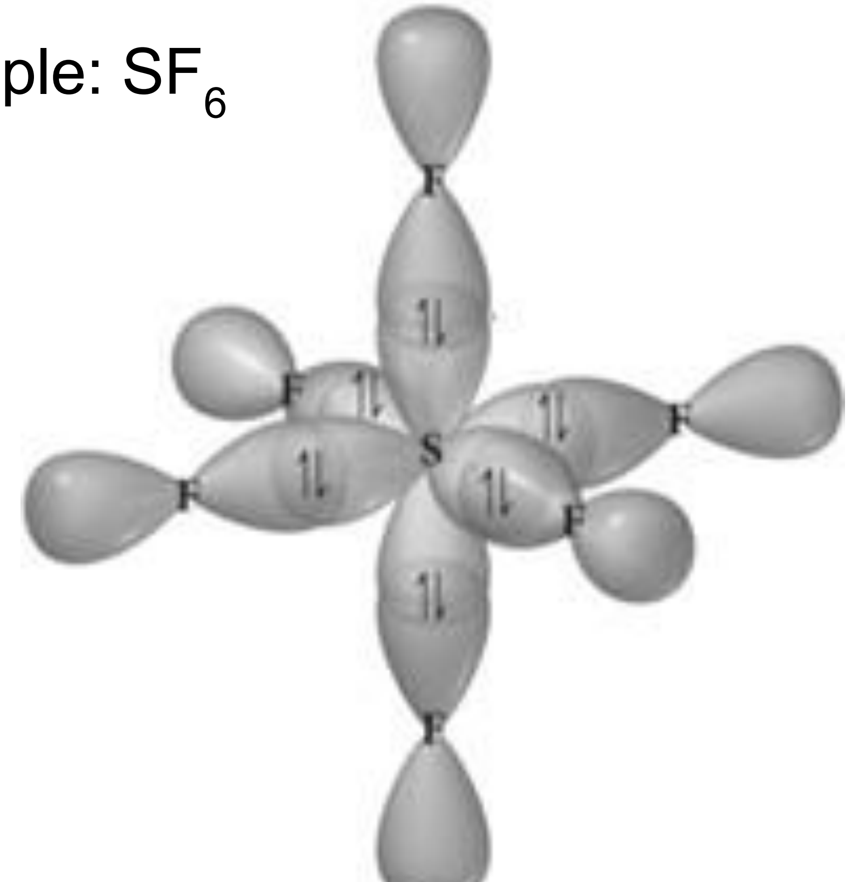
sp^3d^2 hybridisation

The mixing of **one *s*, three *p*** and **two *d*** orbitals to produce **six equivalent sp^3d^2 hybrid orbitals** arranged in the **octahedral geometry.**



Octahedral

Example: SF_6



DRAWING AN ORBITAL OVERLAPPING DIAGRAM

01 Draw the **Lewis Structure**

02 Determine the **electron-pair geometry**

03 Predict the **molecular geometry**

04 Write the electron configuration of the **central atom (ground state of valence e)**

05 **Promote the electron** to possible hybridisation state (excited state). In some cases, excitation of electrons is not necessary.

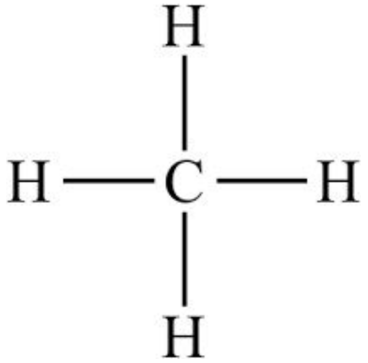
06 Hybridise the orbitals accordingly (hybridised state).

07 State a brief explanation.

EXAMPLE 1: CH_4

C = 4 é
4 x H = 4 é

Total = 8 é

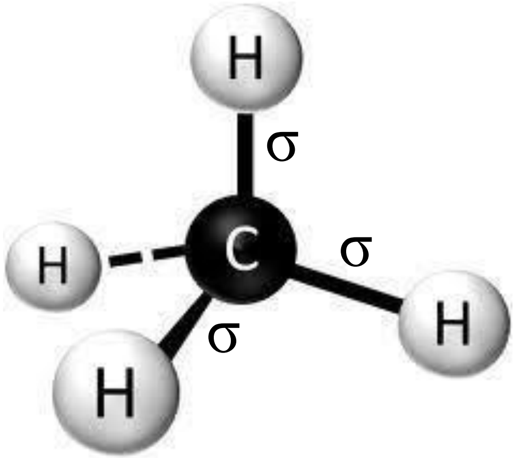


Lewis Structure

Electron-pair geometry: **tetrahedral**

Molecular geometry: **tetrahedral**

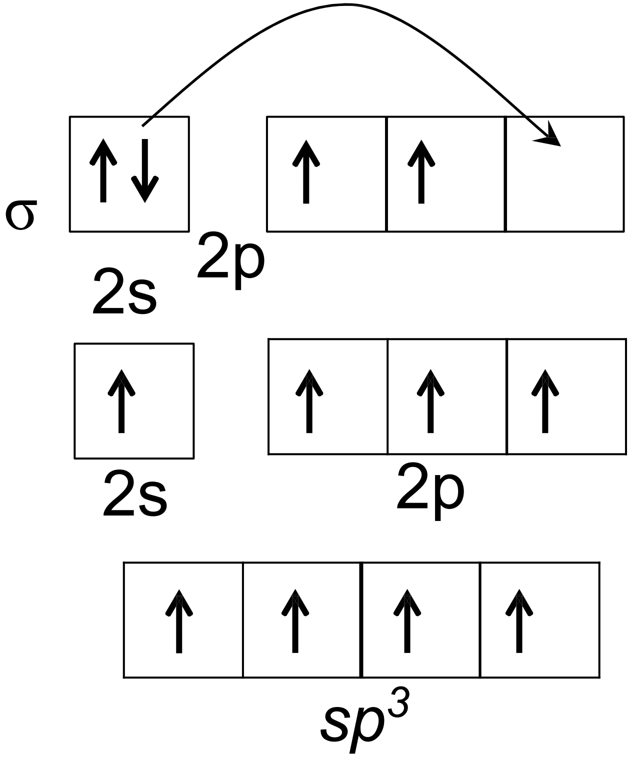
Type of hybridisation: ***sp*³**



C (ground state):

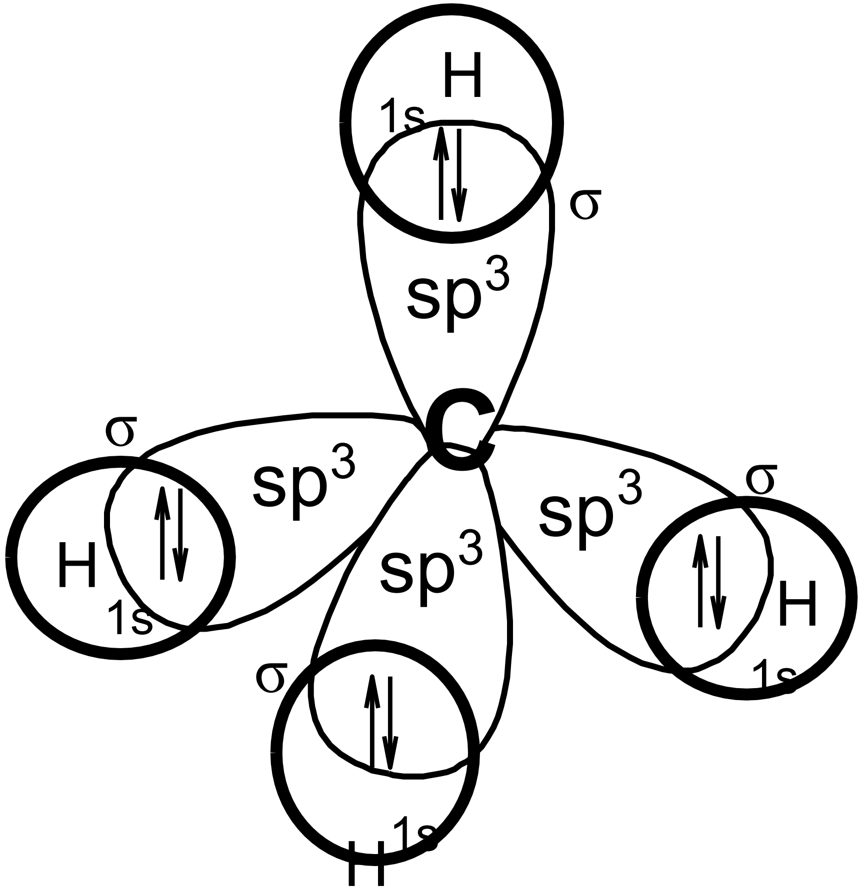
C (excited state):

C (hybridised state):



Four sp^3 hybrid orbitals

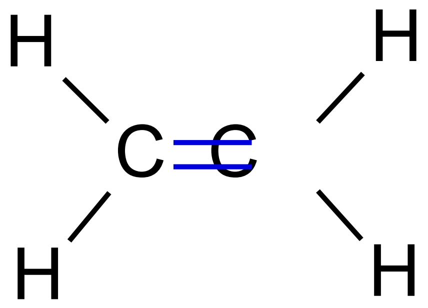
Orbital Overlapping Diagram for CH_4



Four sp^3 hybrid orbitals are formed: tetrahedral shape.

EXAMPLE 2: C_2H_4

2 x C	=	8 é
4 x H	=	4 é
Total	=	12 é



Total electron-pair at C = 3
Electron-pair geometry: **trigonal planar**
Molecular geometry: **trigonal planar**
Type of hybridisation : ***sp²***

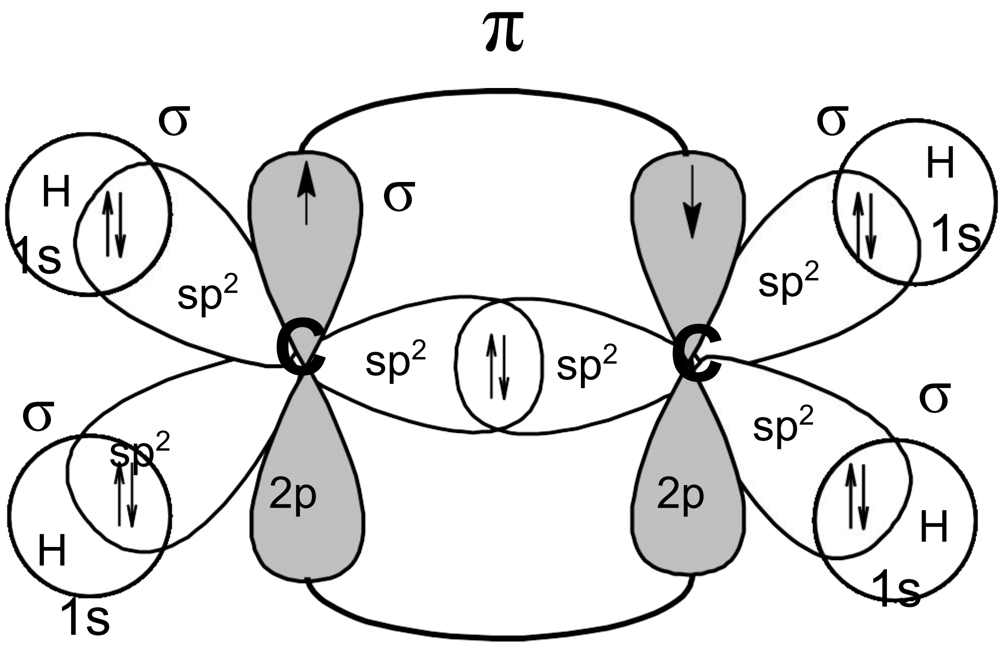
C (ground state): [↑↓] 2s [↑][↑][ ] 2p

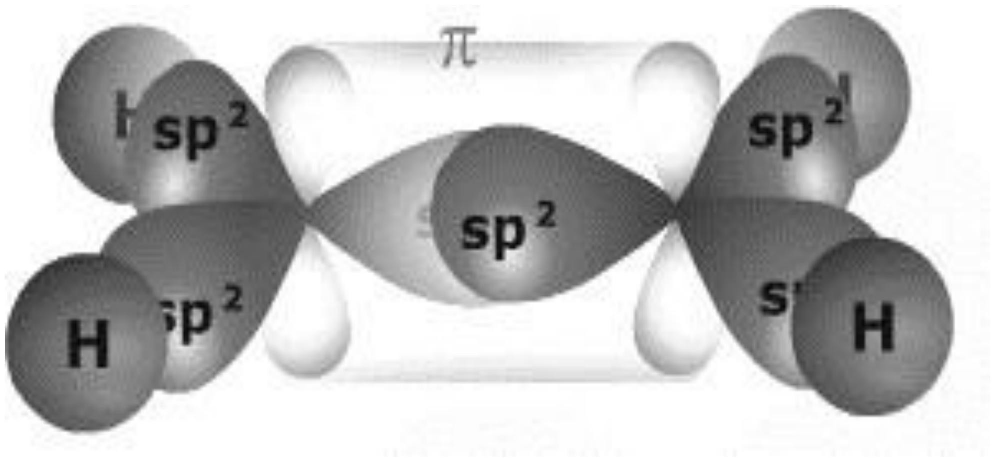
C (excited state): [↑] 2s [↑][↑][↑] 2p

↑

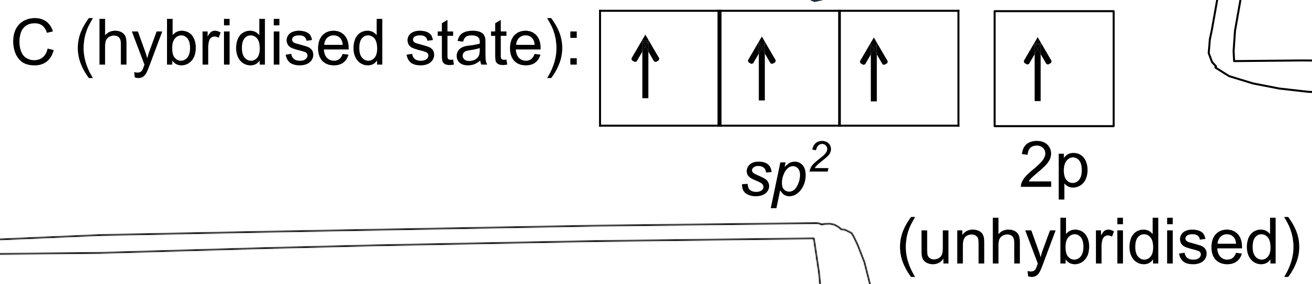
C (hybridised state): [↑][↑][↑] sp^2 [↑] 2p (unhybridised for pi bond) ⇐ unhybridised 2p orbital

Orbital Overlapping Diagram for C_2H_4





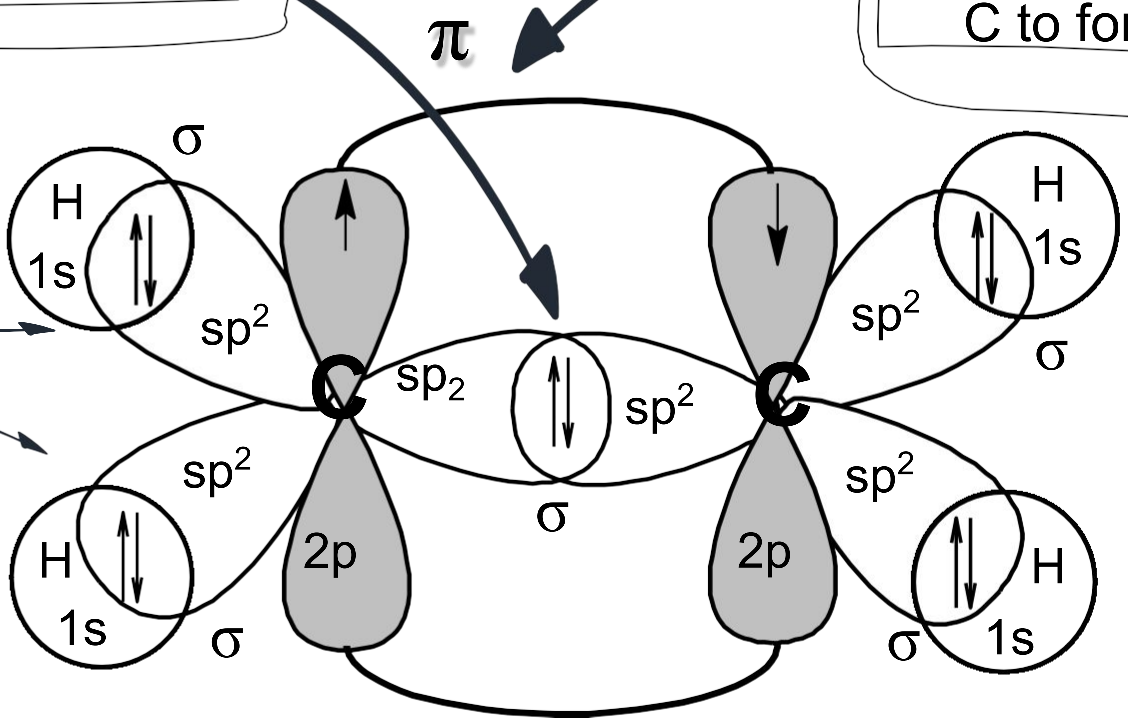
Three sp^2 hybrid orbitals of C (each contains an unpaired electron)



sp^2 orbital of C atom overlaps with sp^2 orbital of adjacent C atom to form σ bond.

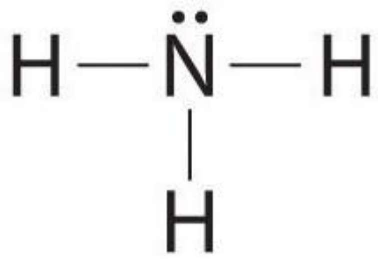
The unhybridised *2p* orbital of C which contains an unpaired electron **overlap side-by-side** with similar orbital of the adjacent C to form a π **bond.**

sp^2 orbital of C atom overlaps with *1s* orbital of H atom to form σ bond.



EXAMPLE 3: NH_3

Lewis Structure



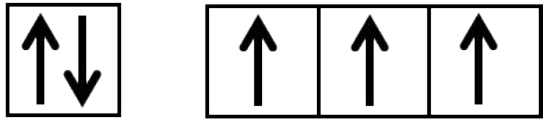
Total electron-pair at N = 4

Electron-pair geometry: **tetrahedral**

Molecular geometry: **trigonal pyramidal**

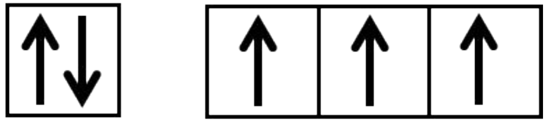
Type of hybridisation : ***sp^3***

N (ground state) :



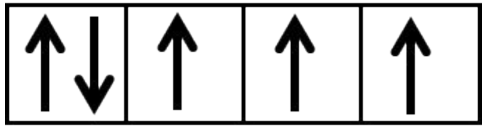
2s 2p

N (excited state) :

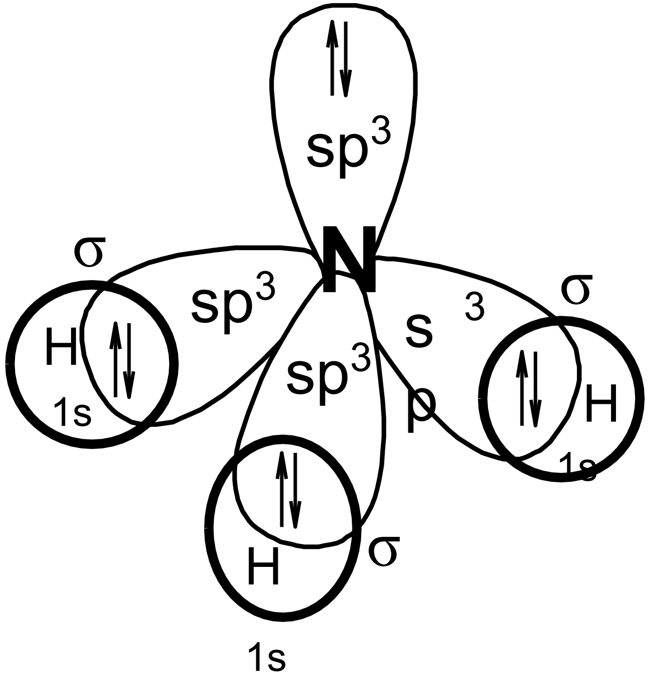


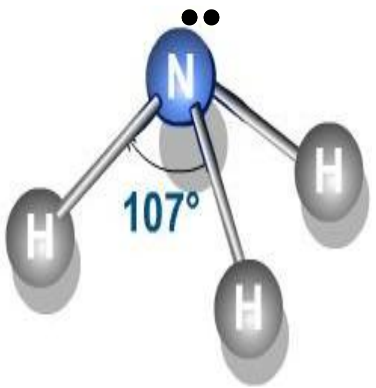
2s 2p

N (hybridised state) :

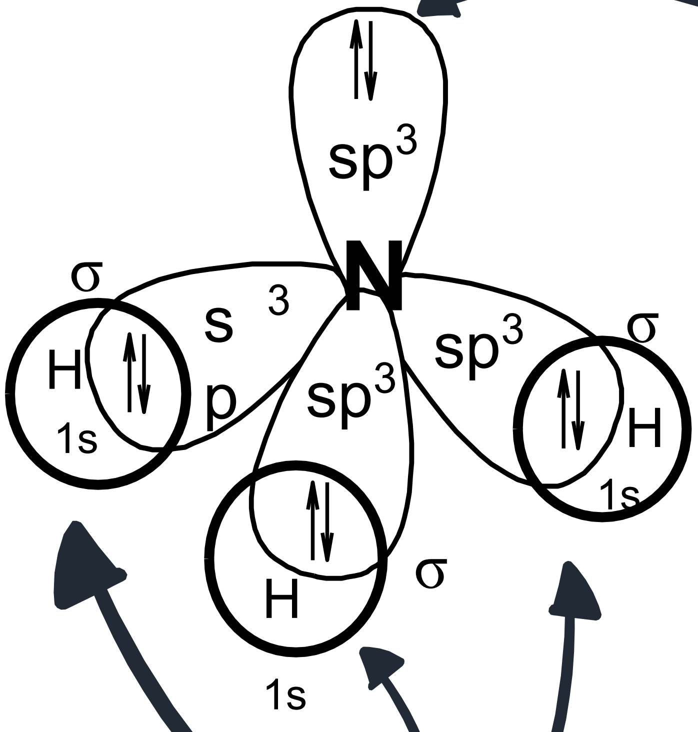


sp^3





The **fourth hybrid orbital** accommodates the **lone pair ē** at the nitrogen

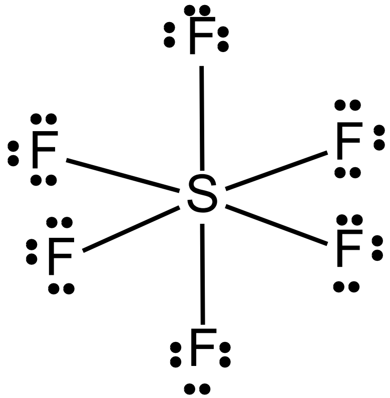


Three out of four *sp³* hybridised orbitals form **covalent N-H bonds**

***sp³* orbitals** overlap with ***s* orbital** of **three** hydrogen atoms to form **three sigma bonds.**

EXAMPLE 4: SF_6

Lewis Structure



Total electron-pair at S = 6

Electron-pair geometry: **octahedral**

Molecular geometry: **octahedral**

Type of hybridisation : ***sp^3d^2***

S (ground state) :

3s	3p			3d				
↑↓	↑↓	↑	↑					

S (excited state) :

3s	3p			3d				
↑	↑	↑	↑	↑	↑			

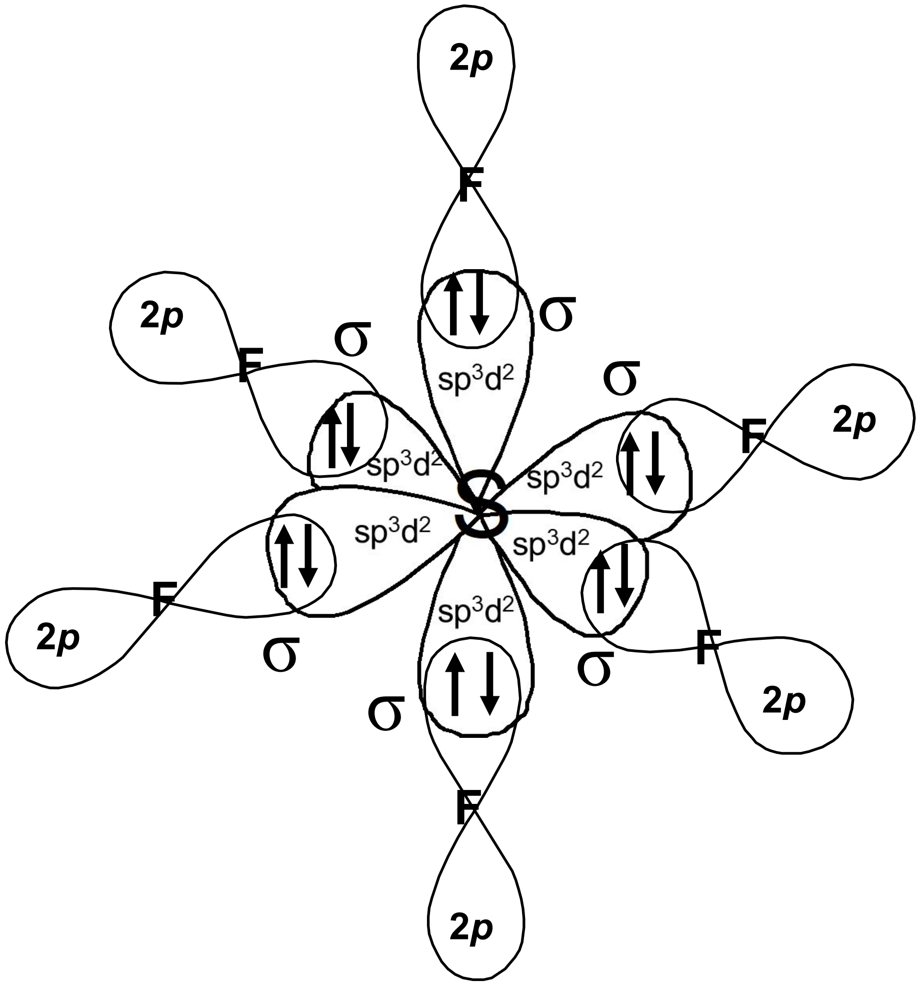
S (hybridised state) :

sp^3d^2					
↑	↑	↑	↑	↑	↑

F :

2s	2p		
↑↓	↑↓	↑↓	↑

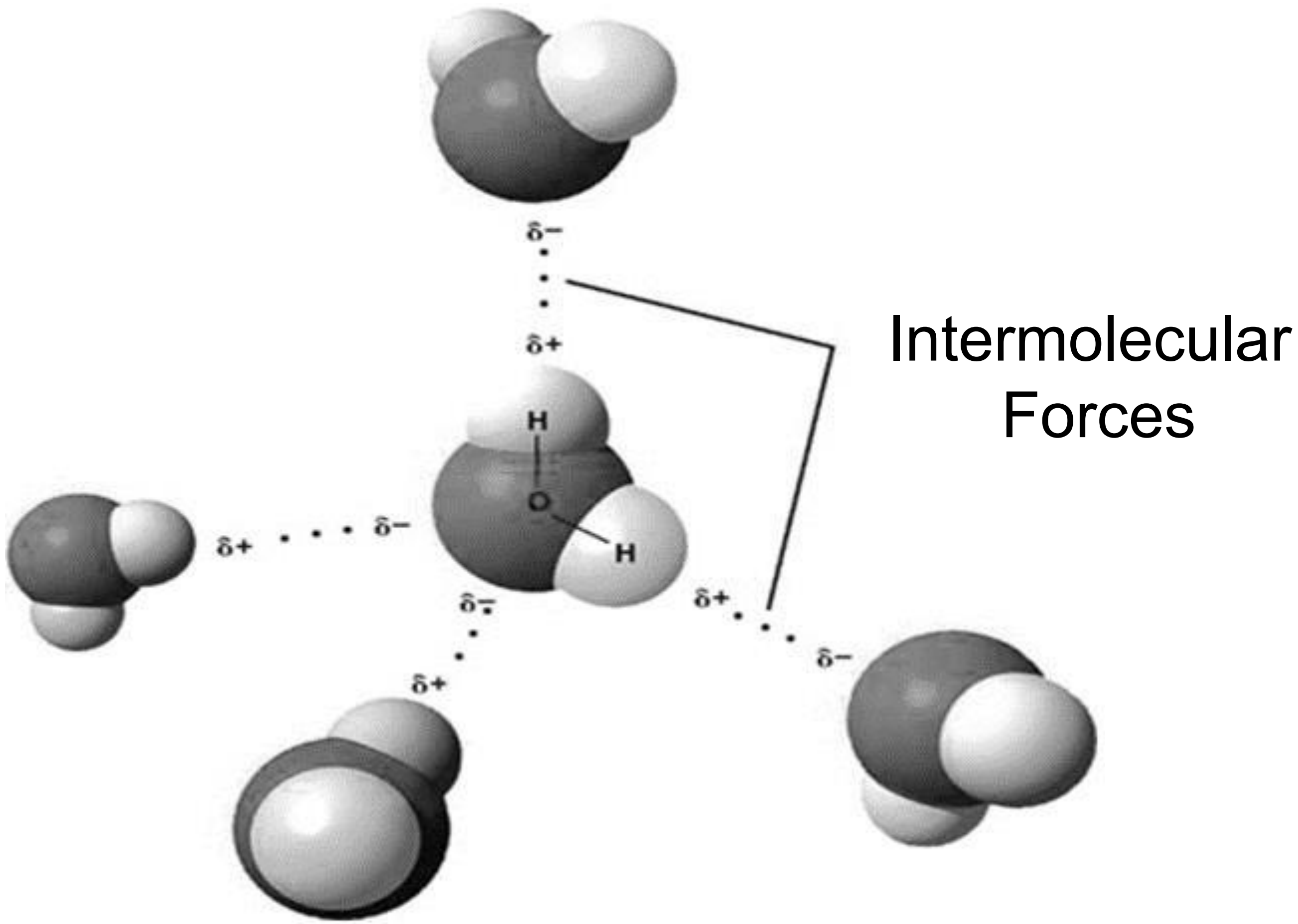
Orbital Overlapping Diagram for SF_6



4.4
Intermolecular Forces

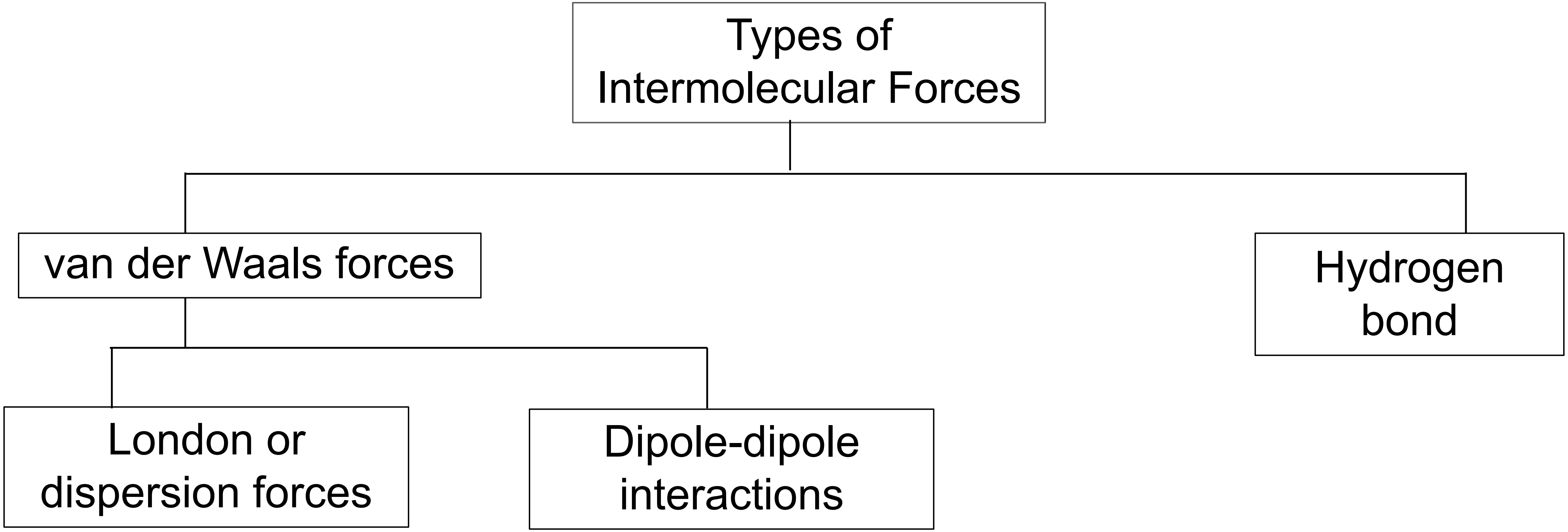
INTERMOLECULAR FORCES

- Forces of attraction between molecules



INTERMOLECULAR FORCES

- ✔ The strength of intermolecular forces are responsible for the physical properties of simple molecular compounds.



- ✔ **Relative strength of intermolecular forces**:

  London @ dispersion forces **<** dipole-dipole interactions **<** Hydrogen bond

INTERMOLECULAR FORCES: **Dipole-dipole interactions**

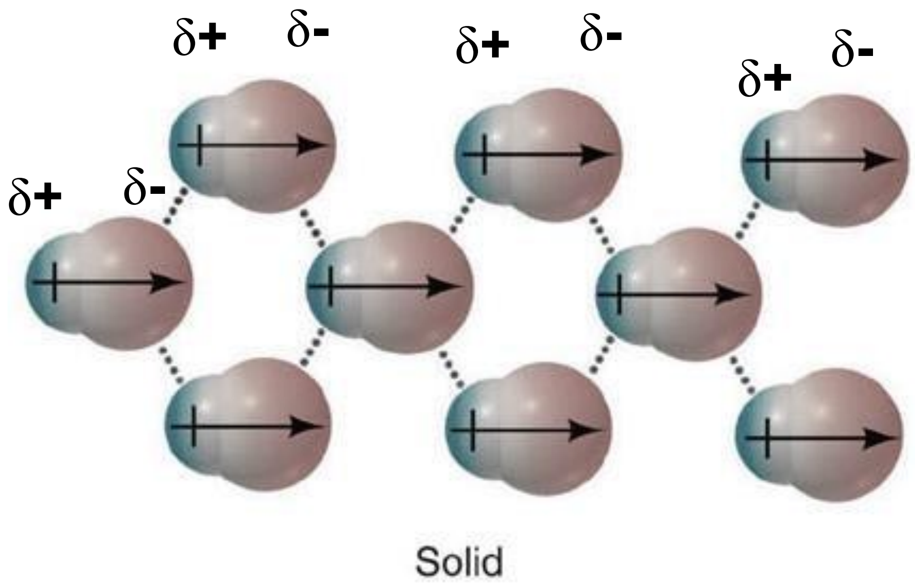
Polar covalent molecules have **permanent dipoles**.

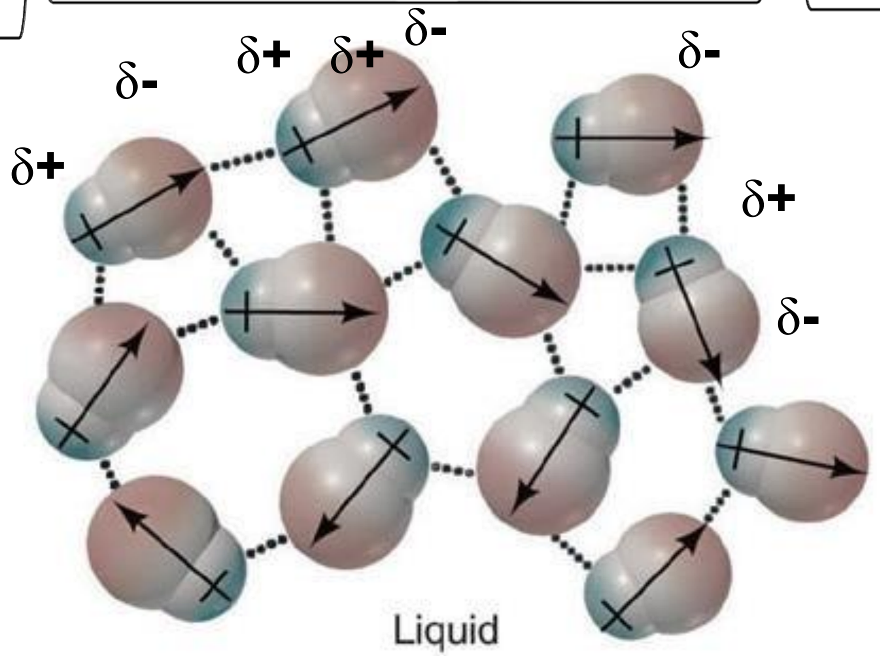
Dipole-dipole interactions are the **attractive forces between polar covalent molecules**.

The molecules will **orientate themselves** so that the **opposite charges** attract each other effectively.

The molecules must be **closed to each other** for the **dipole-dipole interactions** to be **significant.** The interactions decrease as the distance between molecules increases.

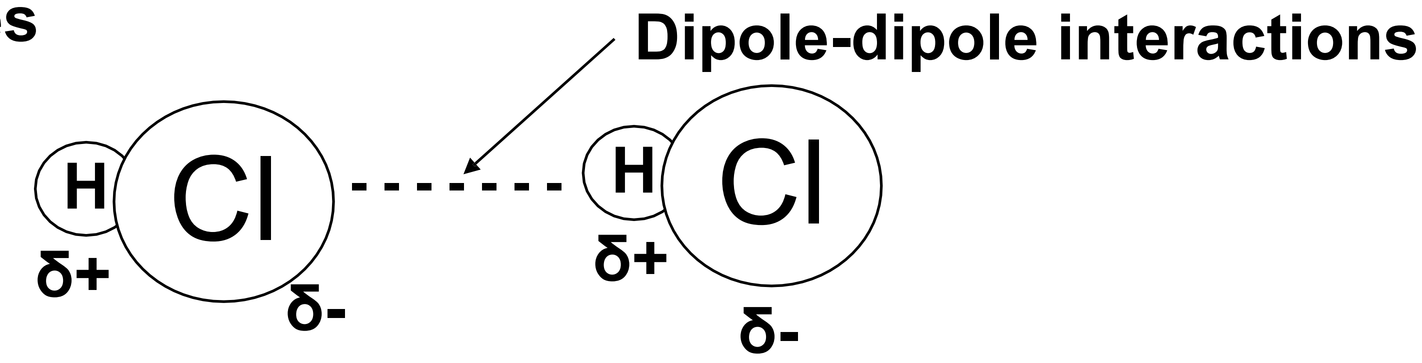
Strength of **dipole-dipole interactions** increases with an **increase in polarity of the molecule**.





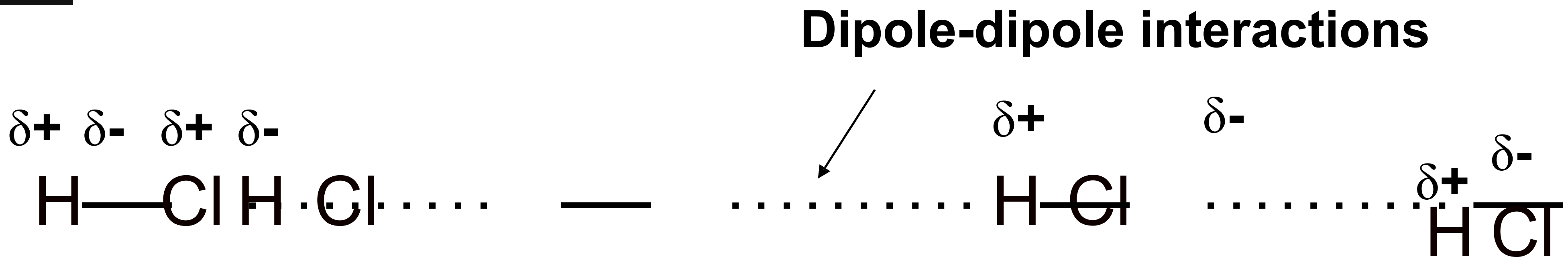
INTERMOLECULAR FORCES: **Dipole-dipole Interactions**

Example: **HCl molecules**



How do the dipole-dipole interactions occur ?

Polar molecules attract one another when the ***partial positive charge*** on one molecule is near the ***partial negative charge*** on the other molecule.

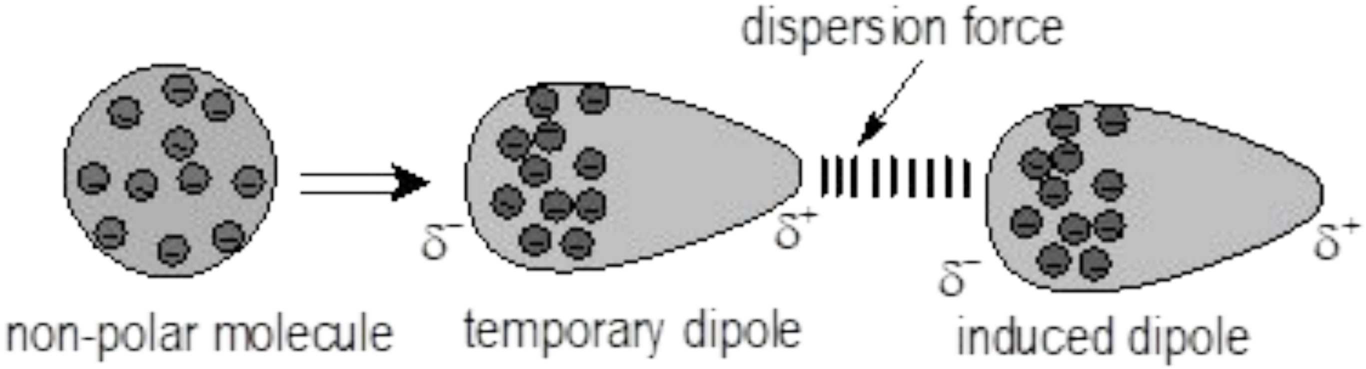


INTERMOLECULAR FORCES: **London forces or Dispersion forces**

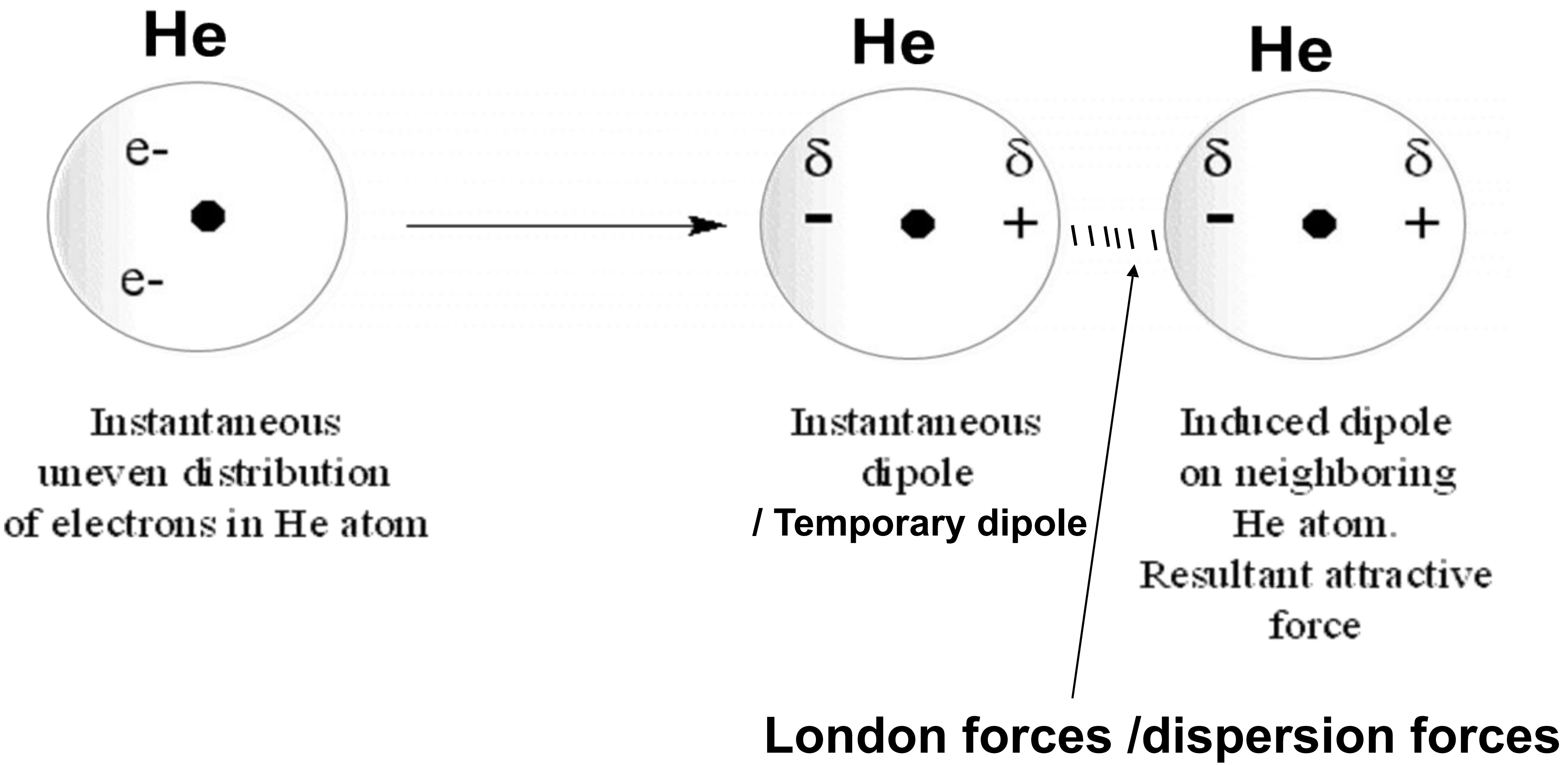
1 Due to **random movements of electrons** in a **non-polar molecule**, there is a high possibility that at any given instant, the electron density might be higher at one end compared to the other, thus giving rise to an **instantaneous dipole** or **temporary dipole.**

2 This **temporary dipole** will **induce a similar dipole** in a neighbouring molecule, causing both the molecules to be attracted to one another.

3 The attractions between the temporary dipoles and induced dipoles are called **London forces** or **dispersion forces**.

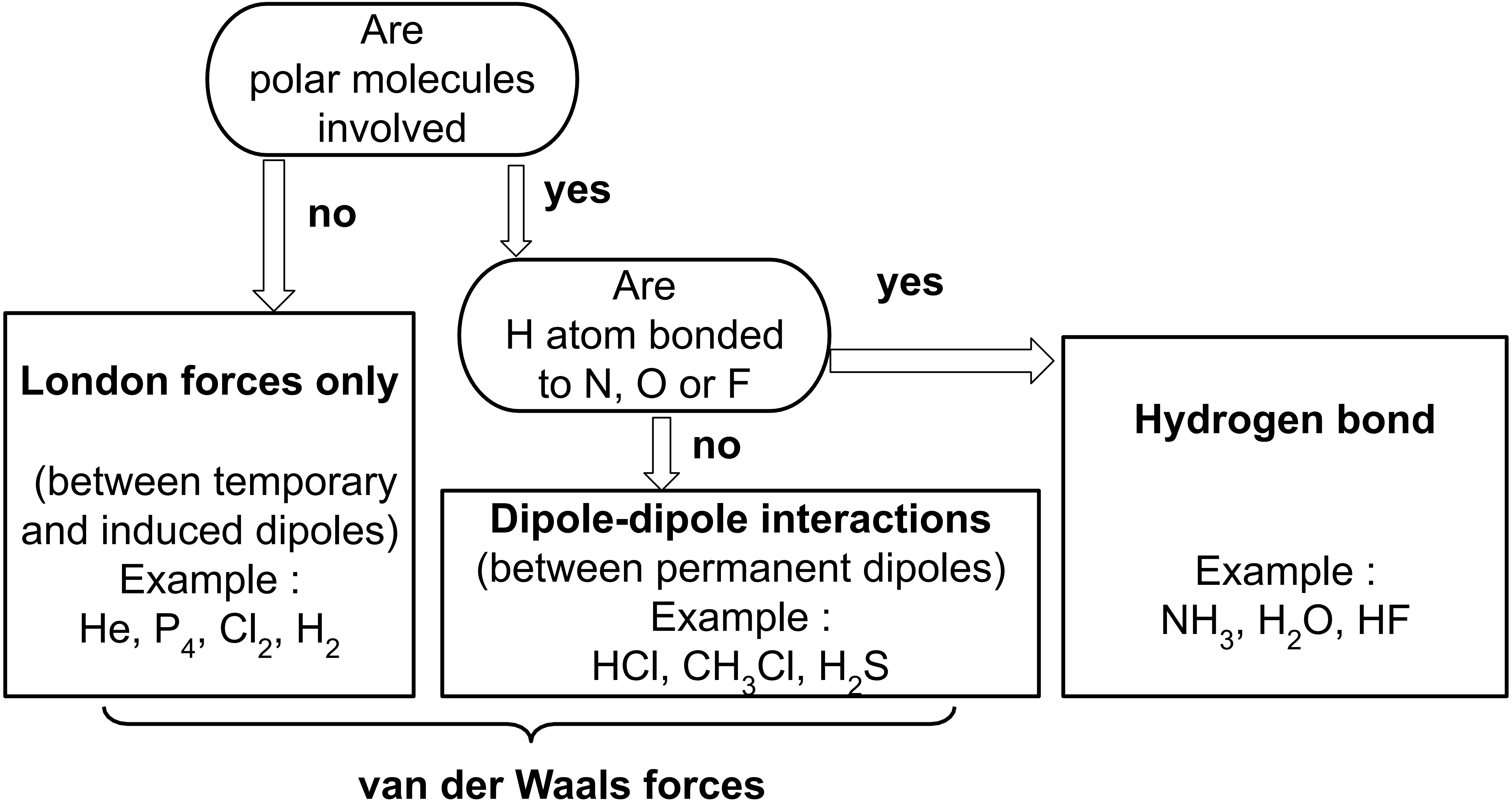


Example:


He
e-
e-
Instantaneous
uneven distribution
of electrons in He atom
He
δ
-
δ
+
Instantaneous
dipole
/ Temporary dipole
He
δ
-
δ
+
Induced dipole
on neighboring
He atom.
Resultant attractive
force
London forces /dispersion forces

- ✔ The temporary dipoles may last just a tiny fraction of second. However, in the next instant, the electrons in different molecule may create a new temporary dipole that can induce another dipole in its neighbour.
- ✔ **London forces** exist between **all covalent molecules**, regardless of polar or nonpolar molecules.
- ✔ The **strength of London forces** depends on the **polarisability of the molecule.**
- ✔ **Polarisability** is **a measure of how easily an electron cloud is distorted,** thus is a measure of the ease with which the **instantaneous** and **induced dipole** can be formed.
- ✔ **Polarisability increases** with the **increased number of electrons, and** the number of electrons increases with **increased molecular mass / size.**

STEPS IN DETERMINING THE TYPES OF INTERMOLECULAR FORCES



Example:

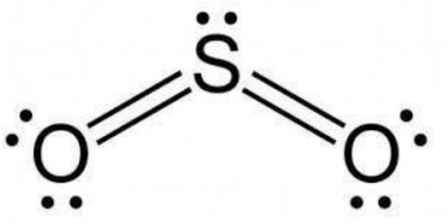
State the type(s) of intermolecular forces exist between each of the following molecules?

i) $\boxed{\mathbf{HBr}}$

δ+ δ-
H-B
r

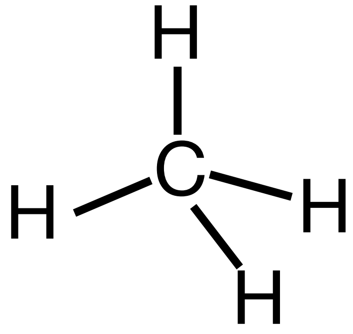
HBr is a **polar molecule**.
- Dipole-dipole interactions.
- London forces.

ii) $\boxed{\mathbf{SO_2}}$

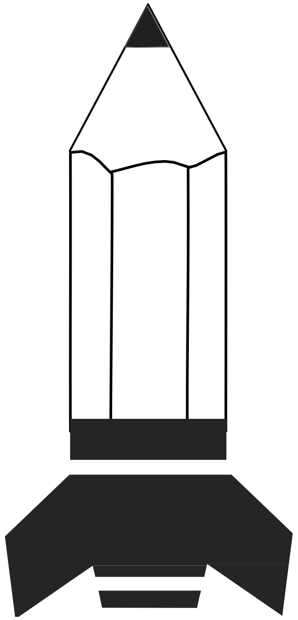


SO_2 is a **polar molecule**.
- Dipole-dipole interactions.
- London forces

iii) $\boxed{\mathbf{CH_4}}$



CH_4 is **non-polar molecule**.
- London forces.



Factors that influence the strength of van der Waals forces :

i. Molecular size

ii. Molecular shape

iii. Molecular polarity

i) Molecular Size

- The **larger** the **molecular size**, the **higher the number of electrons**, **polarisability increases**.

- The **strength of London forces** therefore tends to **increase with increasing molecular size**.

- The **stronger the London forces**, the h**igher the boiling point** and the **melting point** of the molecules.

Example :

Boiling point of sulphur higher than that of phosphorus and chlorine. Explain.

Molecular formula	P_4	S_8	Cl_2
Boiling point (K)	553	718	238

- ✔ Phosphorous, sulphur and chlorine are non-polar molecules.

- ✔ Sulphur molecule is the **biggest in size.** Therefore, it has **strongest London forces@ van der Waals forces** between molecules

- ✔ Therefore, **sulphur has the highest boiling point** than phosphorus and chlorine.

ii) Molecular Shape

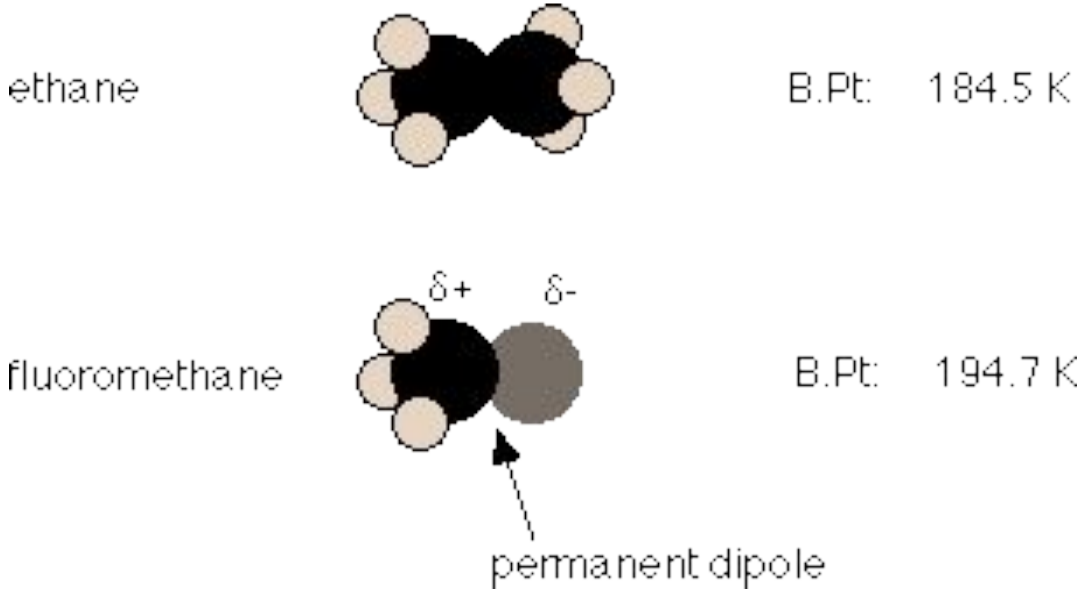
- The **shapes of the molecules** influence the magnitude of London forces.
- The strength of van der Waals **increases with the total surface area of the molecule.**

Example:

Compound	Boiling point
CH_3-CH_2-CH_2-CH_3 Butane	-0.5˚C
CH_3CHCH_3 | CH_3 2-methylpropane	-11.7˚C

- Butane and 2-methylpropane have same molecular size.
- **Butane** is a **straight chain** molecule **has greater surface area** than 2-methylpropane which is branching molecule.
- The **greater** the **surface area**, the **stronger** the **van der Waals forces**, the **higher** the **boiling point**.

iii) Molecular Polarity



- Both molecules have comparable molecular mass. That means that the strength of London forces in both molecules should be much the same.

- van der Waals (**dipole-dipole interactions**) between the **polar fluoromethane** molecule is **stronger** than the non-polar ethane molecule **(London forces).**
- The **higher the polarity** of a molecule, the **stronger the van der Waals forces**, therefore the **higher the boiling point.**

INTERMOLECULAR FORCES: **Hydrogen Bond**

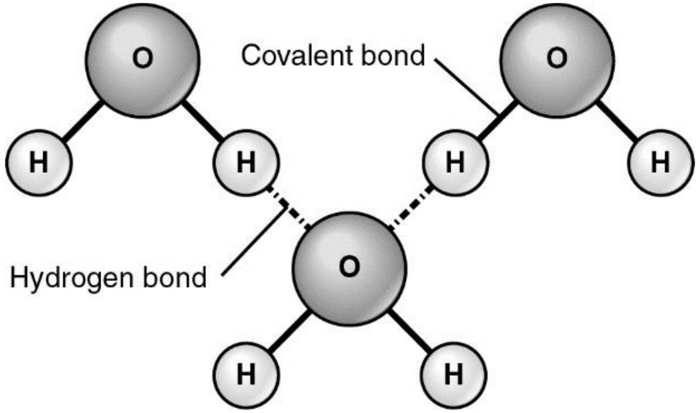
- Is an **attraction between a hydrogen atom attached to a highly electronegative atom (F, O and N)** and **a nearby small electronegative atom (F,O and N) in another molecule.**

- Is **stronger intermolecular forces** than either London forces or dipole-dipole interactions.

- Involving molecules with **polar N-H, O-H and F-H bond**.

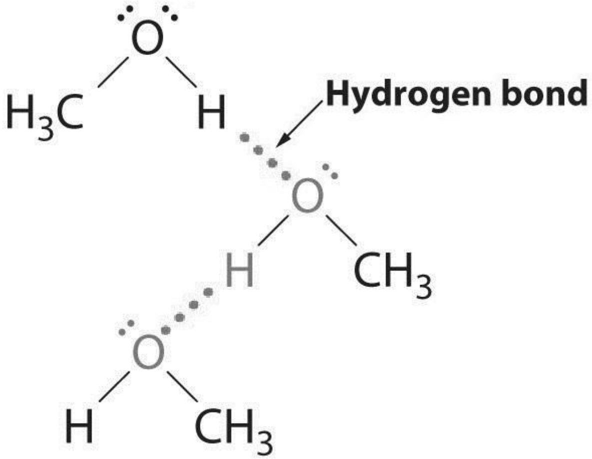
- The **electrostatic attraction** between these molecules will be **greater than those between the polar molecules**.

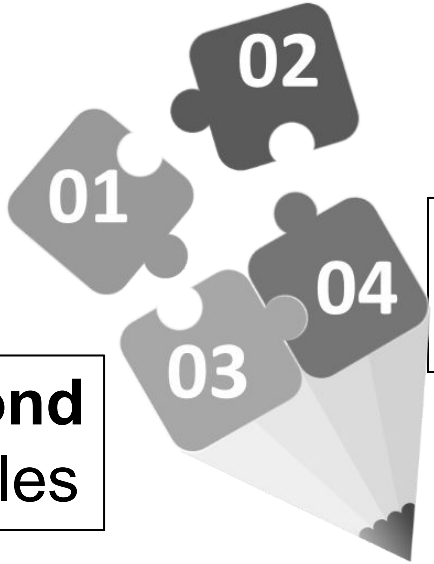
HYDROGEN BOND BETWEEN MOLECULES



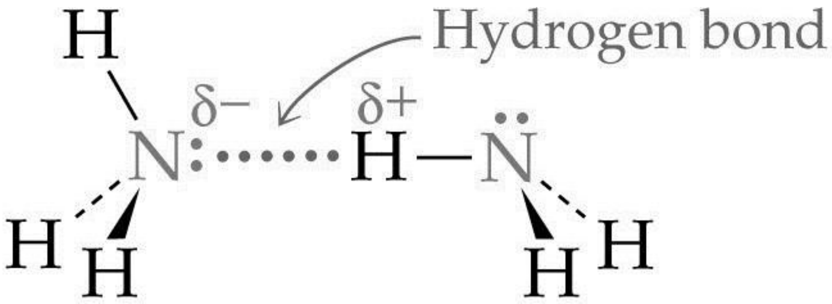
Hydrogen bond
between **water** molecules

Hydrogen bond
between **CH_3OH**
molecules

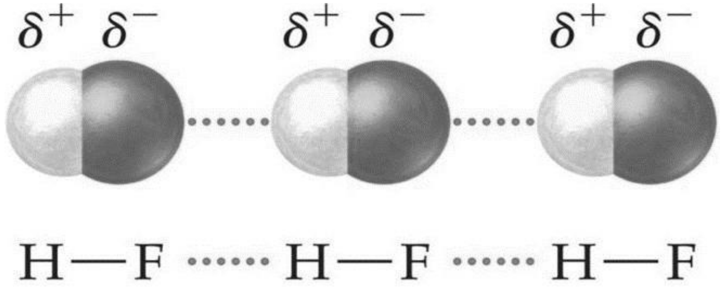




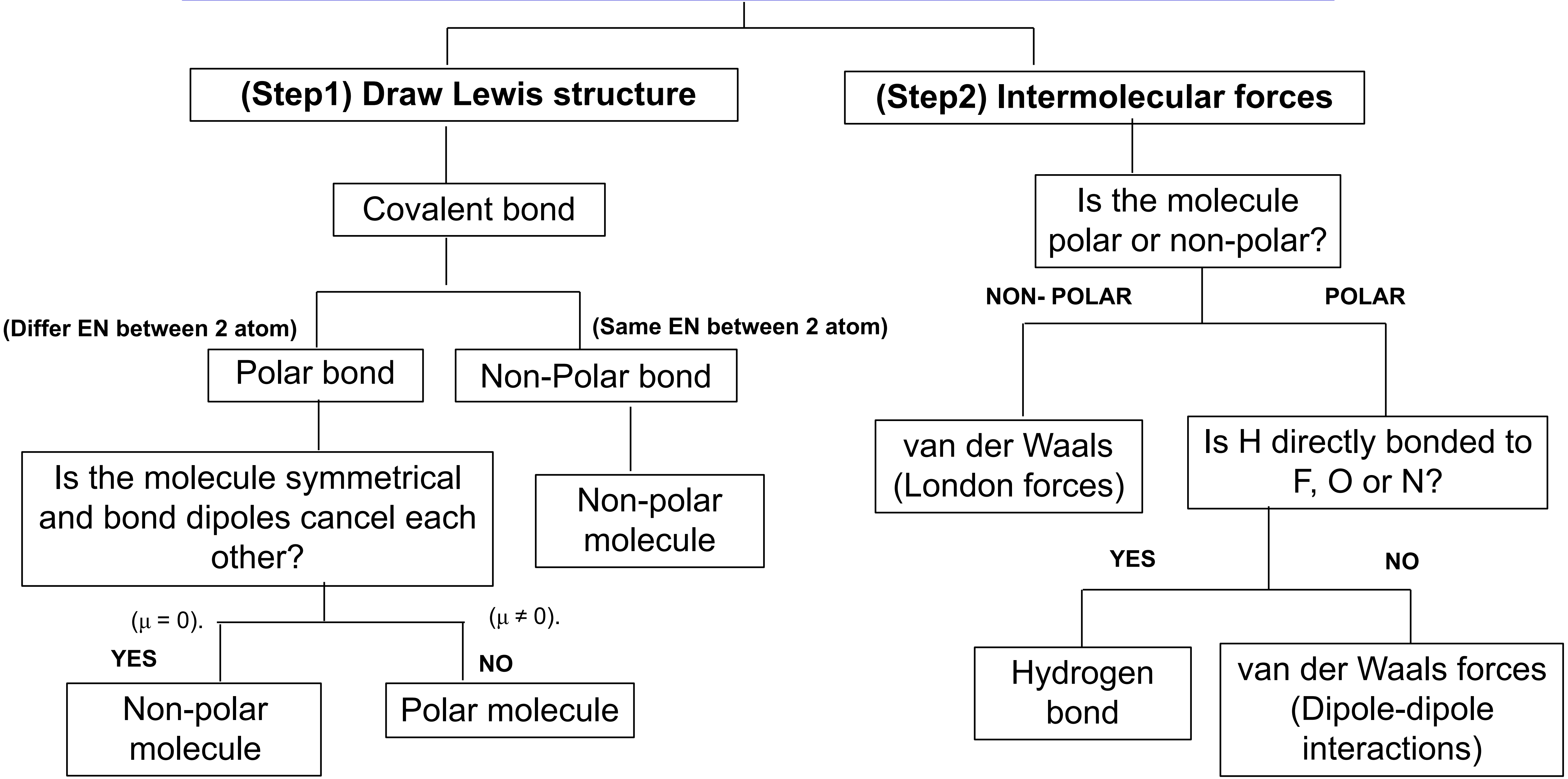
Hydrogen bond
between **NH_3** molecules



Hydrogen bond
between **HF** molecules



DETERMINATION OF INTERMOLECULAR FORCES



EFFECT OF HYDROGEN BOND

BOILING POINT

- Boiling point of H_2O is higher than NH_3 because the **hydrogen bonds between water molecules are stronger** due to the **higher electronegativity of the oxygen atom compared to nitrogen atom** in NH_3.
- Boiling point of H_2O is higher than HF
- ˙Although F is more electronegative than O and the hydrogen bond between HF molecules are stronger than the hydrogen bonds in H_2O molecules, each **HF molecule form less hydrogen bond compared to H_2O molecules.**

SOLUBILITY

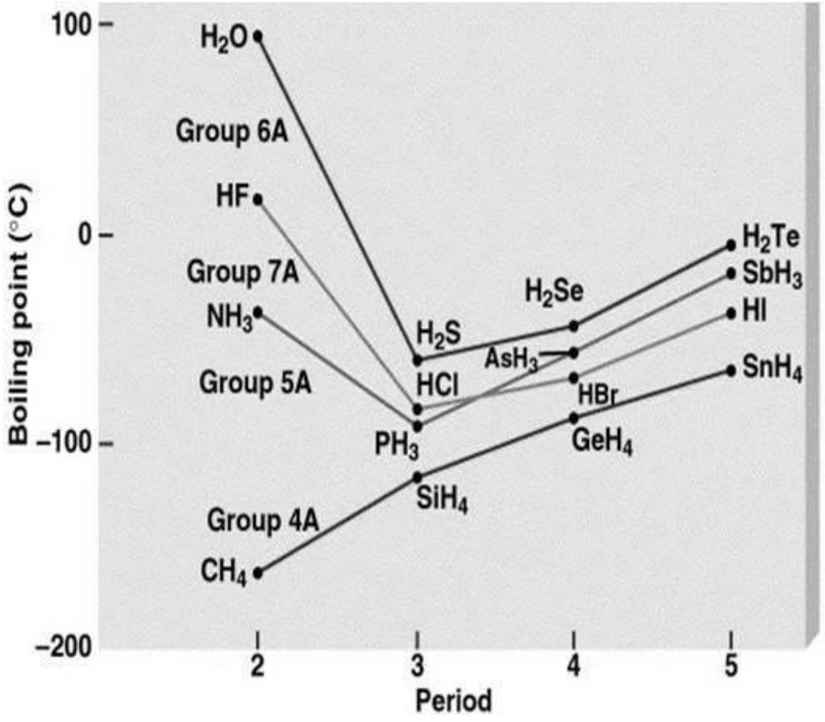
- Water is **polar solvent**.
- **Polar molecules with -OH, -NH_2 and -COOH** groups are **soluble in water** because these groups can **form hydrogen bonds with water molecules**.
- The **non-polar** molecules such as **hydrocarbons**, fats are **insoluble in water because incapable to form hydrogen bond with water molecules.**
- **Solubility of hydrocarbons in water decreases with increasing molecular size** becomes (**hydrophobic** area) **larger**.

" Like dissolves like"

DENSITY

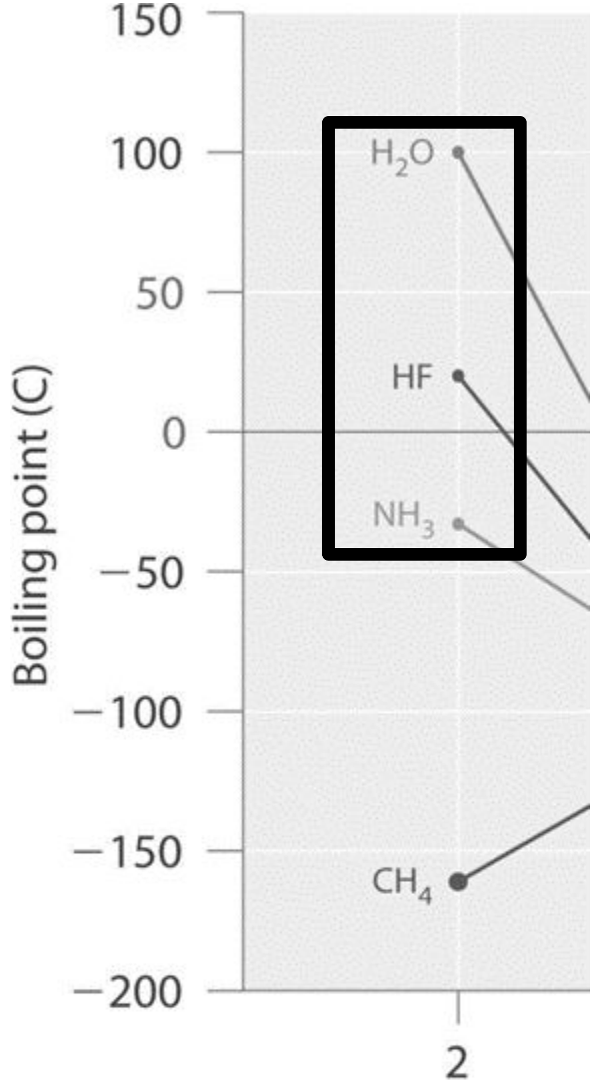
- When **cooled to a solid**, the water molecules **arrange** themselves to **form tetrahedral arrangement via hydrogen bonds to four other H_2O** molecules in a **repeating 3D network**.
- This gives **rise to open structure in ice** with **large spaces between the H_2O** molecules and consequently **ice has lower density than its liquid (water).**
- Therefore, **ice floats on water.**

Effects of hydrogen bond: **Boiling Point**



Halides	HF	HCl	HBr	HI
Molecular mass	20	37	81	128

- Boiling points of a series of compounds generally increase as the molecules get larger.
- This is due to the increase in the strength of London forces.
- However, for hydrides of Group 17 elements, HF is an exception because of the **stronger intermolecular attractive forces @ force of attraction between HF molecules** resulting from **hydrogen bonds acting between the HF molecules.** Thus, it has highest boiling point.



- Boiling point of H_2O is higher than NH_3 because the **hydrogen bonds between water molecules are stronger due to the higher electronegativity of the oxygen atom compared to nitrogen atom** in NH_3.
- Boiling point of H_2O is higher than HF.
- Although F is more electronegative than O and the hydrogen bond between HF molecules are stronger than the hydrogen bonds in H_2O molecules, each **HF molecule can form less hydrogen bonds compared to H_2O molecules.**

Effects of hydrogen bonding : **Solubility**

Water is a good solvent due to its polarity.

When an **ionic or polar** compound enters water, it is **surrounded by water molecules**

The **partially negative dipoles of the water are attracted to positively charged components of the solute**, and vice versa for the positive dipoles.

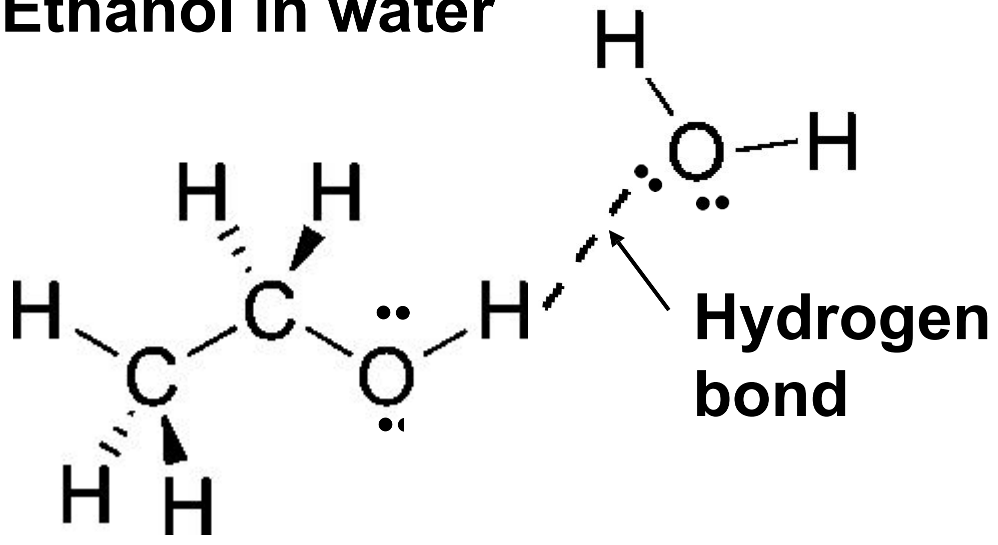
The relatively small size of water molecules typically allows many water molecules to surround one molecule of solute.

In general, **ionic and polar** substances such as **acids, alcohols,** and **salts** are **easily soluble in water**, and **non-polar substances** such as **fats and oils are not.**

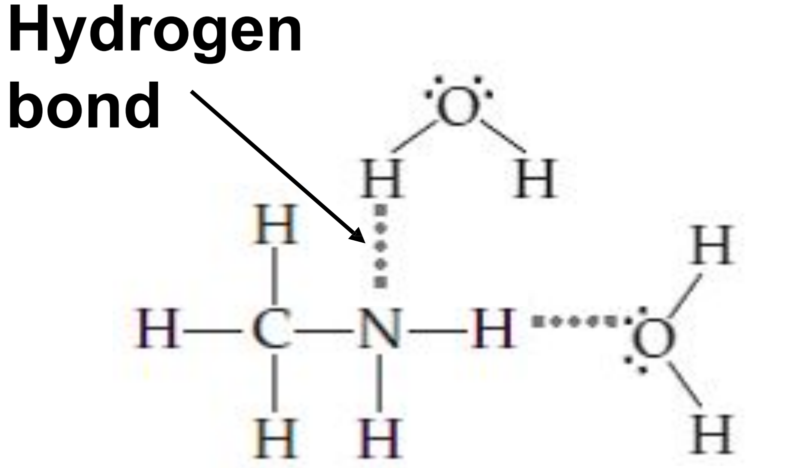
The solubility of a substance is greatly dependent on **its ability to form hydrogen bonds** with water molecules.

Organic molecules with **-OH, $-NH_2$ and -COOH** groups are **soluble in water** because these groups are **able to form hydrogen bonds with water** molecules.

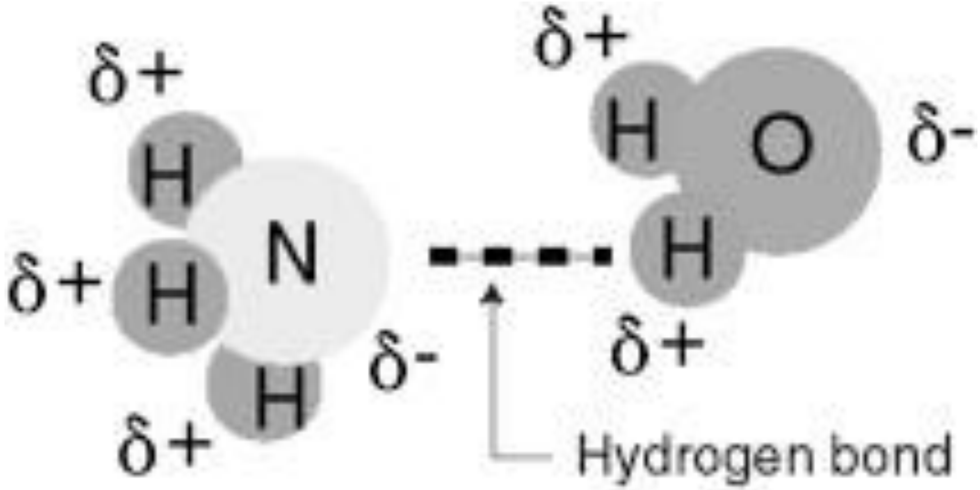
Example 1: Ethanol in water



Example 2: Methylamine in water



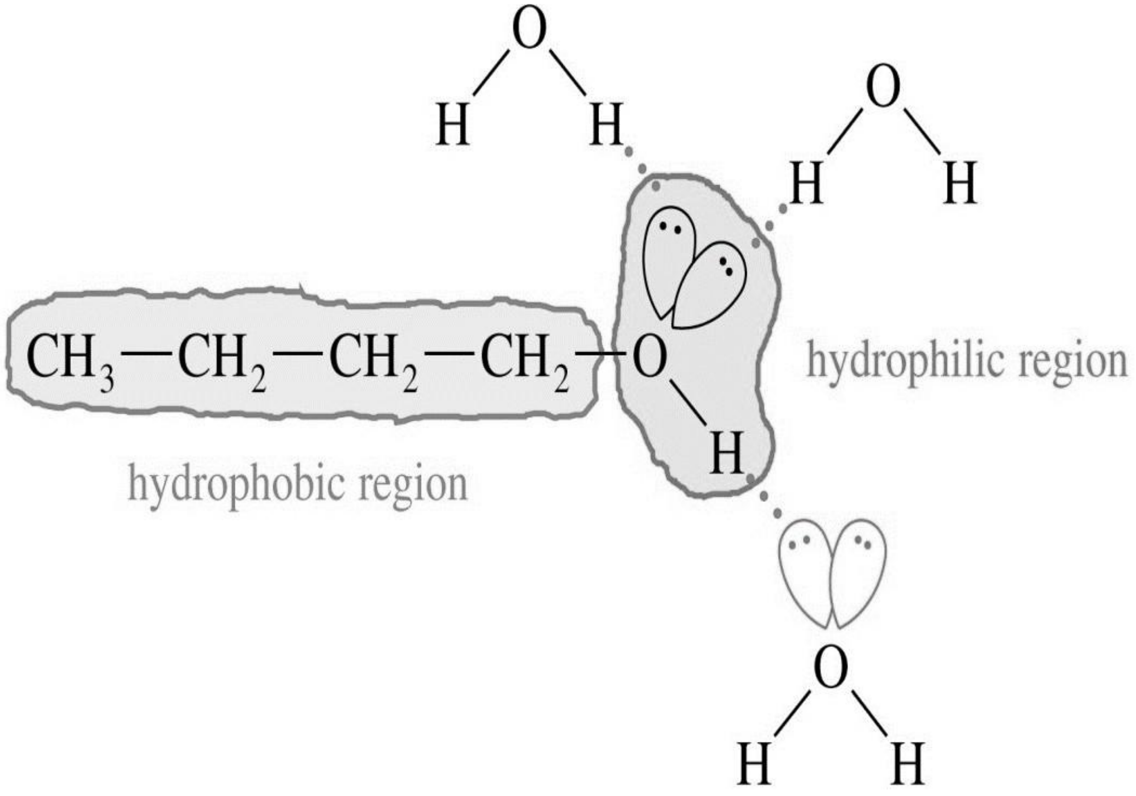
Example 3: Ammonia in water



- However, solubility of substances **decreases with increase in size of the molecules.**

  i.e. solubility is inversely proportional to the molecular size of the molecule (solubility decreases as molecular size increases).

- This is **due to increase in hydrophobic area** of the substances.



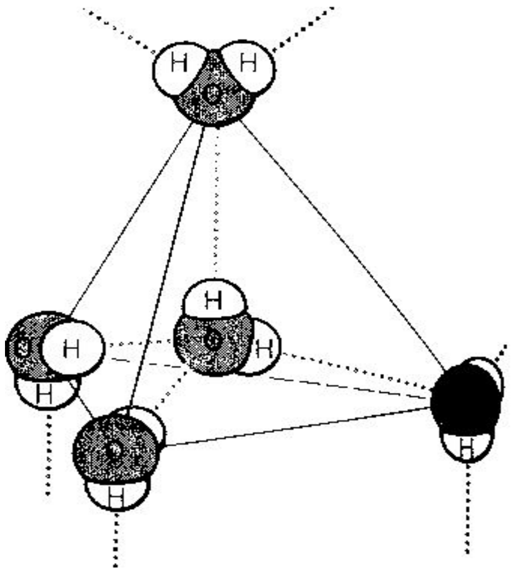
Structural Formula	Name	Molecular Weight	Solubility in Water
CH_3OH	methanol	32	infinite
CH_3CH_2OH	ethanol	46	infinite
$CH_3CH_2CH_2OH$	1-propanol	60	infinite
$CH_3CH_2CH_2CH_2OH$	1-butanol	74	8 g/100 g

Effects of hydrogen bonding: **Density of water compared to ice**

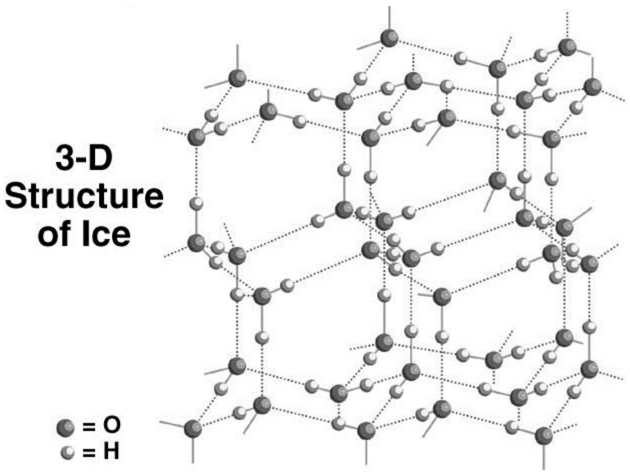
- **H_2O is unusual in its ability** to **form an extensive hydrogen bond network.**

- When **cooled to a solid**, the **water molecules arrange themselves to form tetrahedral arrangement** in such a way to maximize the amount of **hydrogen bond between them.**

- They leave a **relatively large amount of space** between them and **gives rise to an "open" structure**.



Hydrogen bond (dotted lines) between water molecules in ice.



- This arrangement of molecules **has greater volume (less dense) than liquid water**, thus water expands when frozen.

- This **“open” structure of the ice accounts for the fact that ice is less dense than water at 0°C**.

- **Water has a higher density than ice**.

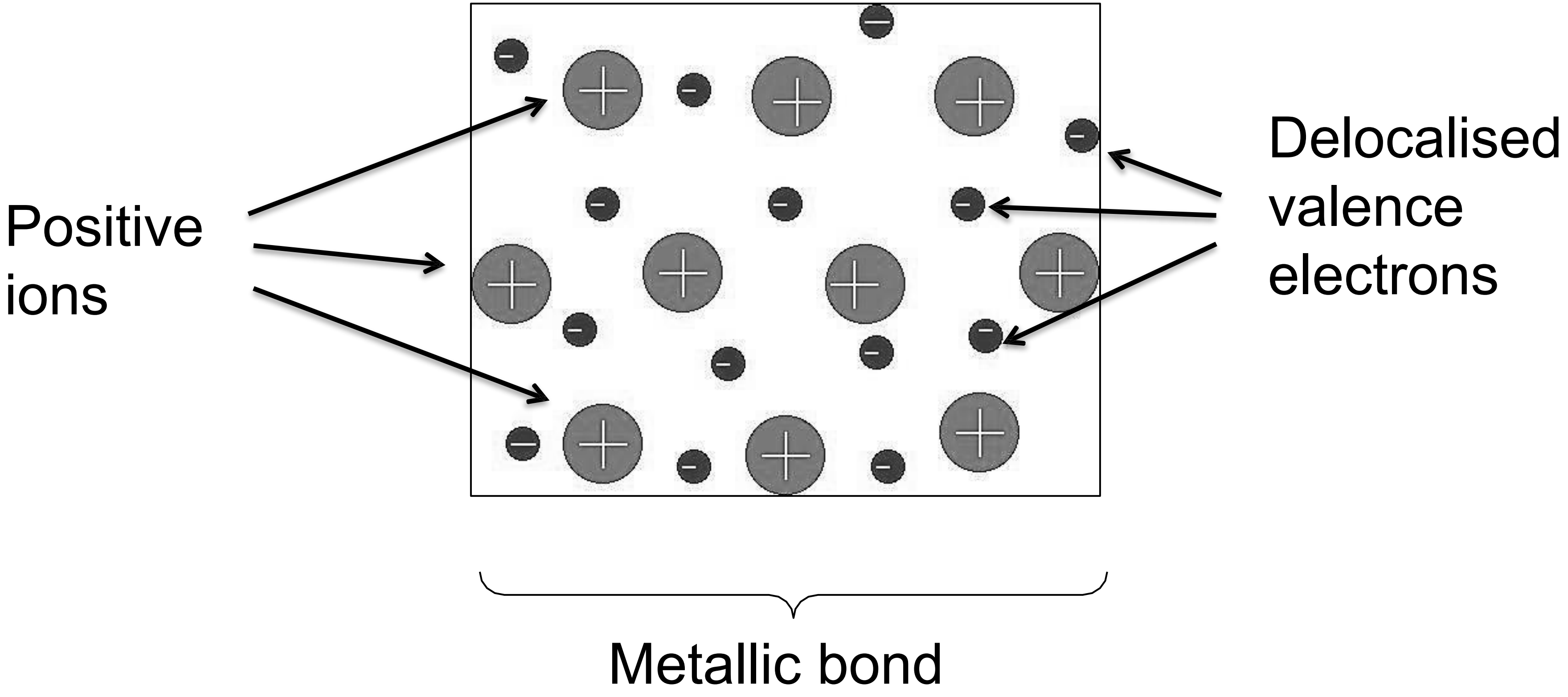
Ice is less dense than water

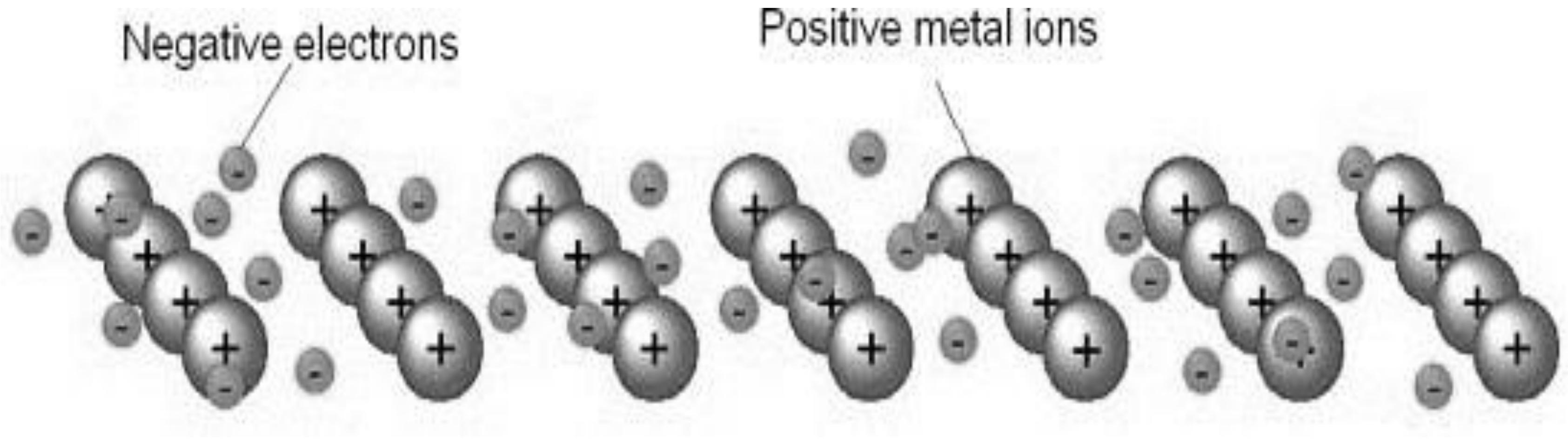
4.5
Metallic Bond

METALLIC BOND

- Defined as the **electrostatic force** between the **positively charged metallic ions** and the **‘sea’ of its delocalised valence electrons**.

The Electron Sea Model



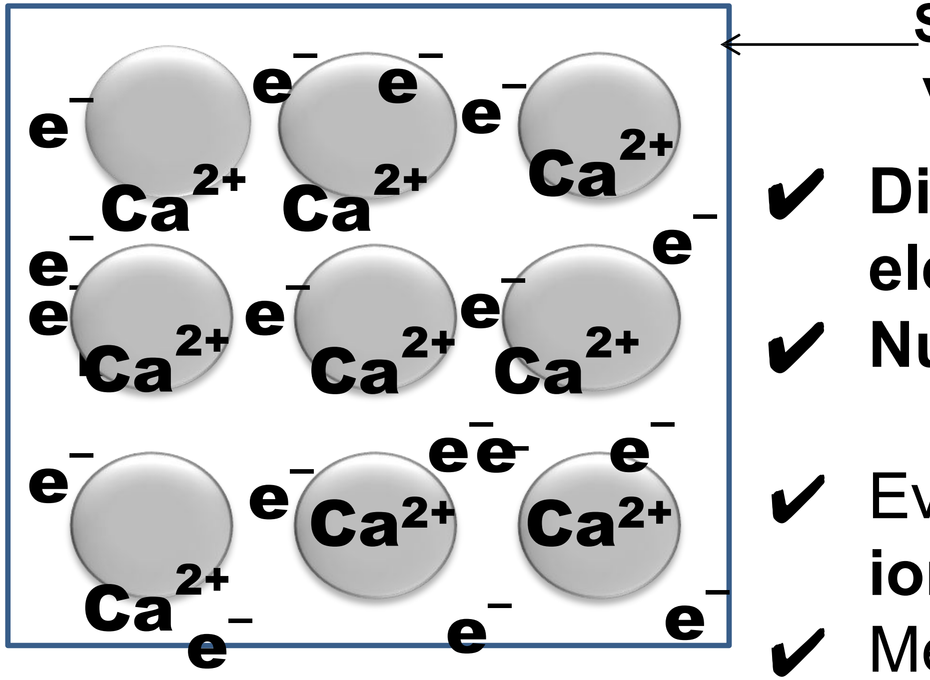


- In a metallic bond, metals atoms can be imagined as an array of **positive ions** immersed in a sea of **delocalised valence electrons**.

- These delocalised valence electrons are **not bound to individual ions** and they can therefore serve to bind large numbers of metal ions together.

PSPM 2022/2023

Calcium is a silvery-white metal that tarnishes easily in the air. By using the electron sea model, illustrate the formation of metallic bonds in calcium. Explain the electrical conductivity exhibited by calcium.



- ✔ **Diagram** with **label sea of delocalised valence electrons.**
- ✔ **Number** of **delocalised electrons equal with charge.**

- ✔ Every **Ca atom releases two electrons** and forms **Ca^{2+} ion.**
- ✔ Metallic bond forms from the **electrostatic attraction between positive ions with negatively charge sea of valence electrons.**
- ✔ Calcium can conduct electricity due to **free electrons/ delocalised electrons@ mobile electrons@ free moving electrons.**

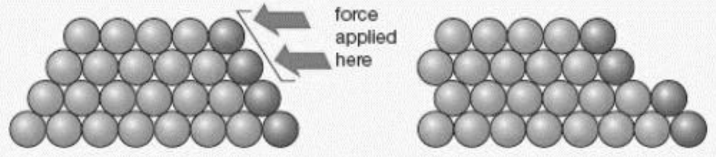
PROPERTIES OF METAL

i) Malleability and Ductility

- In metallic bonding, the delocalised valence electrons do not belong to any particular metal ion.
- If sufficient force is applied to the metal, one layer of atoms can slide over one another without disrupting the metallic bonding.
- This means that metallic bond is flexible and strong.
- Therefore, metal can **be hammered into sheets (malleable)** or **drawn into wires (ductile) without breaking.**

ii) Thermal and Electrical Conductivity

- The **delocalised valence electrons** in metals are **mobile** and **free to move away** from a negative electrode to a positive electrode when a metal is subjected to an electrical potential.
- Hence, metals have high electrical conductivity.
- The mobile delocalised valence electrons can also conduct heat by carrying kinetic energy from a hot part of the metal lattice to a cold one; high thermal conductivity.



Metals change shape when an applied force causes layers of atoms to slip over each other

BAND THEORY OF ELECTRICAL CONDUCTIVITY

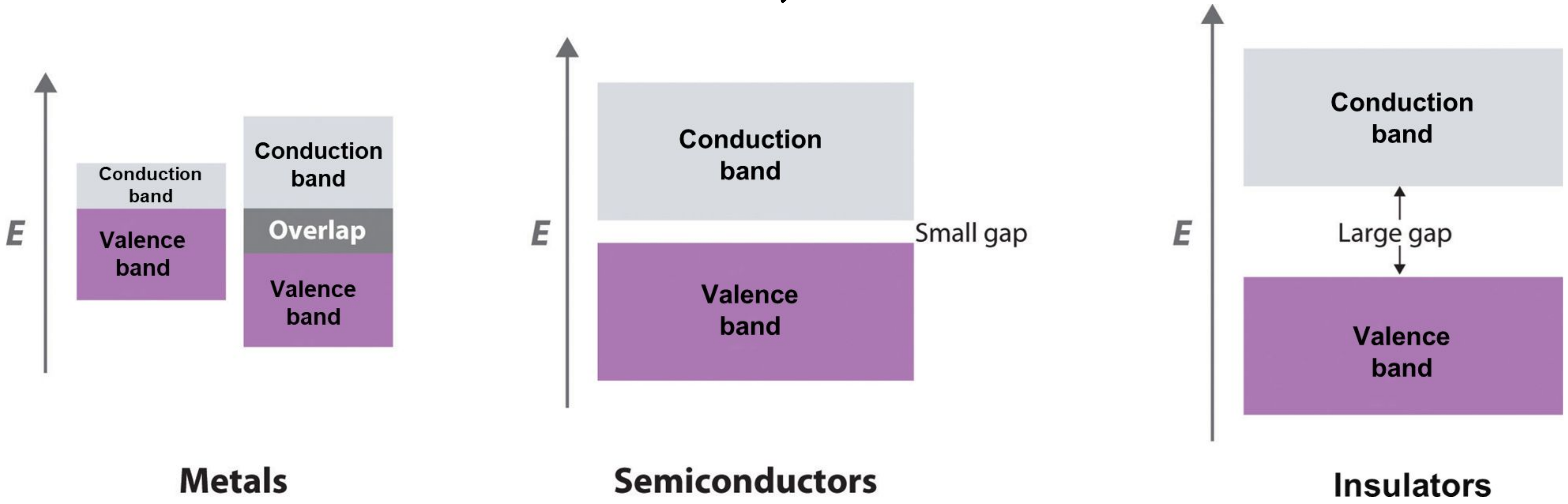
Band theory of solids

- The delocalised electrons move freely through "band" formed by overlapping molecular orbitals.
- The electronic structure of a bulk solid is referred to as a **band structure**.
- When **valence band** (lower energy) and **conduction band** (higher energy) **overlapping**, **allowing electrons to flow through the metals** with minimal applied voltage.
- Electrons jump from valence band to conduction band even at ordinary temperature and if this happens then the solid conducts electricity.
- Conductivity depends on the gap between the valence band and conduction band.

IMPORTANT TERMS

- **Band**:
  An array of closely spaced molecular orbitals occupying a continuous range of energy
- **Band gap**:
  The energy gap between a fully occupied valence band and an empty conduction band.
- **Conduction band**:
  A band of unoccupied molecular orbitals lying higher in energy than the occupied valence band
- **Valence band**:
  A band closely spaced bonding molecular orbitals that is essentially fully occupied by electron

Band structure of conductors, semiconductors and insulators



Conductors	Semiconductors	Insulators
There is no band gap between their valence band and conduction bands, since they overlap. There is a continuous availability of electrons in these closely spaced orbitals.	have a small energy gap between the valence band and the conduction band. Electrons can make the jump up to the conduction band, but not with the same ease as they do in conductors.	The band gap between the valence band the conduction band is so large that electrons cannot make the energy jump from the valence band to the conduction band.

- All **metals are good electrical conductors** because the **conduction band and valence band overlap**, allowing electrons to flow with minimal applied voltage.
- In Period 3, the **number of delocalised valence electrons increases from Na, Mg to Al.** More electrons can move and carry the charge through the metallic structure. Thus, the **electrical conductivity increases** from Na, Mg to Al.
- Si in Period 3 is **metalloid** with **electrical conductivity lower than metal**, but **higher than the non-metals**. There is a **small energy gap between conduction band and valence band.**
- P, S and Cl are **non-metals** of Period 3. They do not have delocalised electrons and there is a **large gap between the conduction band and valence band.** Therefore, they are insulators which **cannot conduct electricity.**

Factors that Affect the Strength of Metallic Bonds

- The **strength** of **metallic bonds** depends on the **number of valence electrons.**

$$\text{Strength of metallic bond} \propto \frac{\text{Number of valence electron per atom}}{\text{Ionic radius}}$$

- Thus, metallic bond is **weaker** in **sodium** (**1** valence e^-) compared to
magnesium (**2** valence e^-) and **aluminium** (**3** valence e^-).

Effect of Strength of Metallic Bonding on Melting and Boiling Point of Metals

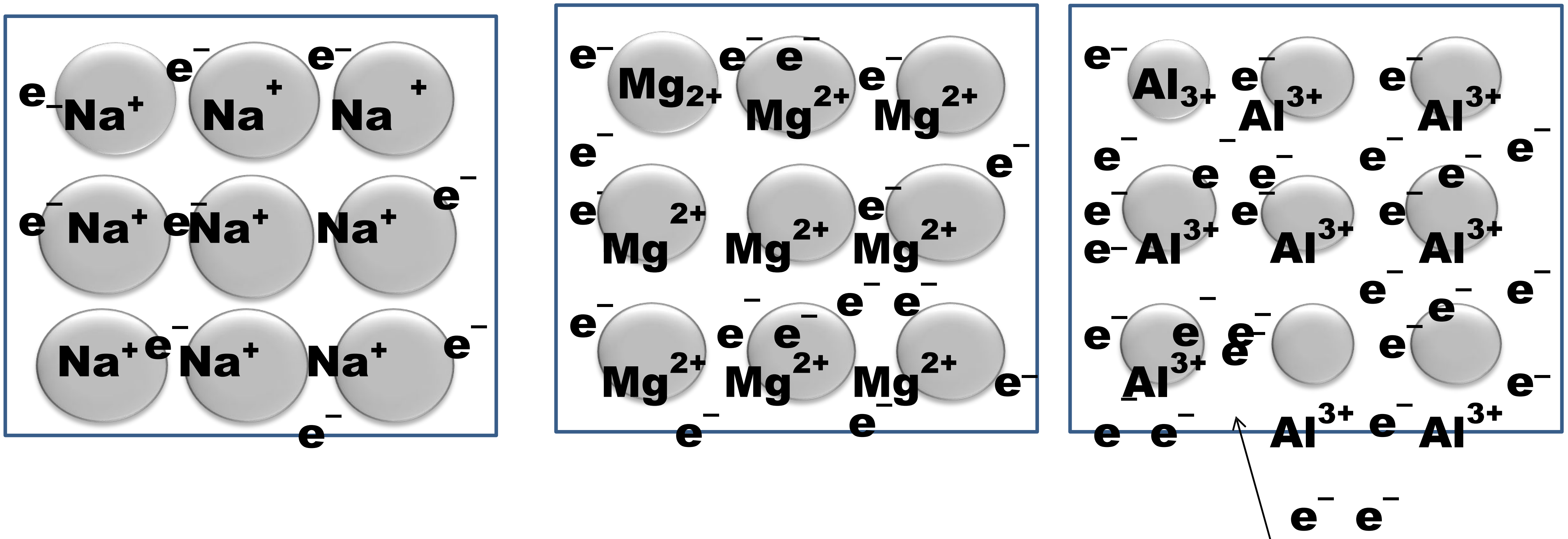
Element	Na	Mg	Al
Melting point (^0C)	98	650	660
Boiling point (^0C)	892	1107	2450
Number of valence e^-	1	2	3

- Na, Mg and Al are metals in Period 3.
- **Across period 3 from Na, Mg to Al**,
  number of of valence electrons increases
  ionic radius of metallic ion decreases

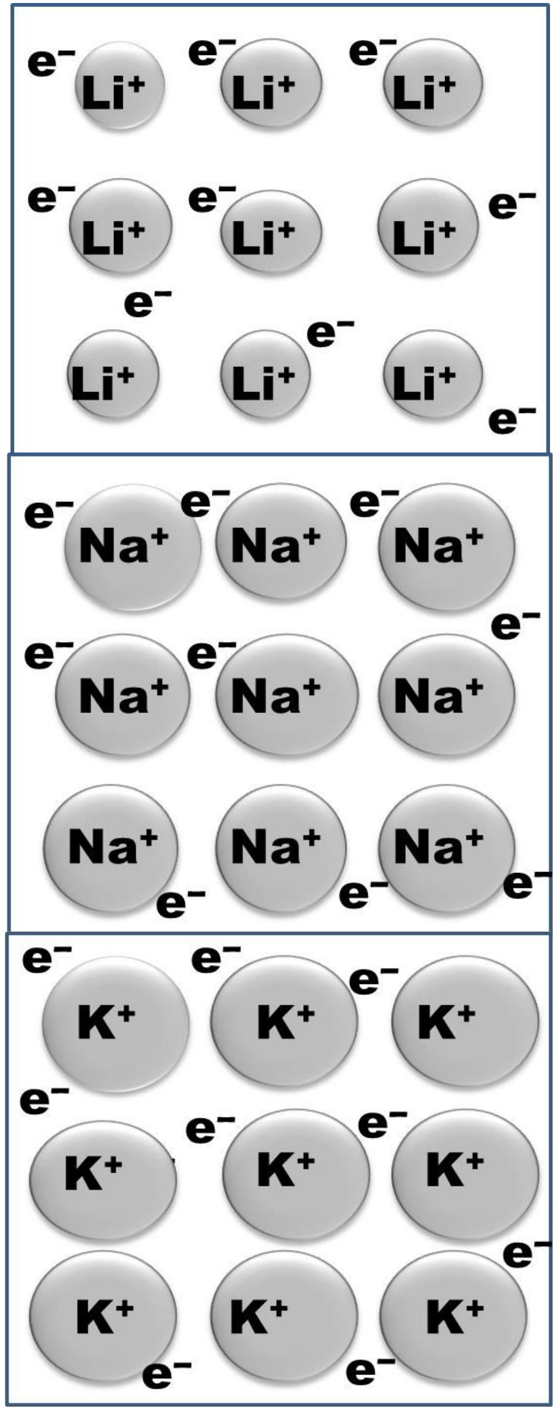
  As a result, the strength of metallic bond also increases
- The **stronger** the metallic bonds in the metal lattice, the **higher the melting and boiling points** of the metal.

The Electron Sea Model

Across Period 3



Sea of delocalised valence electrons



Trend Down Group 1

Element	Melting Point	Boiling Point (^oC)
Li	181	1330
Na	98	890
K	63	760
Rb	39	688
Cs	28	671

- ❖ Li, Na and K are metals in Group 1 that have 1 valence electron.
- ❖ Down the group, from Li to Cs,

  ionic radius increases

  ⇒ The strength of metallic bond decreases

∴ The melting and boiling points decrease.

Trend Down Group 17

Element	Boling point (^oC)	Melting point (^oC)
F_2	-188	-233
Cl_2	-35	-103
Br_2	59	-7.2
I_2	184.4	113

❖ Going down Group 17 (halogen), the boiling point and melting point **increases.**

❖ Going down a group, **size of molecules becomes larger** which lead to a **stronger van der Waals forces** and results in **higher melting and boiling points.**

VARIATION IN MELTING AND BOILING POINTS OF PERIOD 3 ELEMENTS

01
Na, Mg and Al

Metallic bond

- ✔ Na, Mg and Al are **metals**.
- ✔ Strength of metallic bond directly proportional to number of valence electron or ionic radius

02
Si

Covalent bond

- ✔ Each Si is tetrahedrally bonded to four other Si atoms by strong covalent bonds in a **giant covalent network.**
- ✔ High energy is required to overcome the strong covalent bond.

$\Rightarrow$ The highest melting / boiling point compared to other period 3 elements

03
P_4, S_8, Cl_2 and Ar

van der Waals forces

- ✔ Exist as **simple covalent molecules**.
- ✔ The bigger the molecular size, the stronger the van der Waals forces.
- ✔ Melting / boling points: $S_8 > P_4 > Cl_2 > Ar$

Example :

Explain the trend of melting point and boiling for elements in Period 3.

Element	**Na**	**Mg**	**Al**	**Si**	**P**	**S**	**Cl**	**Ar**
Melting Point	98	639	660	1410	44	113	-101	-189
Boiling Point	890	1120	2450	2680	280	445	-34	-186

Answer :

<u>From Na to Al</u>

Na, Mg and Al are **all metals** that form **metallic bond**.
Strength of **metallic bond is directly proportional to the number of valence electron**.
Na has **1 valence electron**, **Mg** has **2 valence electrons** and **Al** has **3 valence electrons.** Hence, the **boiling point** or **melting point increases** from **Na to Mg to Al.**

Si

Si has strong covalent bonds linking the atoms **together** in a **giant covalent network**. **High energy is required** to overcome the strong covalent bonds, so silicone has the highest melting point and boiling point.

From P to Ar

P, S, Cl and Ar exist as covalent molecules of P_4, S_8, Cl_2 and Ar respectively.

Phosphorous (M_r P_4 = 124), Sulphur(M_r S_8= 256), Chlorine (M_r Cl_2= 71) and Argon (M_r Ar = 40), all are **held by van der Waal forces**.

Strength of van der Waals forces depends on the **molecular size**.

The **larger the molecules**, the **stronger** the **van der Waals forces**.

Size of S_8 > P_4 > Cl_2 > Ar. So, the strength of van der Waals forces of

$S_8 > P_4 > Cl_2 >$ Ar.

Therefore, the boiling point and melting points **increase from P to S** but **decreases from S to Ar**.

THE END